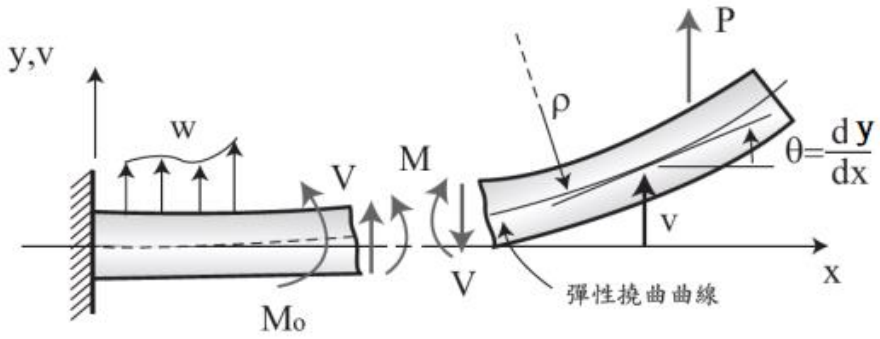
CH12　樑的彎曲變形

12-1.積分法-撓曲線方程式

一、撓曲線方程式

（一）撓曲線方程式



如圖　所示一根任意懸臂樑，以變形前直樑的軸線爲x軸，垂直向上的軸爲y軸，當樑在xy面內發生彎曲時，樑的軸線由直線變爲xy面內的一條光滑連續曲線，稱爲樑的撓曲曲線，樑彎曲後橫截面仍然垂直於樑的撓曲線，樑發生彎曲時，各個截面不僅發生了線位移，而且還產生了角位移。

直樑中橫截面的形心在垂直於 x 軸方向的線位移，稱爲直樑的撓度，用符號 v 表示，在圖示座標系下，撓度向上爲正，向下爲負，各個截面的撓度是截面形心座標 x 的函數，即可表示爲 $y=y(x)$，此式稱之爲撓曲曲線方程式。

(二)撓曲線近似微分方程

對細長梁，梁上的彎矩 M 和相應截面處梁軸的曲率半徑 ρ 均為截面位置 x 的函數，因此，梁的撓曲線的曲率可表為

$$\frac{1}{\rho(x)}=\frac{M(x)}{EI}$$

即梁的任一截面處撓曲線的曲率與該截面上的彎矩成正比，與截面的抗彎剛度 EI 成反比。另外，由微積分知曲線 $y=(x)$ 任一點的曲率為

$$\frac{1}{\rho(x)}=\pm\frac{y''}{\left[1+(y')^2\right]^{\frac{3}{2}}}$$

顯然，上述關係同樣適用於撓曲線。比較上兩式，可得

$$\pm \frac{y''}{\left[1+(y')^2\right]^{\frac{3}{2}}} = \frac{M(x)}{EI}$$

上式稱為撓曲線微分方程。這是一個二階非線性常微分方程，求解是很困難的。而在工程實際中，梁的撓度 y 和轉角 θ 數值都很小，因此，$(y')^2$ 之值和 1 相比很小，可以略去不計，於是，該式可簡化為

$$\pm y'' = \frac{M(x)}{EI}$$

式中左端的正負號的選擇，與彎矩 M 的正負符號規定及座標系的選擇有關。根據彎矩 M 的正負符號規定，當梁的彎矩 $M > 0$ 時，梁的撓曲線為凹曲線，按圖示座標系，撓曲線的二階導函數值 $y'' > 0$ ；反之，當梁的彎矩 $M < 0$ 時，撓曲線為凸曲線，在圖示坐標系中撓曲線的 $y'' < 0$ 。可見，在圖示右手座標系中，梁上的彎矩 M 與撓曲線的二階導數 y'' 符號一致。所以，上式的左端應取正號，即

$$\mathrm{y''} = \frac{\mathrm{M(x)}}{\mathrm{EI}}$$

(三) 撓曲線微分方程求解

上式稱為撓曲線近似微分方程。實踐表明，由此方程求得的撓度和轉角，對工程計算來說，已足夠精確，梁的撓曲線近似微分方程可用直接積分的方法求解，將撓曲近似微分方程

$$y'' = \frac{M(x)}{EI}$$

積分，可得梁的轉角方程為

$$\theta(x) = y' = \int \frac{M(x)}{EI} dx + C$$

再積分一次，即可得梁的撓曲線方程

$$y(x) = \int \left[\int \frac{M(x)}{EI} dx \right] dx + Cx + D$$

式中 C 和 D 為積分常數，它們可由梁的約束所提供的已知位移來確定。

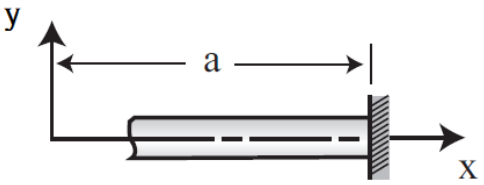
當梁的載荷較多或截面的形狀，尺寸沿梁軸改變時，各段梁的撓曲近似微分方程也不相同。這樣，在求梁的變形時就要分段寫出不同的撓曲近似微分方程，積分後，除了利用梁的邊界條件來確定部分積分常數外，同時還必須考慮撓曲線的光滑連續性，即利用各梁段交界處梁的撓曲線的連續條件來確定其餘的積分常數

(四)內力關係式

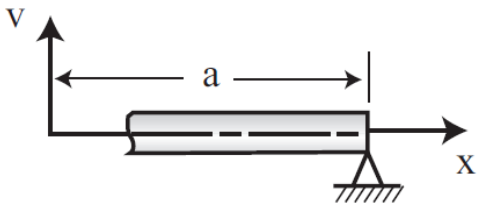
內剪力 V、彎矩 M 與負載密度 w之關係，可由上述式子所構成，在EI爲定值的前提下，可得下列一連串相當有用之關係式。

$$EI\frac{d^4y}{dx^4}=-w(x) \qquad EI\frac{d^3y}{dx^3}=V(x) \qquad EI\frac{d^2y}{dx^2}=M(x)$$

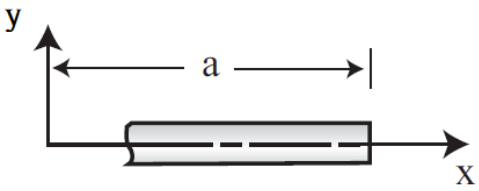
二、撓曲線方程式之邊界條件



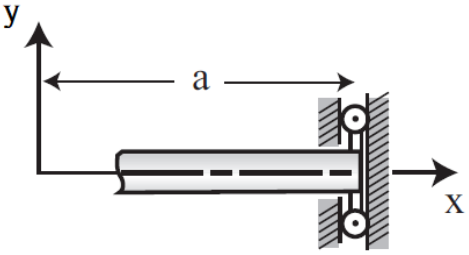
$y(a)=0$
$\theta(a)=0$



$y(a)=0$
$M(a)=0$



$y(a)=0$
$M(a)=0$



$\theta(a)=0$
$y(a)=0$

範例 1

Derive the equation of the deflection curve for a cantilever beam AB supporting a load P at the free end (see figure). Also, determine the deflection δ_B and angle of rotation θ_B at the free end. (Note: Use the second-order differential equation of the deflection curve.)



【105 成大醫工】【105 北科土木】

【中文翻譯】

導出支持一集中負載於自由端之懸臂樑 *AB* 的撓曲曲線方程式（ 見圖 ）。此外，求在自由端之撓曲 δ_B 及旋轉角 θ_B 。(註：使用撓曲曲線的二階微分方程式。)

【解析】

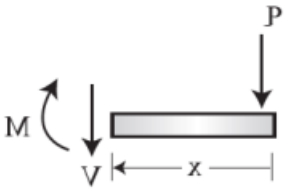
彈性曲線：負載將使樑彎曲成如圖，整根樑的內部彎矩可用單一x座標表示。彎矩函數：從自由物體圖，M作用在正方向，得M＝－Px

斜率與彈性曲線：

$$EI\frac{d^2\upsilon}{dx^2}=-Px \quad \cdots\cdots ①$$

$$EI\frac{d\upsilon}{dx}=-\frac{Px^2}{2}+C_1 \quad \cdots\cdots ②$$

$$EI\upsilon=-\frac{Px^3}{6}+C_{1x}+C_2 \quad \cdots\cdots ③$$



利用邊界條件在x＝L，$d\upsilon/dx=0$ 及在x＝L，$\upsilon=0$ ，②及③式變成

$$0=-\frac{PL^2}{2}+C_1 \qquad 0=-\frac{PL^3}{6}+C_1L+C_2$$

因此，$C_1=PL^2/2$ 及 $C_2=-PL^3/3$ 。將此結果代入②及③式

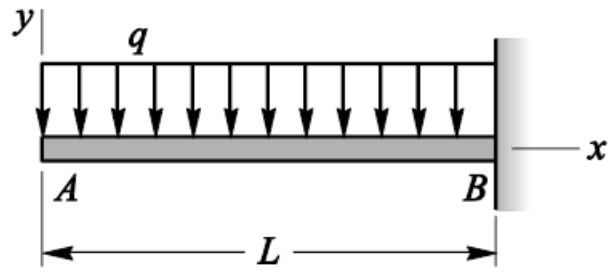
及 $\theta=d\upsilon/dx$ ，得 $\theta=\frac{P}{2EI}\left(L^2-x^2\right)$ $\upsilon=\frac{P}{6EI}\left(-x^3+3L^2x-2L^3\right)$

最大斜率及位移發生在A(x＝0)，即

$$\theta_A=\frac{PL^2}{2EI} \quad \cdots\cdots ④ \qquad \upsilon_A=-\frac{PL^3}{3EI} \quad \cdots\cdots ⑤$$

範例 2

Derive the equations of the deflection curve for a cantilever beam AB carrying a uniform load of intensity q over the span (see figure). Also, determine the deflection B at the end of the beam.



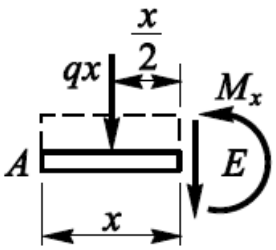
【106 台大應力】

【中文翻譯】

求受有強度 q 作用於樑之一部份之懸臂樑 AB 之撓曲曲線之方程式（ 見圖 ）。此外，求在樑端之撓角 θ_A 及撓曲 δ_A 。

【解析】

(一)內力分析



(二)撓曲線方程式

$$\Sigma M_E = 0 \ ， \ M_x = -(qx)\left(\frac{x}{2}\right) = -\frac{qx^2}{2}$$

$$EIy'' = M_x = -\frac{qx^2}{2} \quad \text{.....①}$$

$$EIy' = -\frac{qx^3}{6} + C_1$$

(三)帶入邊界條件

$BC(1)：x=L$，$y'=0$，$-\frac{qL^3}{6}+C_1=0$，$C_1=\frac{qL^3}{6}$

$$EIy'=-\frac{qx^3}{6}+\frac{qL^3}{6}\text{.....②}$$

$$EIy=-\frac{qx^4}{24}+\frac{qL^3}{6}x+C_2$$

$BC(2)：x=L$，$y=0$，$-\frac{qL^4}{24}+\frac{qL^4}{6}+C_2=0$，$C_2=-\frac{qL^4}{8}$

得 $y=-\frac{qx^4}{24EI}+\frac{qL^3x}{6EI}-\frac{qL^4}{8EI}$，或 $y=-\frac{q}{24EI}(x^4-4L^3x+3L^4)$

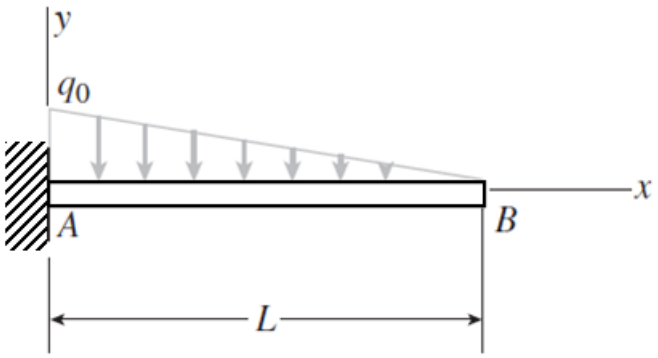
(四)計算撓角及撓度

$$\theta_A=y_A'=\frac{qL^3}{6EI}\text{ (ccw)}$$

$y_A=-\frac{qL^4}{8EI}$，或 $y_A=\frac{qL^4}{8EI}(\downarrow)$

範例 3

A cantilever beam AB supporting a triangularly distributed load of maximum intensity q_o is shown in the figure. Derive the equation of the deflection curve and then obtain formulas for the deflection δ_B and angle of rotation θ_B at the free end. (*Note:* Use the second-order differential equation of the deflection curve.)



【104 成大醫工】【108 台大應力】【108 台大土木】

【中文翻譯】

一懸臂樑 AB 支持一三角形分佈負載，最大強度為 q_0，如圖所示。求撓曲曲線的方程式，並且再獲得在自由端之撓曲 δ_B 及旋轉角 θ_B 的公式。(註：使用撓曲曲線之二階微分方程式。)

【解析】

（一）撓曲線方程式

$$EIv'' = M = -\frac{q_0}{6L}(L-x)^3$$

$$EIv' = \frac{q_0}{24L}(L-x)^4 + C_1$$

B.C. $v'(0) = 0 \quad \therefore C_1 = -\frac{q_0 L^3}{24}$

$$EIv = -\frac{q_0}{120L}(L-x)^5 - \frac{q_0 L^3 x}{24} + C_2$$

(二)帶入邊界條件

B.C. $v(0) = 0 \qquad \therefore\ c_2 = \dfrac{q_0 L^4}{120}$

$$v = -\frac{q_0 x^2}{120\,LEI}\,(10L^3 - 10L^2 x + 5Lx^2 - x^3)$$

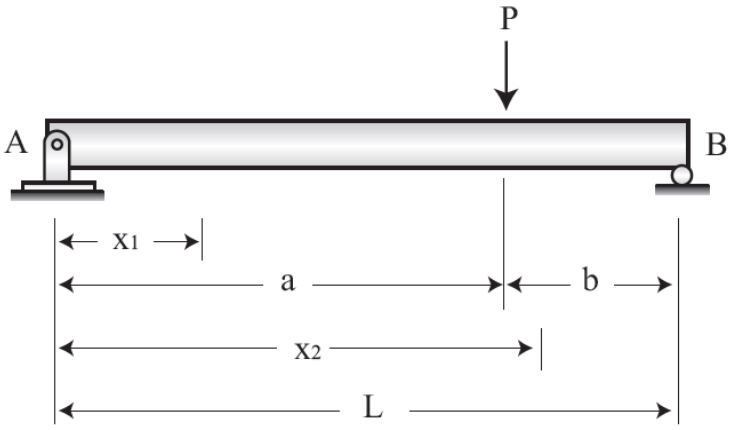
$$v' = -\frac{q_0 x}{24\,LEI}\,(4L^3 - 6L^2 x + 4Lx^2 - x^3)$$

$$\delta_B = -v(L) = \frac{q_0 L^4}{30\,EI}$$

$$\theta_B = -v'(L) = \frac{q_0 L^3}{24\,EI}$$

範例 4

如圖所示簡支樑AB受到集中力P作用，求(1)彈性曲線方程式。(2)求樑之最大旋轉角。(3)求樑之最大撓度。



【104 中央機械】

【解析】

：(1) 求反力並列樑的彎矩方程 $R_A=\frac{b}{L}P \quad R_A=\frac{a}{L}P$

(2) 建立座標系，分兩段列出AB樑的彎矩方程為：

AC段 $M_1(x_1)=\frac{b}{L}Px_1 \qquad (0\le x_1\le a)$

CB段 $M_2(x_2)=\frac{b}{L}Px_2-P(x_2-a) \qquad (a\le x_2\le L)$

(3) 對撓曲線近似微分方程積分，將AC和CB兩段的撓曲線近似微分方程及積分結果，列表如下。

AC段 $(0\le x_1\le a)$	CB段 $(a\le x_2\le L)$
$EIv_1''=\frac{Pb}{L}x_1$	$EIv_2''=\frac{Pb}{L}x_2-P(x_2-a)$
$EIv_1'=\frac{Pb}{2L}x_1^2+C_1$	$EIv_2'=\frac{Pb}{2L}x_2^2-\frac{P}{2}(x_2-a)^2+C_2$
$EIv=\frac{Pb}{6L}x_1^3+C_1x_1+D_1$	$EIv_2=\frac{Pb}{6l}x_2^3-\frac{P}{6}(x_2-a)^3+C_2x_2+D_2$

(4) 確定積分常數C_1、D_1和C_2、D_2，需要連續條件和邊界條件來確定。

A. 當$x_1 = x_2 = a$時，$\theta_1 = \theta_2$、$v_1 = v_2$

$$\frac{Pb}{6L}a^2 + C_1 = \frac{Pb}{6L}a^2 - \frac{P}{2}(L-a)^2 + C_2$$

$$\frac{Pb}{6L}a^3 + C_1 a + D_1 = \frac{Pb}{6L}a^3 - \frac{p}{6}(L-a)^3 + C_2 a + D_2$$

由上兩式解得$C_1 = C_2, D_1 = D_2$

B. 樑在A、B兩端的邊界條件：

$x_1 = 0$時$y_1 = 0$，$x_2 = L$時，$y_2 = 0$

$$D_1 = 0 \qquad \frac{Pb}{6l}L^2 - \frac{P}{6}(L-a)^3 + C_2 L = 0$$

解得$D_1 = D_2 = 0$ $\qquad C_1 = C_2 = -\frac{Pb}{6L}(L^2 - b^2)$

(5) 樑AC和CB段的轉角方程和撓曲線方程列於下表：

AC段 $0 \le x_1 \le a$	CB段 $a \le x_2 \le L$
$\theta_1(x_1) = -\frac{Pb}{6EIL}(L^2 - b^2 - 3x_1^2)$	$\theta_2(x_2) = -\frac{Pb}{6EIL}\left[(L^2 - b^2 - 3x_1^2) + \frac{3L}{b}(x_2 - a)^2\right]$
$v_1(x_1) = -\frac{Pbx_1}{6EIL}(L^2 - b^2 - x_1^2)$	$v_2(x_2) = -\frac{Pb}{6EIL}\left[(L^2 - b^2 - x_2^2) + \frac{L}{b}(x_2 - a)^3\right]$

(6) 求樑的最大撓度和轉角

A. 在樑的左端截面的旋轉角為 $\theta_A = \theta_1(x_1)\big|_{x_1=a} = -\dfrac{Pab(L+b)}{6EIL}$

B. 在樑右端截面的轉角為 $\theta_B = \theta_2(x_2)\big|_{x_{21}=a} = \dfrac{Pab(L+a)}{6EIL}$

當 $a > b$ 時，可以斷定 θ_B 為最大轉角。

C. 為了確定撓度為極值的截面，先確定 C 截面的轉角

$$\theta_C = \theta_1(x_1)\big|_{x_1=a} = \frac{Pab}{3EIL}(a-b)$$

若 $a > b$，則旋轉角 $\theta_C > 0$，AC段撓曲線為光滑連續曲線，

而 $\theta_A < 0$，當轉角從截面 A 到截面 C 連續地由負值變為正值時，

AC段內必有一截面轉角為零。為此，令 $\theta_1(x_1) = 0$，即

$$-\frac{Pb}{6EIL}(L^2 - b^2 - 3x_0^{\,2}) = 0 \Rightarrow x_0 = \sqrt{\frac{L^2-b^2}{3}}$$

x_0 的轉角為零，亦即撓度最大的截面位置，由AC段的撓曲線方程可

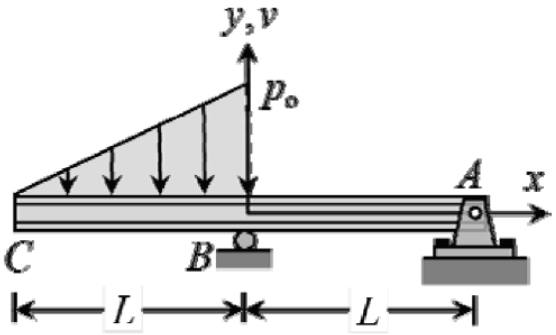
求得AB梁的最大撓度為 $v_{max} = \dfrac{Pb}{9\sqrt{3}EIL}\sqrt{(L^2-b^2)^3}$

範例 5

如圖四所示，長度為2L的彈性伸臂（over－hanging）梁ABC，端點A受鉸接支撐（hinge support），中點B受滾柱支撐（roller support），BC 段承受線性分布荷載（distributed loading），梁的重量可忽略不計，撓曲剛度（flexural rigidity）EI為常數。試求：

(一) 梁AB段的撓度曲線（deflection curve）。

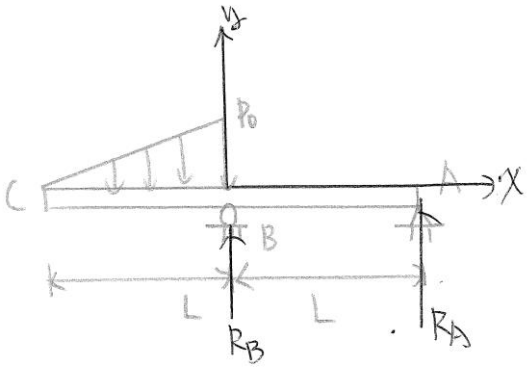
(二) 梁在支撐點A、B處的傾角（slope）。



【106 地特四等】

【解析】

(一)

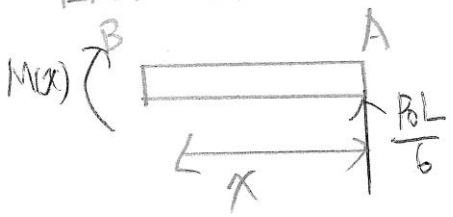


先求支承反力 R_A

$\circlearrowleft \sum M_D = 0$

$$\frac{1}{2} \times L \times P_0 \times \frac{L}{3} + R_A L = 0 \Rightarrow R_A = \frac{-P_0 L}{6}$$

由AB段



$$M(x)=\frac{-P_0L}{6}\cdot x$$

$$y''=\frac{M(x)}{EI}=\frac{1}{EI}M(x)=\frac{-1}{EI}\left(\frac{P_0L}{6}\cdot x\right.$$

$$y'=\frac{P_0L}{EI}\times\frac{-1}{12}x^2+C_1$$

$$y=\frac{P_0L}{EI}\times\frac{-1}{36}x^3+C_1x+C_2$$

由邊界條件 $y(0)=0$, $y(L)=0$

$$C_2=0\ ,\ C_1=-\frac{P_0L^3}{36EI}$$

故 $y=\frac{-P_0L}{36EI}x^3+\frac{P_0L^3}{36EI}x$

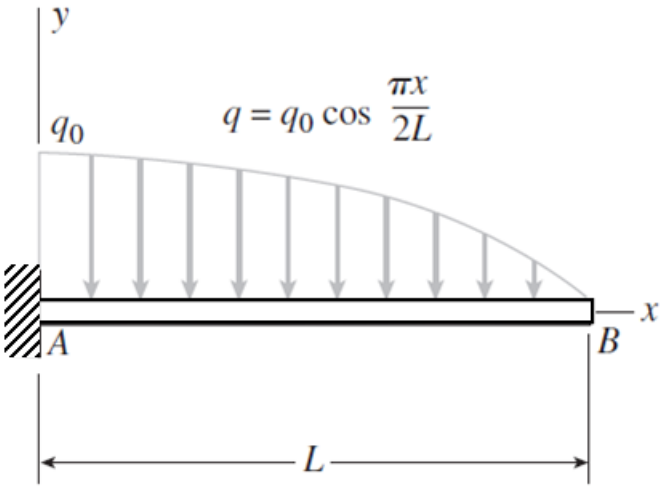
(三) $y'=\frac{-P_0L}{12EI}x^2+\frac{P_0L^3}{36EI}\Rightarrow\theta_A=y'(0)=\frac{P_0L^3}{36EI}$ (↻)

$$\theta_B=y'(L)=-\frac{P_0L^3}{18EI}$$

$$\theta_B=\frac{P_0L^3}{18EI}\ (\circlearrowleft)$$

範例 6

The distributed load acting on a cantilever beam AB has an intensity q given by the expression $q = q_0 \cos \pi x / 2L$, where q_0 is the maximum intensity of the load (see figure). Derive the equation of the deflection curve, and then determine the deflection B at the free end. Use the fourth-order differential equation of the deflection curve (the load equation).



【104 成大航太】

【中文翻譯】

用於懸臂樑 AB，具有 $q = q_0 \cos \pi x / 2L$ 之分佈負載，其中 q_0 為最大強度 (見圖)。導出撓曲曲線之方程式，且再求在自由端之撓曲 δ_B。使用撓曲曲線之四階微分方程式 (負載方程式)。

【解析】

$$EIv'''' = -q = -q_0 \cos \frac{\pi x}{2L}$$

$$EIv''' = -q_0 \left(\frac{2L}{\pi} \right) \sin \frac{\pi x}{2L} + C_1$$

$$EIv''' = V \quad EIv'''(L) = 0 \quad \therefore C_1 = \frac{2q_0 L}{\pi}$$

$$EIv'' = q_0 \left(\frac{2L}{\pi} \right)^2 \cos \frac{\pi x}{2L} + \frac{2q_0 Lx}{\pi} + C_2$$

$$EIv'' = M \quad EIv''(L) = 0 \quad \therefore\ C_2 = -\frac{2q_0L^2}{\pi}$$

$$EIv' = q_0\left(\frac{2L}{\pi}\right)^3 \sin\frac{\pi x}{2L} + \frac{q_0Lx^2}{\pi} - \frac{2q_0L^2x}{\pi} + C_3$$

$$v'(0) = 0 \quad \therefore C_3 = 0$$

$$EIv = -q_0\left(\frac{2L}{\pi}\right)^4 \cos\frac{\pi x}{2L} + \frac{q_0Lx^3}{3\pi} - \frac{q_0L^2x^2}{\pi} + C_4$$

$$v(0) = 0 \quad \therefore\ C_4 = \frac{16q_0L^4}{\pi^4}$$

$$v = -\frac{q_0L}{3\pi^4EI}\left(48L^3\cos\frac{\pi x}{2L} - 48L^3 + 3\pi^3Lx^2 - \pi^3x^3\right)$$

$$\delta_B = -v(L) = \frac{2q_0L^4}{3\pi^4EI}(\pi^3 - 24)$$

12-2.積分法-奇異函數法(singular funtion)

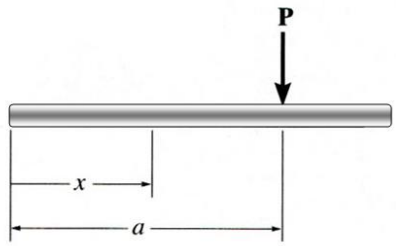
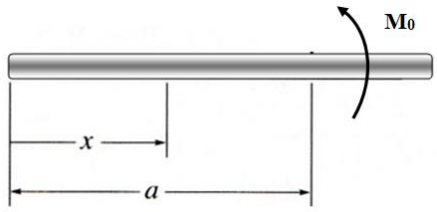
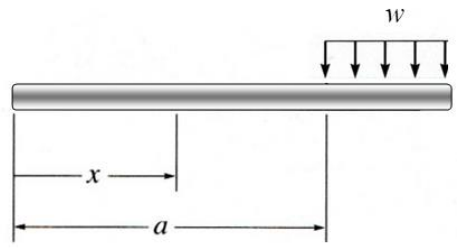
一、 奇異函數的定義

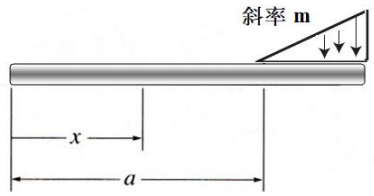
奇異函數法為不連續函數求解樑彈性曲線的方法，此法在解決包含數個負載之樑或軸的問題時特別有效，因為僅使用邊界條件就可計算出兩個積分常數，而相容條件自動滿足，我們將討論利用單一表示式或者由作用在樑上的負載，或內彎矩，來求出多重負載樑的彈性曲線方程式，描述負載的情況。其一般式可寫成：

$$< x-a >^n = \begin{cases} 0 & \text{for } x < a \\ (x-a)^n & \text{for } x \geq a \end{cases}$$
$$n \geq 0$$

x 為樑上一點的座標位置，而 a 為樑上產生不連續的位置

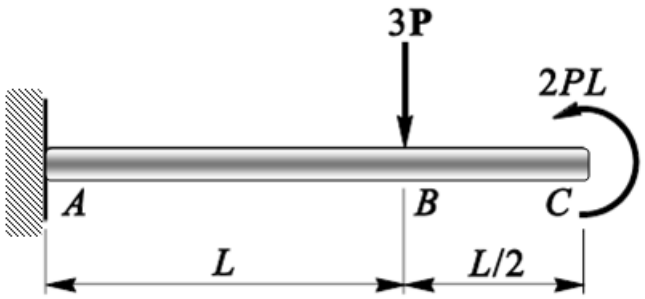
二、奇異函數描述各負載

負載	負載函數 $w = w(x)$	剪力 V $\mathrm{V} = -\int \mathrm{w}dx$	彎矩 $\mathrm{M} = \int \mathrm{V}dx$
1. 集中負載 	$w = \mathrm{P}\langle x-a\rangle^{-1}$	$V = -\mathrm{P}\langle x-a\rangle^{0}$	$M = -\mathrm{P}\langle x-a\rangle^{1}$
2. 彎矩 	$w = \mathrm{M}_0\langle x-a\rangle^{-2}$	$V = -\mathrm{M}_0\langle x-a\rangle^{-1}$	$M = -\mathrm{M}_0\langle x-a\rangle^{0}$
3. 均布負載 	$w = \mathrm{w}_0\langle x-a\rangle^{0}$	$V = -\mathrm{w}_0\langle x-a\rangle^{1}$	$M = -\frac{\mathrm{w}_0}{2}\langle x-a\rangle^{2}$

4. 均變負載 	$w = \mathrm{m}\langle x-a\rangle^{1}$	$V = -\frac{\mathrm{m}}{2}\langle x-a\rangle^{2}$	$M = -\frac{\mathrm{m}}{6}\langle x-a\rangle^{3}$

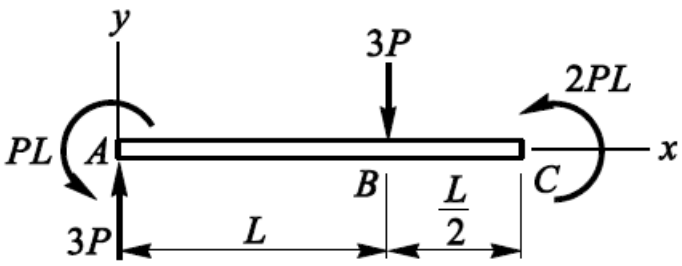
範例 7

圖中懸臂梁，試求 B、C 兩點之撓度。



【解析】

(一)內力分析



(二)由奇異函數法

$$EIy'' = -PL\langle x-0\rangle^0 + 3P\langle x-0\rangle^1 - 3P\langle x-L\rangle^1$$

$$= -PL + 3Px - 3P\langle x-L\rangle^1$$

$$EIy' = -PLx + \frac{3P}{2}x^2 - \frac{3P}{2}\langle x-L\rangle^2 + C_1 \quad \cdots\cdots ①$$

$$EIy = -\frac{PL}{2}x^2 + \frac{P}{2}x^3 - \frac{P}{2}\langle x-L\rangle^3 + C_1x + C_2 \quad \cdots\cdots ②$$

$BC(1)：x=0$，$y'=0$，得 $C_1=0$；$BC(2)：x=0$，$y=0$，得 $C_2=0$

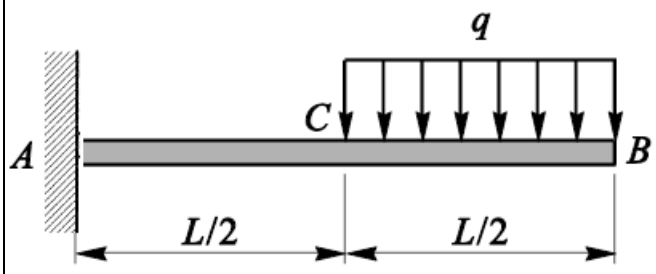
B 點撓度：$x=L$，由公式②，$EIy_B = -\frac{PL}{2}(L^2) + \frac{P}{2}(L^3) = 0$，$y_B = 0$

C 點撓度：$x=\frac{3L}{2}$，由公式②

$$EIy_C = -\frac{PL}{2}\left(\frac{3L}{2}\right)^2 + \frac{P}{2}\left(\frac{3L}{2}\right)^3 - \frac{P}{2}\left(\frac{L}{2}\right)^3 = \frac{PL^3}{2}，\ y_C = \frac{PL^3}{2EI}(\uparrow)$$

範例 8

A cantilever beam AB supporting a uniform load of intensity q acting over one-half of the length (see figure). Also, obtain formulas for the deflections B and C at points B and C, respectively.



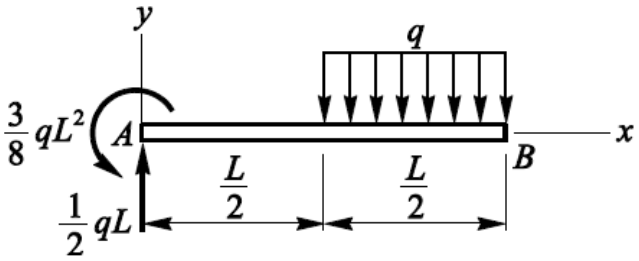
【機械高考】【99 清大動機】【中央土木】

【解析】

強度 q 作用於樑的一半長度上之均勻負載的懸臂樑 AB，試求在分別於點 B 及 C 之撓曲 δ_B 及 δ_C 的公式

【解析】

(一)內力分析



(二)由奇異函數法

$$EIy'' = -\frac{3}{8}qL^2 <x-0>^0 + \frac{1}{2}qL <x-0>^1 - \frac{q}{2}\left\langle x-\frac{L}{2}\right\rangle^2$$

$$= -\frac{3}{8}qL^2 + \frac{qL}{2}x - \frac{q}{2}\left\langle x-\frac{L}{2}\right\rangle^2$$

$$EIy' = -\frac{3qL^2}{8}x + \frac{qL}{4}x^2 - \frac{q}{6}\left\langle x-\frac{L}{2}\right\rangle^2 + C_1 \quad \cdots\cdots ①$$

$$EIy = -\frac{3qL^2}{16}x^2 + \frac{qL}{12}x^3 - \frac{q}{24}\left\langle x-\frac{L}{2}\right\rangle^4 + C_1x + C_2 \quad \cdots\cdots ②$$

$BC(1)：x=0，\ y'=0$，代入①式得 $C_1=0$

$BC(2)：x=0，\ y=0$，代入②式得 $C_2=0$

B 點$(x=L)$之斜角與撓度，由①及②式

$$EIy_B'=EI\theta_B=-\frac{3qL^2}{8}(L)+\frac{qL}{4}(L^2)-\frac{q}{6}\left(\frac{L}{2}\right)^3=-\frac{7qL^3}{48}$$

$$\theta_B=-\frac{7qL^3}{48EI}\ ，或\ \theta_B=\frac{7qL^3}{48EI}(\text{cw})$$

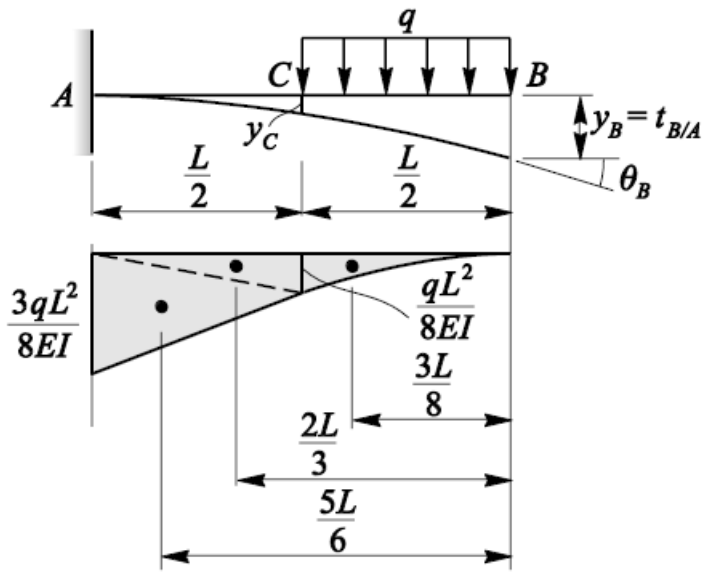
$$EIy_B=-\frac{3qL^2}{16}(L^2)+\frac{qL}{12}(L^3)-\frac{q}{24}\left(\frac{L}{2}\right)^4=-\frac{41qL^4}{384}$$

$$y_B=-\frac{41qL^4}{384EI}\ ，\ y_B=\frac{41qL^4}{384EI}(\downarrow)$$

同理

$$\delta_C=\frac{7qL^4}{192EI}$$

另解:力矩面積法



$$\theta_B = \frac{1}{3}\left(\frac{qL^2}{8EI}\right)\left(\frac{L}{2}\right)+\frac{1}{2}\left(\frac{qL^2}{8EI}\right)\left(\frac{L}{2}\right)+\frac{1}{2}\left(\frac{3qL^2}{8EI}\right)\left(\frac{L}{2}\right)$$

$$= \frac{7qL^3}{48EI}(\text{cw})$$

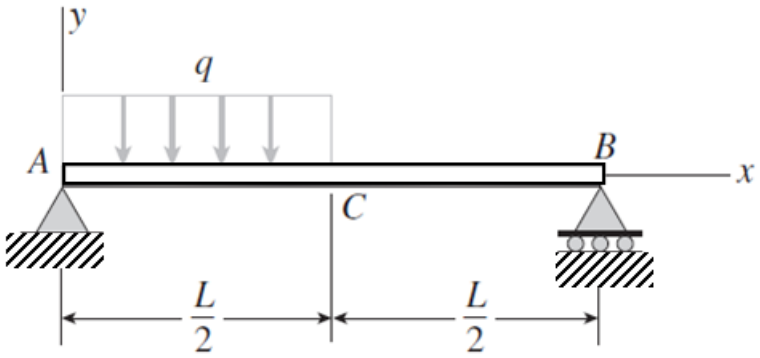
$$y_C = t_{C/A} = \left(\frac{qL^3}{32EI}\right)\left(\frac{L}{6}\right)+\left(\frac{3qL^3}{32EI}\right)\left(\frac{L}{3}\right)$$

$$= \frac{7qL^4}{192EI}(\downarrow)$$

$$y_B = t_{B/A} = \left(\frac{qL^3}{48EI}\right)\left(\frac{3L}{8}\right)+\left(\frac{qL^3}{32EI}\right)\left(\frac{2L}{3}\right)+\left(\frac{3qL^3}{32EI}\right)\left(\frac{5L}{6}\right) = \frac{41qL^4}{384EI}(\downarrow)$$

範例 9

Derive the equations of the deflection curve for a simple beam AB with a uniform load of intensity q acting over the left-hand half of the span (see figure). Also, determine the deflection δ_C at the midpoint of the beam.



【台大工科】

【中文翻譯】

導出支持強度 q 作用於樑的一半長度之均勻負載的簡樑 AB 之撓曲曲線的方程式。此外，求在樑中點的撓曲 δ_C。

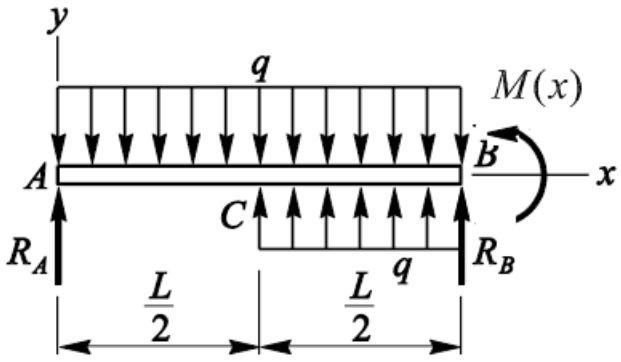
【解析】

(一)內力分析

由靜力平衡式可得支承反力

$$R_A = \frac{3}{8}qL(\uparrow),\ R_B = \frac{1}{8}qL(\uparrow)$$

(二)由奇異函數法修正負載



$$EIy'' = R_A x - \frac{q}{2}x^2 + \frac{q}{2}\left\langle x - \frac{L}{2}\right\rangle^2$$

$$EIy' = \frac{R_A}{2}x^2 - \frac{q}{6}x^3 + \frac{q}{6}\left\langle x - \frac{L}{2}\right\rangle^3 + C_1 \ldots\ldots ①$$

$$EIy = \frac{R_A}{6}x^3 - \frac{q}{24}x^4 + \frac{q}{24}\left\langle x - \frac{L}{2}\right\rangle^4 + C_1 x + C_2 \ldots\ldots ②$$

(三)帶入邊界條件

$$y(0) = 0 \Rightarrow C_2 = 0$$

$$y(L) = 0 \Rightarrow C_1 = -\frac{9}{384}qL^3$$

$$\therefore y = \frac{1}{EI}[\frac{1}{16}qLx^3 - \frac{q}{24}x^4 + \frac{q}{24} < x - \frac{L}{2} >^4 - \frac{9}{384}qL^3 x]$$

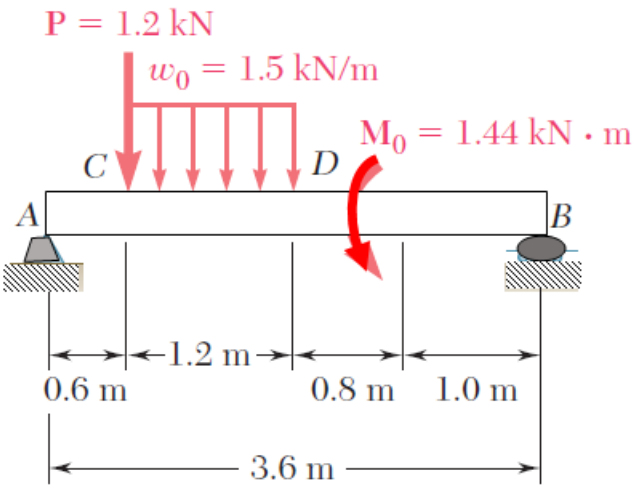
(四)梁的撓度計算

梁中點 C 之撓度

$$\delta_C = y(\frac{L}{2}) = -\frac{5qL^4}{768EI}$$

範例 10

請利用奇異函數法列出下面梁之彎矩方程式

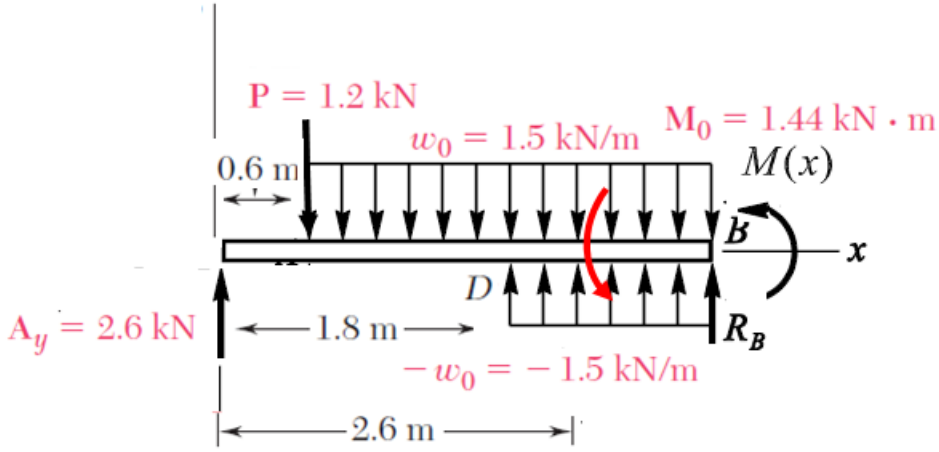


【解析】

(一)內力分析

Ay=2.6KN

(二)由奇異函數法修正負載



$$M(x) = -0.75\langle x - 0.6\rangle^2 + 0.75\langle x - 1.8\rangle^2 + 2.6x - 1.2\langle x - 0.6\rangle^1 - 1.44\langle x - 2.6\rangle^0$$

$$
\begin{aligned}
EI\theta = & -0.25\langle x-0.6\rangle^3 + 0.25\langle x-1.8\rangle^3 \\
& + 1.3x^2 - 0.6\langle x-0.6\rangle^2 - 1.44\langle x-2.6\rangle^1 + C_1 \\
EIy = & -0.0625\langle x-0.6\rangle^4 + 0.0625\langle x-1.8\rangle^4 + 0.4333x^3 \\
& - 0.2\langle x-0.6\rangle^3 - 0.72\langle x-2.6\rangle^2 + C_1 x + C_2
\end{aligned}
$$

12-3.力矩面積法

一、 力矩面積法第一定理:

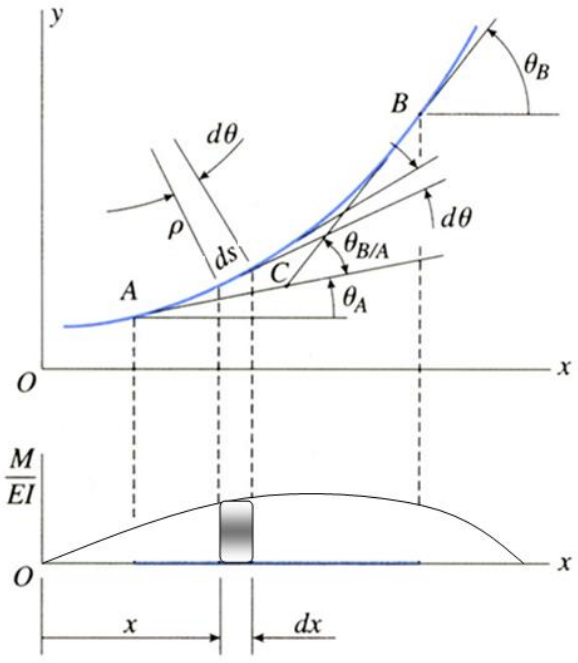
(一) 在分析直樑變形時，若無須知道其變形曲線函數，僅是要求得某些位置的撓度與偏位角度，可力用「$\frac{M}{EI}$圖」下面積來進行分析，此方法稱之為力矩面積法。

(二) 撓曲曲線之曲率k，其定義為$k=\frac{1}{\rho}=\frac{d\theta}{ds}$，若考慮樑之變形單純由彎矩所造成，則曲率與彎矩具有$k=-\frac{M}{EI}$之關係，此外，撓曲曲線之斜率則為$\frac{dy}{dx}=\tan\theta$。今假設樑之變形非常微小，亦即$ds\approx dx$、$\theta\approx\tan\theta$(θ之單位為徑度)，則可得

$\frac{d\theta}{dx}=\frac{M}{EI}\Rightarrow d\theta=\frac{M}{EI}dx$，兩邊取積分，可得$\theta=\int\frac{M}{EI}dx$。

(三) 由樑上選取A、B二點，由A點積分至B點，此積分值為直樑位於A點的旋轉角度與B點的旋轉角度差，表示彈性直樑上任意兩點的切線夾角差(B點與A點的斜率差)，等於「$\frac{M}{EI}$圖」曲線下的面積(由A點至B點)，此即為力矩面積法第一定理。

$$\theta_B-\theta_A=\theta_{B/A}=\int_A^B\frac{M}{EI}dx$$



二、 力矩面積法第二定理:

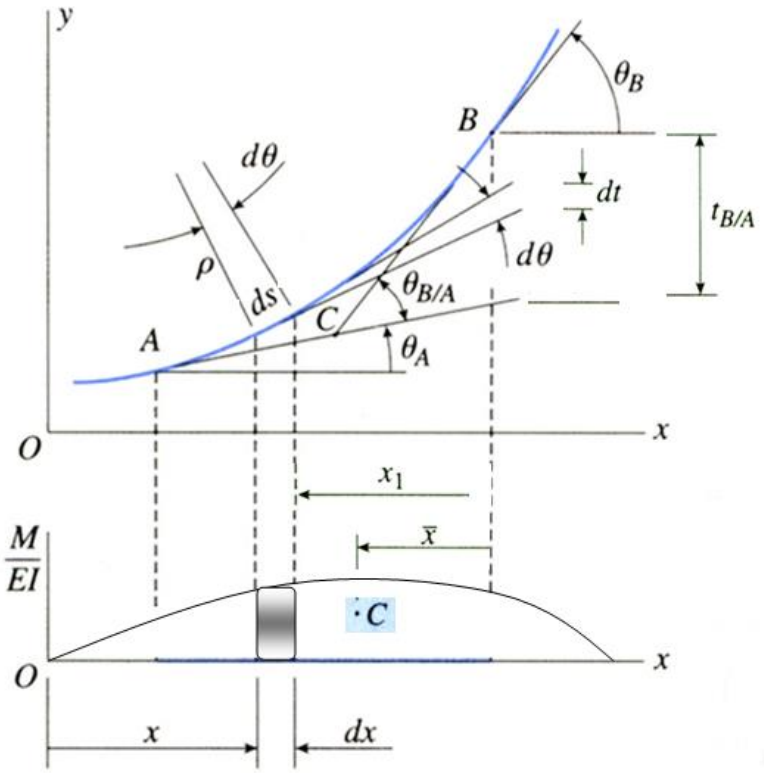
(一) 承第一定理所述，若由A點的切線與B點位移方向的交點算起，偏位Δ_{BA}為A點拉出切於v(x)之線，切線延伸至B點處時，此線與B點位移的垂直高度差，稱之為「切

線偏移量」，其等於由 A 端至 B 端 $\frac{M}{EI}$ 圖形曲線下的面積(由 A 點至 B 點)對 B 點的一次矩，可表示為：

$$\boxed{t_{B/A}=\int_A^B \frac{M}{EI}xdx}$$

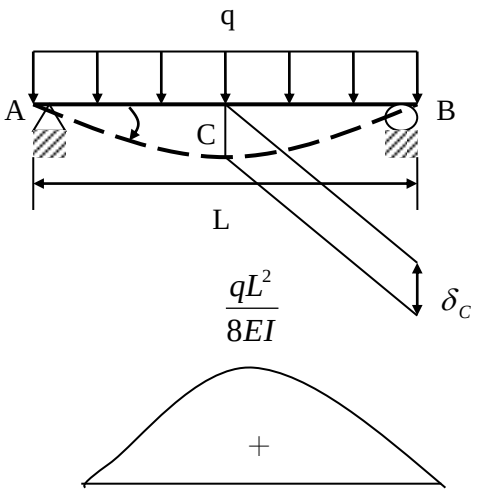
(二) 由於面積的形心 $\bar{x}\int dA=\int xdA \Rightarrow \boxed{t_{B/A}=\bar{x}\int_A^B \frac{M}{EI}dx}$，式中 $\bar{x}$ 表 A 點至曲線下區域形心的距離。

(三) 彈性直樑上一點(A)的切線相對於從另一點(B)所延伸的位移方向切之垂直偏移量，等於由 A 端至 B 端 $\frac{M}{EI}$ 圖形曲線下的面積(由 A 點至 B 點)對 B 點的一次矩，此稱之為力矩面積法第二定理。



三、 基本撓度變形與旋轉角:

圖式	力矩面積法



1.力矩面積法第一定理：

$$\theta_A - \theta_C = \theta_A - 0 = (\frac{2}{3})(\frac{1}{2})(\frac{1}{8})\frac{qL^3}{EI}$$

$$\boxed{\theta_A = \frac{qL^3}{24EI}}\text{(順時針)}$$

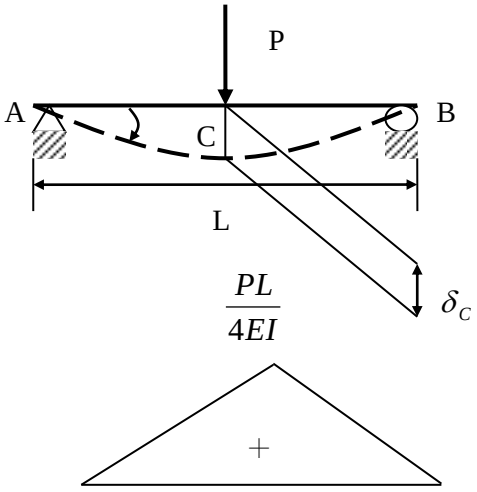
$$\boxed{\theta_B = \frac{qL^3}{24EI}}\text{(逆時針)}$$

2.力矩面積法第二定理：

$$\delta_C = \Delta_{AC} = (\frac{1}{24})(\frac{1}{2})(\frac{5}{8})\frac{qL^4}{EI}$$

(由C→A之面積矩)

$$\boxed{\delta_C = \frac{5qL^4}{384EI}(\downarrow)}$$



1.力矩面積法第一定理：

$$\theta_A - \theta_C = \theta_A - 0 = (\frac{1}{2})(\frac{1}{4})(\frac{1}{2})\frac{PL^2}{EI}$$

$$\boxed{\theta_A = \frac{PL^2}{16EI}}\text{(順時針)}$$

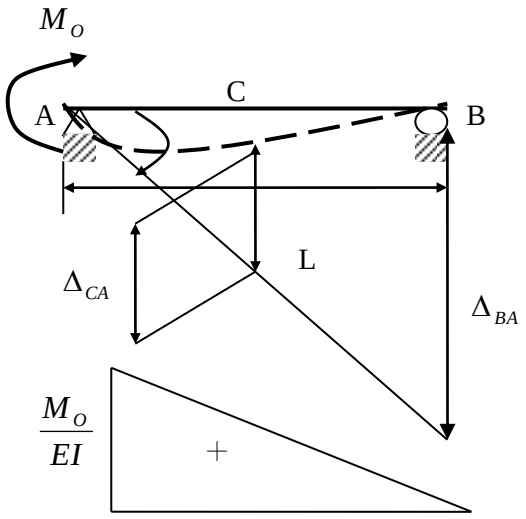
$$\boxed{\theta_B = \frac{PL^2}{16EI}}\text{(逆時針)}$$

2.力矩面積法第二定理：

$$\delta_C = \Delta_{AC} = (\frac{1}{16})(\frac{1}{2})(\frac{2}{3})\frac{PL^3}{EI}$$

(由C→A之面積矩)

$$\boxed{\delta_C = \frac{PL^3}{48EI}(\downarrow)}$$



1.力矩面積法第二定理：

$$\theta_A = \frac{\Delta_{BA}}{L} = (\frac{1}{2})(1)(1)(\frac{2}{3})\frac{M_O L}{EI}$$

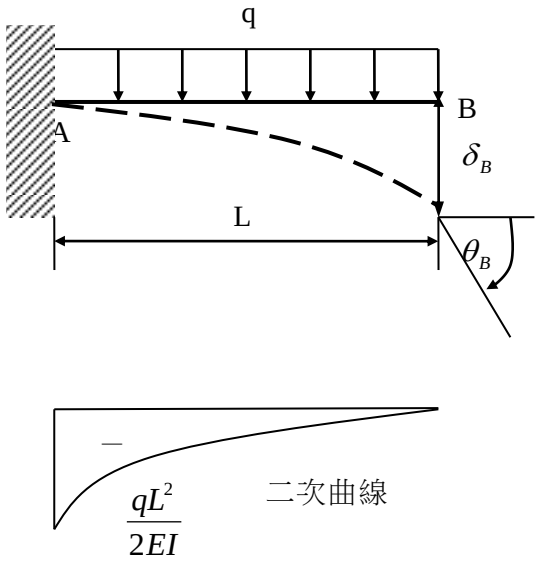
$$\boxed{\theta_A = \frac{M_O L}{3EI}}\text{(順時針)}$$

同理$\boxed{\theta_B = \frac{\Delta_{AB}}{L} = \frac{M_O L}{6EI}}$(逆時針)

$$\delta_C = \frac{\Delta_{BA}}{2} - \Delta_{CA}$$

$$= (\frac{1}{6}) - (\frac{3}{8}\times\frac{1}{6}\times\frac{\frac{1}{2}+2}{\frac{1}{2}+1})\frac{M_O L^2}{EI}$$

$$\boxed{\delta_C = \frac{M_O L^2}{16EI}(\downarrow)}$$



1.力矩面積法第一定理：

$$\theta_A = 0$$

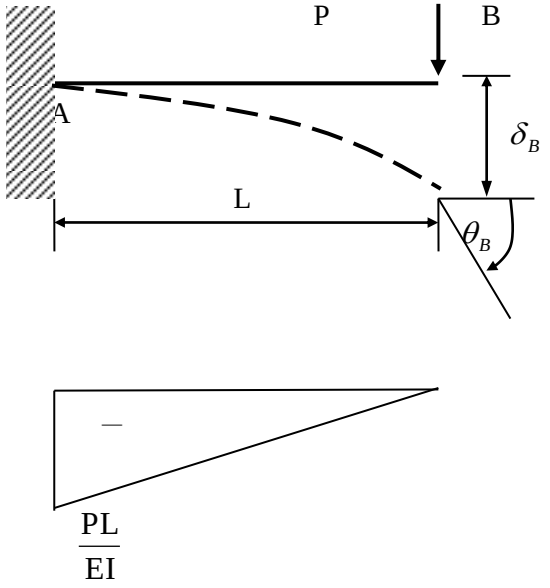
$$\theta_B - 0 = (\frac{1}{2})(1)(\frac{1}{3})\frac{qL^3}{EI}$$

$$\boxed{\theta_B = \frac{qL^3}{6EI}}\text{(順時針)}$$

2.力矩面積法第二定理：

$$\delta_B = \Delta_{BA} = (\frac{1}{6})(\frac{3}{4})\frac{qL^4}{EI}$$

$$\boxed{\delta_B = \frac{qL^4}{8EI}(\downarrow)}$$



1.力矩面積法第一定理：

$\theta_A = 0$

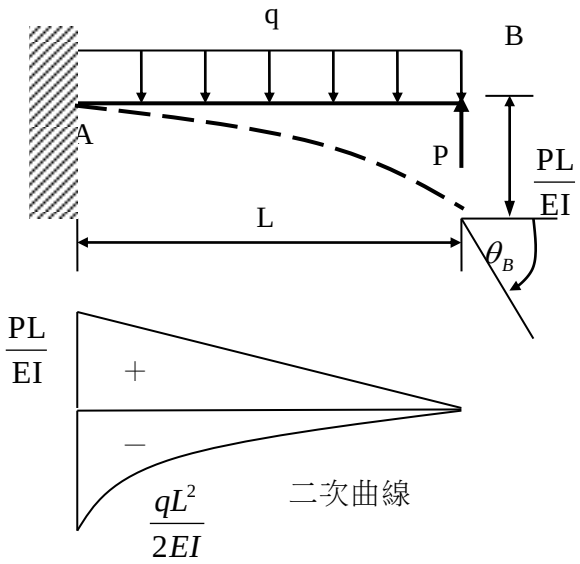
$\theta_B - 0 = (1)(1)(\frac{1}{2})\frac{PL^2}{EI}$

$\boxed{\theta_B = \frac{PL^2}{2EI}}$(順時針)

2.力矩面積法第二定理：

$\delta_B = \Delta_{BA} = (\frac{1}{2})(\frac{2}{3})\frac{PL^3}{EI}$

$\boxed{\delta_B = \frac{PL^3}{3EI}\ (\downarrow)}$



3.力矩面積法第一定理：

疊加的方式繪製彎矩圖求解

$\theta_A = 0$

$\theta_B - 0 = (\frac{1}{2})(1)(\frac{1}{3})\frac{qL^3}{EI} - \frac{PL^2}{2EI}$

$\boxed{\theta_B = \frac{qL^3}{6EI} - \frac{PL^2}{2EI}}$(取順時針為正)

4.力矩面積法第二定理：

$\delta_B = \Delta_{BA} = (\frac{1}{6})(\frac{3}{4})\frac{qL^4}{EI} - \frac{PL^3}{3EI}$

$\boxed{\delta_B = \frac{qL^4}{8EI} - \frac{PL^3}{3EI}\ (\downarrow)}$

四、 溫度彎曲:

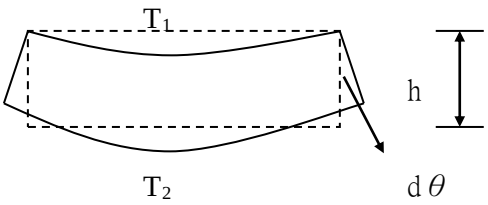
(一) 樑的頂部與樑的底部溫度不同(底部溫度>頂部溫度)：

$hd\theta = \alpha\,(T_2 - T_1)dx(T_2 > T_1)$

$\boxed{K_T = \left(\frac{\alpha\Delta T}{h}\right)}$

(二) 若樑受彎矩 M 之力，且樑的頂部與樑的底部溫度不同(底部溫度>頂部溫度)：

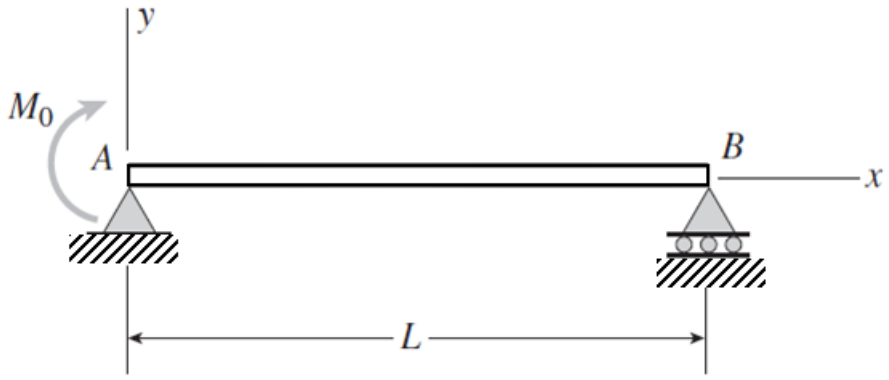
樑彎曲＝彎矩彎曲+溫差彎曲$\Rightarrow K = -\left(\frac{M}{EI} + \frac{\alpha\Delta T}{h}\right)$



溫度彎曲

範例 11

Derive the equation of the deflection curve for a simple beam AB loaded by a couple M_0 at the left-hand support (see figure). Also, determine the maximum deflection $\delta_{\max}$.



【103 台大工科】

【中文翻譯】

導出受有力偶 M_0 於左端支撐（ 見圖 ）之簡樑 AB 的撓曲曲線方程式（ 見圖 ）。此外，也求最大撓曲 $\delta_{\max}$ 。

【解析】

$$EIv'' = M = M_0\left(1 - \frac{x}{L}\right)$$

$$EIv' = M_0\left(x - \frac{x^2}{2L}\right) + C_1$$

$$EIv = M_0\left(\frac{x^2}{2} - \frac{x^3}{6L}\right) + C_1 x + C_2$$

B.C. $v(0) = 0 \quad \therefore C_2 = 0$

B.C. $v(L) = 0 \quad \therefore C_1 = -\dfrac{M_0 L}{3}$

$$v = -\frac{M_0 x}{6\,LEI}(2L^2 - 3Lx + x^2)$$

$$v' = -\frac{M_0}{6\,LEI}(2\,L^2 - 6\,Lx + 3\,x^2)$$

$$v' = 0$$

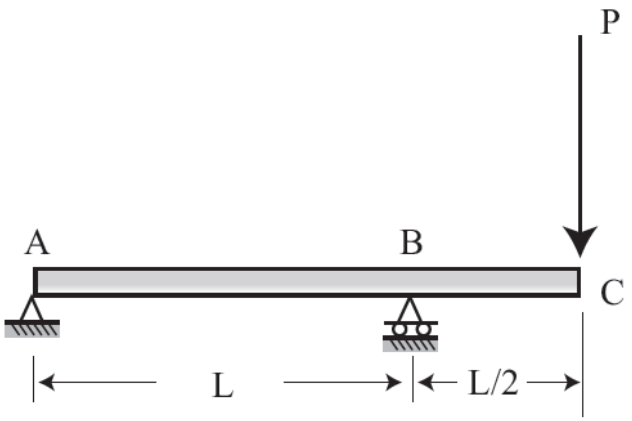
$$x_1 = L\left(1 - \frac{\sqrt{3}}{3}\right)$$

$$\delta_{\max} = -(v)_{x=x_1}$$

$$= \frac{M_0 L^2}{9\sqrt{3}EI}$$

範例 12

如圖所示，試求在C點之撓度。(EI為樑的撓度剛性)。(95年機械地特四等)



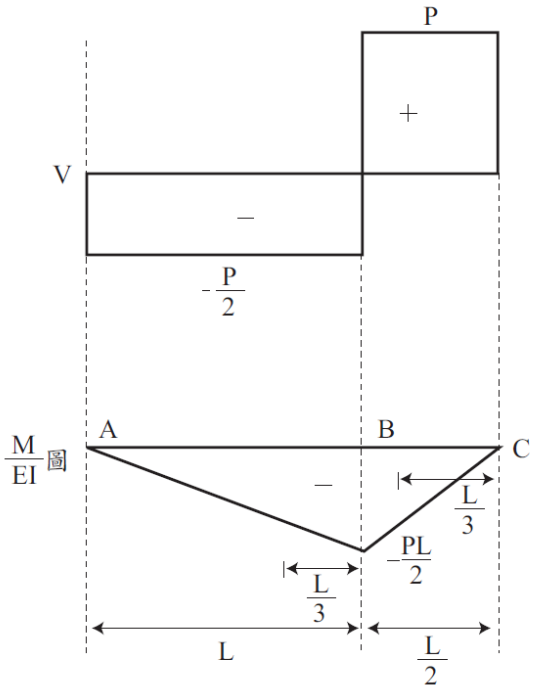
【台大機械】【101 交大機械】

【解析】

答：(1) 先求支承反力

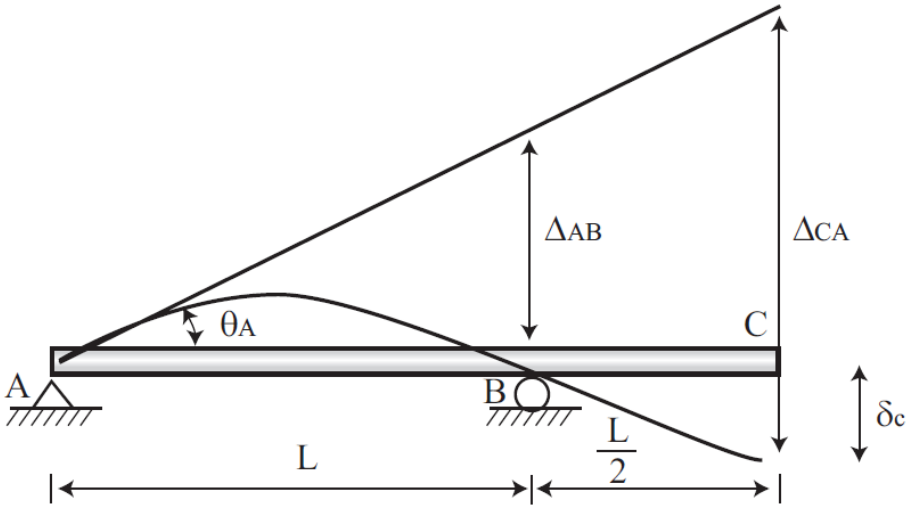
$$\Sigma M_A = 0 \Rightarrow R_B = \frac{3}{2}P(\uparrow) \quad \Sigma F_y = 0 \Rightarrow R_A = \frac{1}{2}P(\downarrow)$$

(2) 畫剪力圖及彎矩圖：



$\Rightarrow \frac{1}{2}\times\frac{2}{3}=\frac{L}{3}$(BC段的$\frac{M}{EI}$圖，距C點$\frac{L}{3}$)

(3) 樑之變形如下所示：



由力矩面積法第二定律

$$\Delta_{CA}=\frac{-1}{EI}\left[\frac{PL}{2}\times L\times\frac{1}{2}\times\left(\frac{L}{3}+\frac{L}{2}\right)+\frac{PL}{2}\times\frac{L}{2}\times\frac{1}{2}\times\frac{L}{3}\right]$$

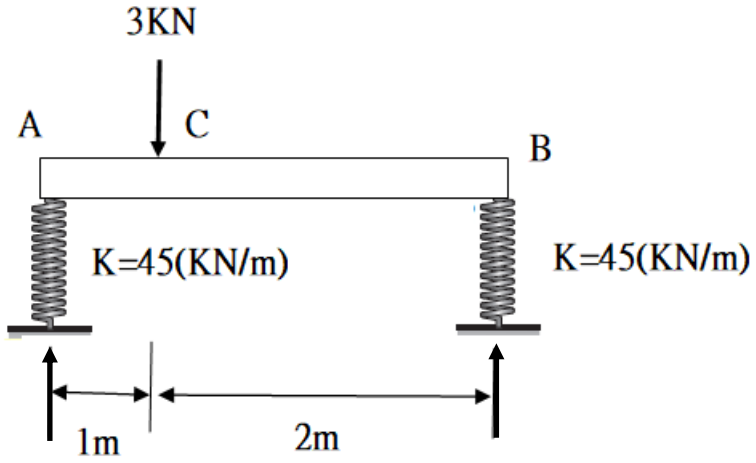
大三角形面積×面積矩 ＋ 小三角形面積×面積矩$=\frac{-PL^3}{4EI}(\downarrow)$

$$\Delta_{BA}=\frac{-1}{EI}\times\left[\frac{PL}{2}\times L\times\frac{1}{2}\times\frac{L}{3}\right]=\frac{-PL^3}{12EI}(\downarrow)$$

$\delta_C=\Delta_{CA}-\frac{3}{2}\times\Delta_{BA}=\frac{-PL^3}{8EI}$ 負號表示向下$=\frac{PL^3}{8EI}(\downarrow)$。

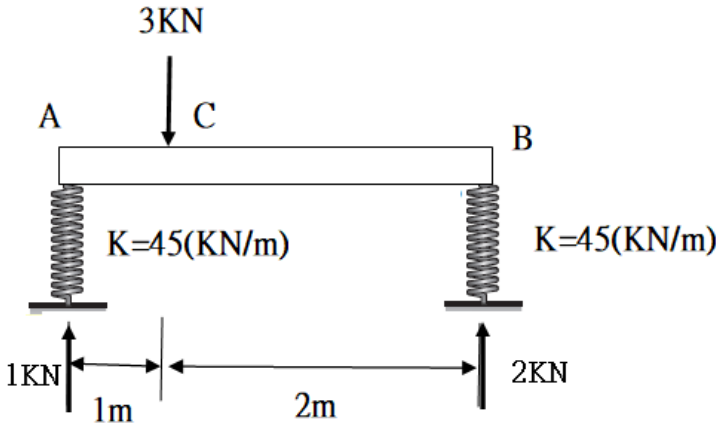
範例 13

如圖所示支鋼棒以兩根彈簧在 A、B 兩端支撐，每一根彈簧之勁度 K=45(KN/m)且開始時並未伸張，若此棒在 C 處施加一 3KN 之力，試求此力的鉛直位移。忽略棒的重量且 E=200GPa、$I = 4.6875 \times 10^{-6} m^4$

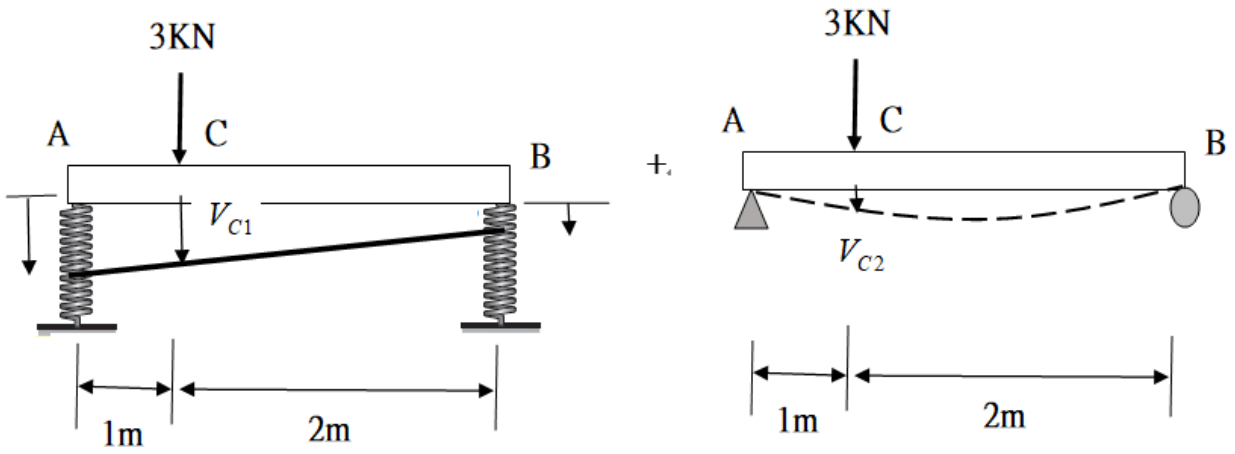


【解析】

(一)內力分析



(二)變形分析



$$(\upsilon_C)_1 = 0.0222\ \text{m} + \frac{2}{3}[0.0444\ \text{m} - 0.0282\ \text{m}] = 0.0370\ \text{m} \downarrow$$

$$(\upsilon_C)_2 = \frac{Pab}{6EIL}(L^2 - b^2 - a^2)$$

$$= \frac{(3\ \text{kN})(1\ \text{m})(2\ \text{m})[(3\ \text{m})^2 - (2\ \text{m})^2 - (1\ \text{m})^2]}{6(200)(10^6)\ \text{kN/m}^2\,(4.6875)(10^{-6})\ \text{m}^4\,(3\ \text{m})}$$

$$= 0.001422\ \text{m} = 1.422\ \text{mm} \downarrow$$

將兩個位移分量相加，則得

$$(+\downarrow)\quad \upsilon_C = 0.0370\ \text{m} + 0.001422\ \text{m} = 0.0384\ \text{m} = 38.4\ \text{mm} \downarrow$$

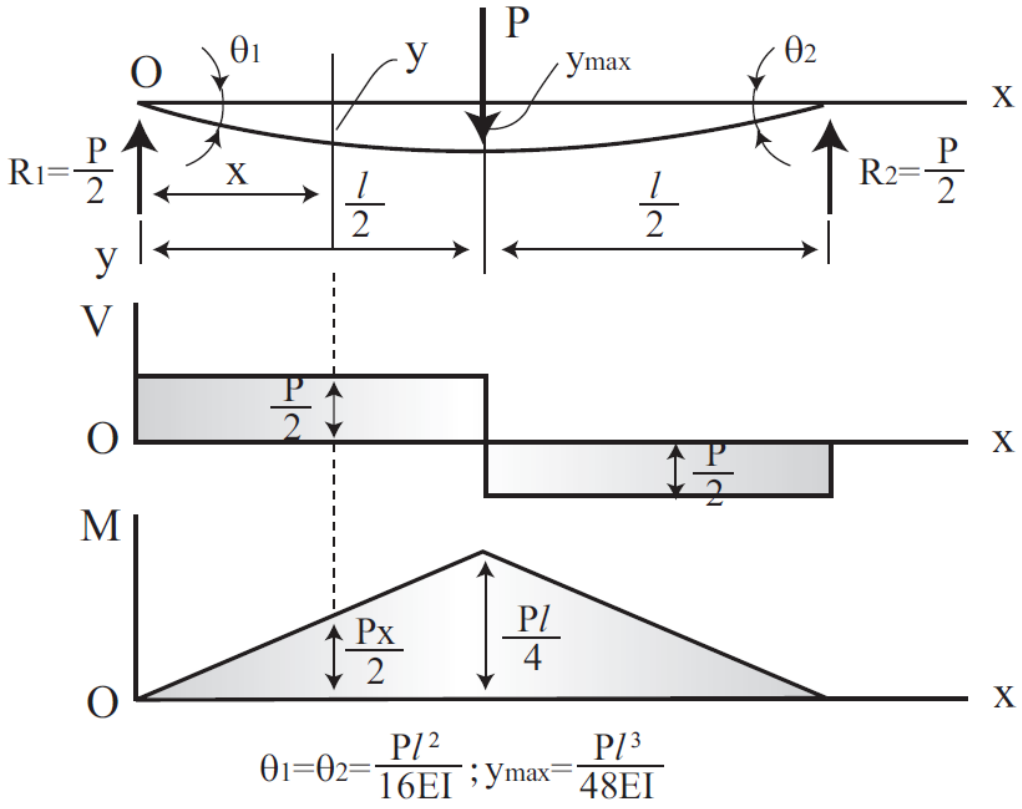
12-4.重疊原理

一、 重疊原理:

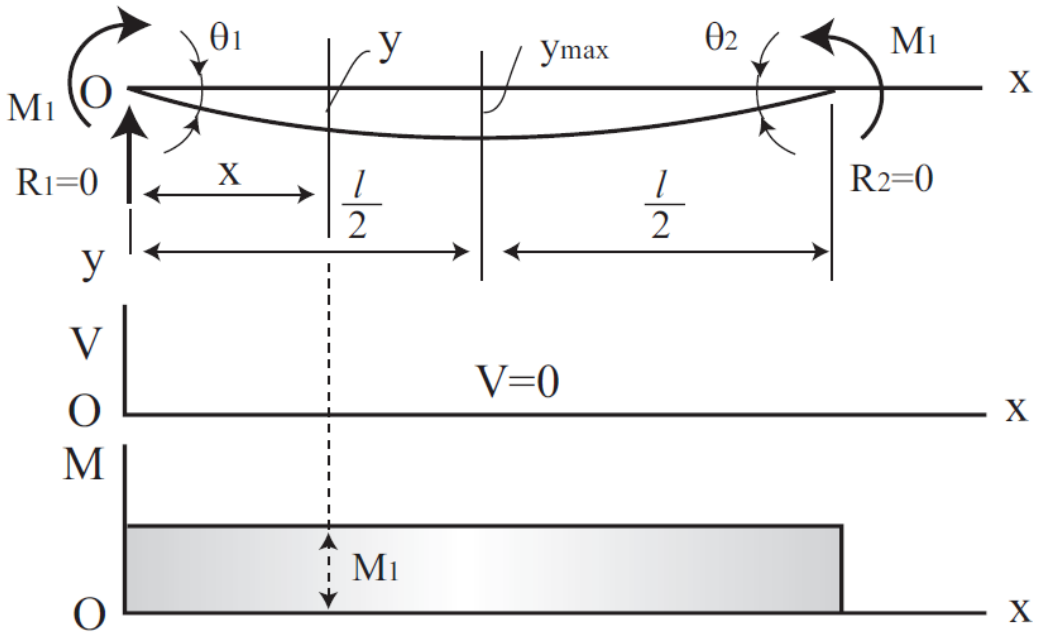
積分法及力矩面積法是求梁變形的基本方法，利用此方法求任意截面的撓度和轉角時，必須先求出梁的撓曲線方程和轉角方程。當梁上同時作用若干個載荷，而且只需要求出某些特定截面（如撓度為最大，或轉角為最大的截面）的撓度和轉角時（不需要知道撓曲線方程和轉角方程），這些方法就顯得繁瑣。在這種情況下，用疊加法求梁的變形要方便得多。線上彈性小變形前提下，構件的支反力、內力、應力和變形都可以用疊加法的方法計算，彎曲變形時，梁的撓度與轉角都與載荷成線性關係。因此，可以用疊加法計算梁的彎曲變形，當梁上有幾個載荷共同作用時，可以分別計算梁在每個載荷單獨作用時的變形，然後進行疊加，即可求得梁在幾個載荷共同作用時的總變形。應用疊加法求梁的變形時，若已知梁在簡單載荷作用時的變形，是很方便的。因此將梁在簡單載荷作用下的變形匯總在下表中，以便直接查用

線性疊加法

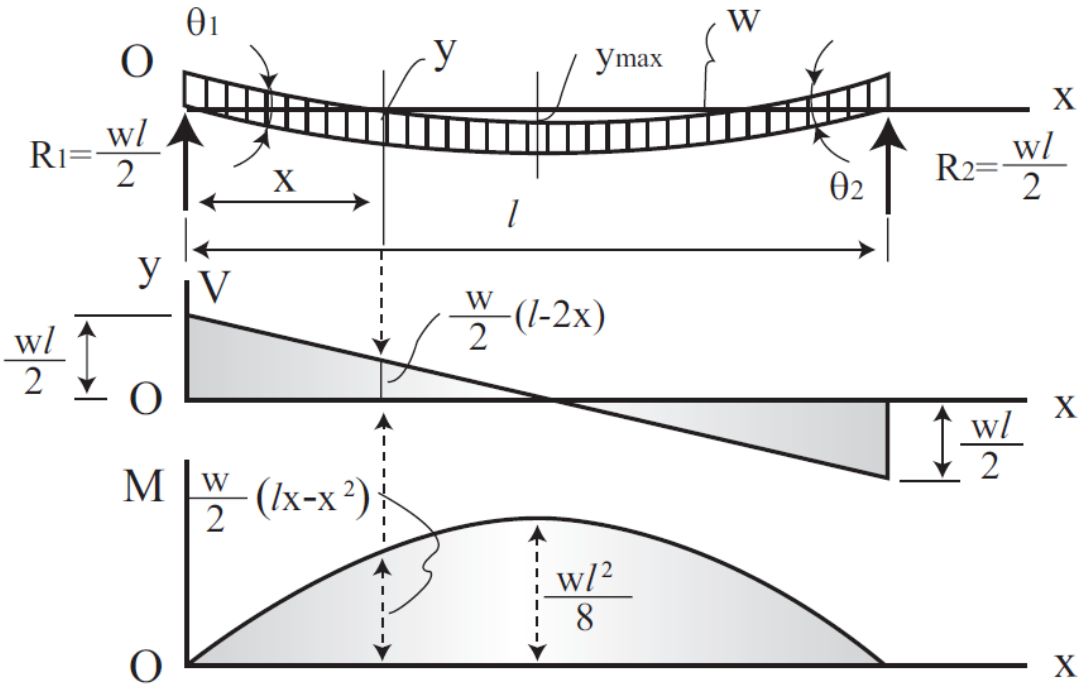
(1) 樑的基本撓度



$$\theta_1=\theta_2=\frac{Pl^2}{16EI}\ ;\ y_{max}=\frac{Pl^3}{48EI}$$

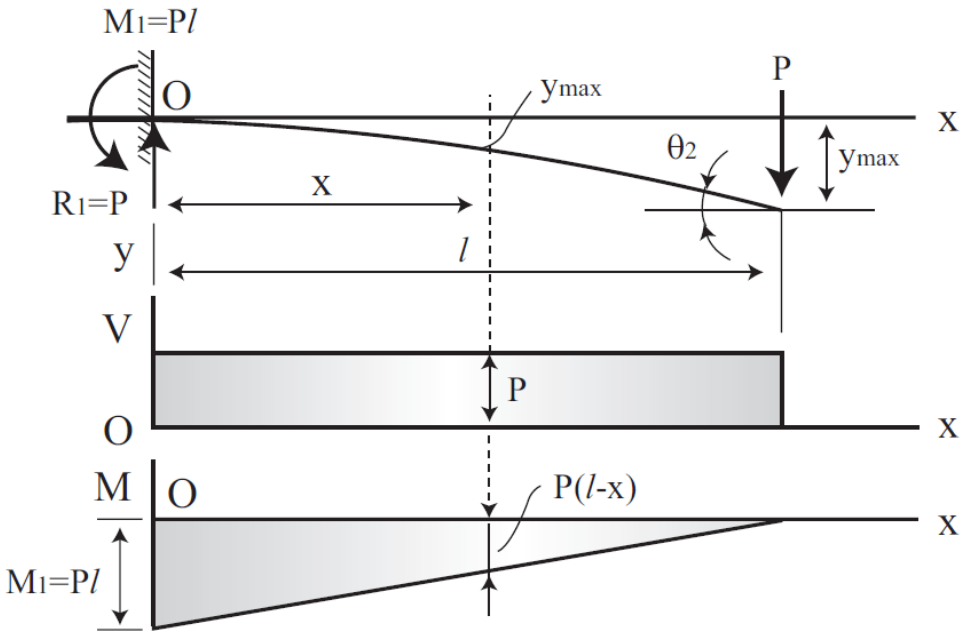


$$\theta_1 = \theta_2 = \frac{M_1 l}{2EI}; y = \frac{M_1}{2EI}(lx - x^2); y_{max} = \frac{M_1 l_2}{8EI}$$

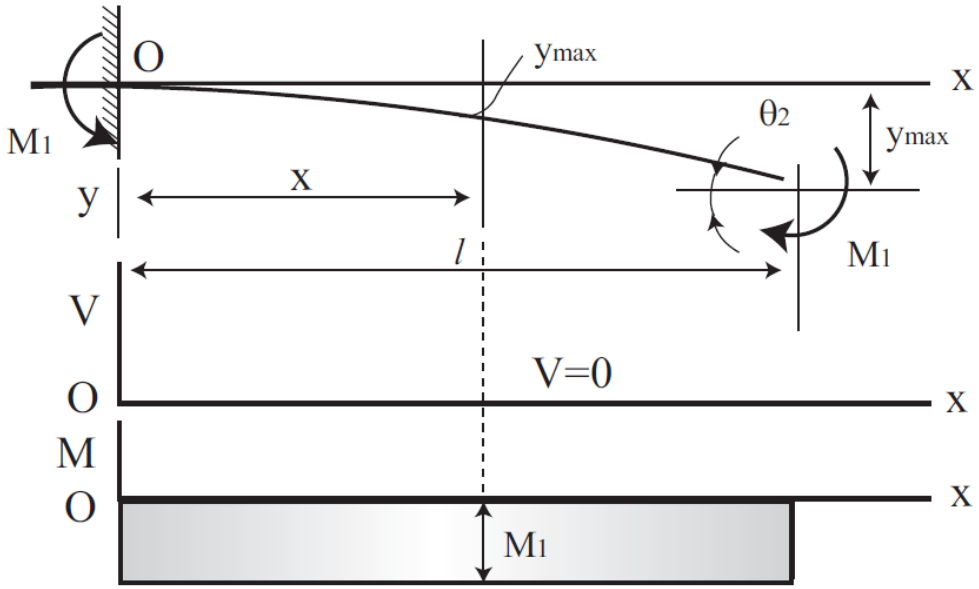


$$\theta_1 = \theta_2 = \frac{wl^3}{24EI}\ ; y_{max} = \frac{5wl^4}{384EI}$$

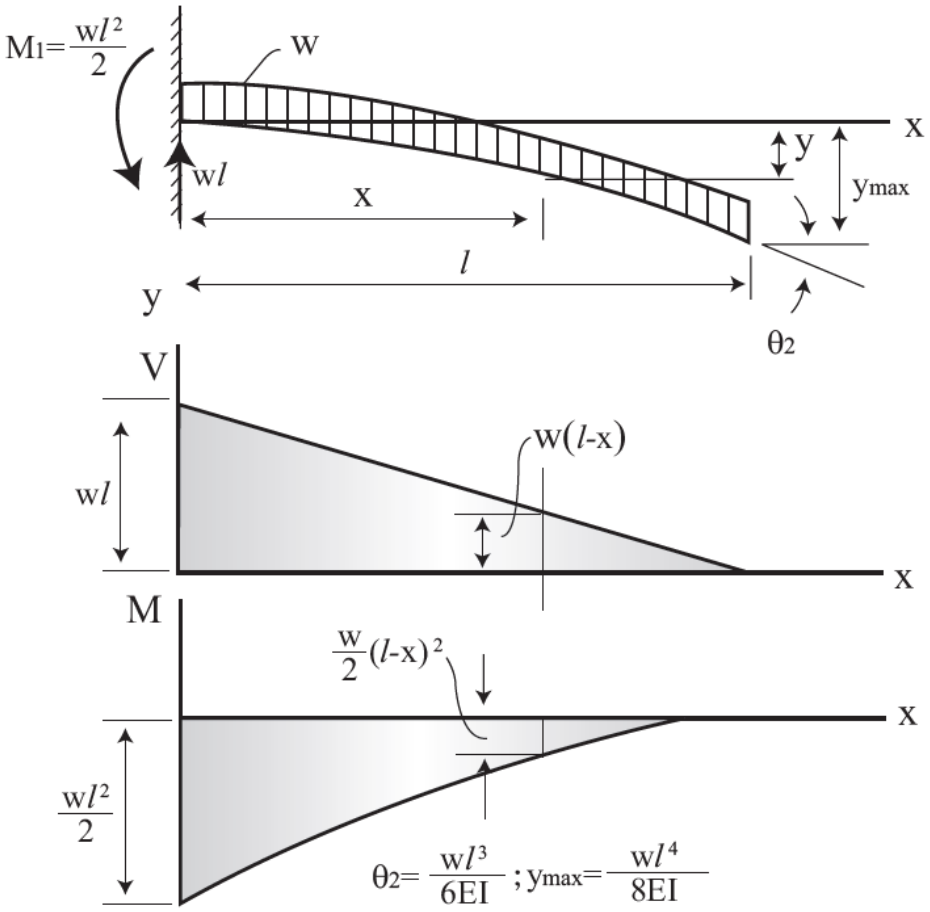
$$y = \frac{wx}{24EI}\left(l^3 - 2lx^2 + x^3\right)$$



$$\theta_2 = \frac{Pl^2}{2EI} \; ; \; y = \frac{P}{6EI}\left(3lx^2 - x^3\right) \; ; \; y_{max} = \frac{Pl^3}{3EI}$$

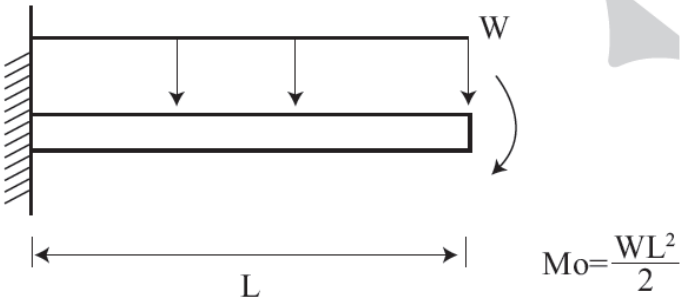


$$\theta_2 = \frac{M_1 l}{EI} \; ; \; y = \frac{M_1 x^2}{2EI} \; ; \; y_{max} = \frac{M_1 l^2}{2EI}$$


M1=wl²/2
w
wl
x
y
ymax
l
θ2
V
wl
w(l-x)
M
w/2(l-x)²
wl²/2
θ2= wl³/6EI ; ymax= wl⁴/8EI

範例 14

某左端固定之懸臂樑，長度為L、撓性剛度為EI。其上承受往下均布負載w作用，且其右端同時承受一順時針彎矩 $M_o = wL^2/2$ 作用。試求(1)樑最大之旋轉角（撓角），（以w、L、EI表之）。



(2)樑最大之撓度，（以w、L、EI表之）。(98年地方特考四等)

【解析】

答：(1) 利用基本撓度公式

A. 均布負載W作用之撓角 $M_0 = \frac{WL^2}{2}$　$\theta_1 = \frac{WL^3}{6EI}$ (↻)

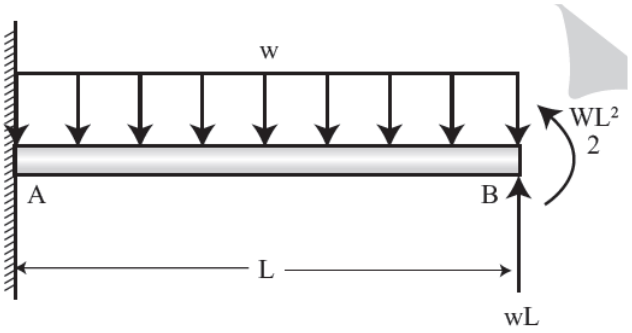
B. 彎矩 M_0 作用之撓角 $\theta_2 = \frac{M_0 L}{EI} = \frac{WL^3}{2EI}$ (↻)　$\theta = \theta_1 + \theta_2 = \frac{2WL^3}{3EI}$ (↻)

(2) A.均布負載W作用之撓度 $\delta_1 = \frac{WL^4}{8EI}$ (↓)

B. 彎矩 M_0 作用之撓度 $\delta_2 = \frac{M_0 L^2}{2EI} = \frac{WL^4}{4EI}$　$\delta = \delta_1 + \delta_2 = \frac{3WL^4}{8EI}$

範例 15

如圖所示之懸臂樑，楊氏係數（Young's modulus）為E，斷面面積矩以I表示，試求其在右側自由端B點處之垂向位移。(98年關務三等)



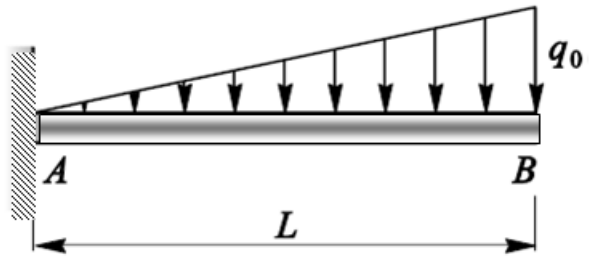
【解析】

答：利用基本撓度公式

$$\delta_B = \frac{WL^4}{8EI} - \frac{(WL)\times L^3}{3EI} - \frac{(\frac{WL^2}{2})\times L^2}{2EI} = -0.4583\times(\frac{L^4}{EI})(\uparrow)$$

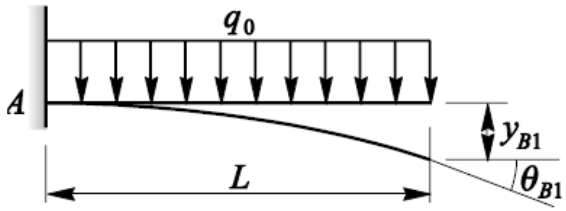
範例 16

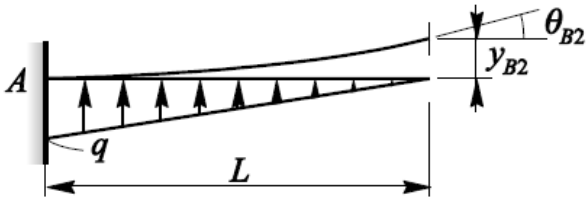
圖中懸臂梁承受均變負荷，試求 B 點之斜角與撓度。



【108 北科土木】

【解析】





$$y_{B1}=\frac{q_0L^4}{8EI}(\downarrow)\cdot\ \theta_{B1}=\frac{q_0L^3}{6EI}(\text{cw})$$

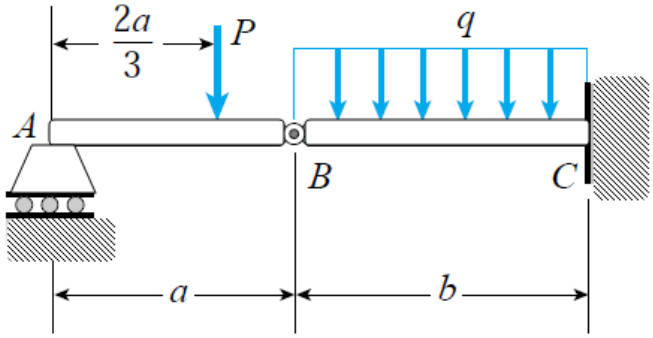
$$y_{B2}=\frac{q_0L^4}{30EI}(\uparrow)\cdot\ \theta_{B2}=\frac{q_0L^3}{24EI}(\text{ccw})$$

$$y_B=y_{B1}-y_{B2}=\frac{11q_0L^4}{120EI}(\downarrow)$$

$$\theta_B=\theta_{B1}-\theta_{B2}=\frac{q_0L^3}{8EI}(\text{cw})$$

範例 17

A compound beam ABC has a roller support at A, an internal hinge at B, and a fixed support at C (Fig.). Segment AB has length a and segment BC has length b. A concentrated load P acts at distance $\frac{2a}{3}$ from support A and a uniform load of intensity q acts between points B and C. Determine the deflection δ_B at the hinge and the angle of rotation θ_A at support A. (Note: The beam has constant flexural rigidity EI.)

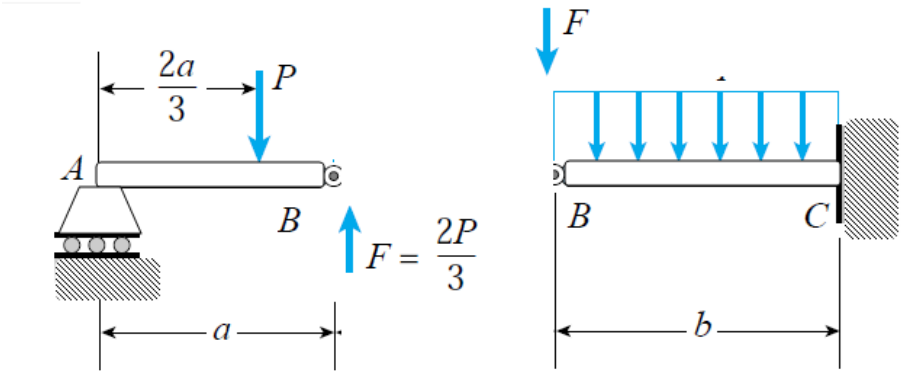


【中文翻譯】

複合樑 ABC 有滾子支撐於 A，內有樞鈕於 B，且一固定支撐於 C。一集中負載 P 作用於從支撐 A 距離 $2a/3$ 處，及一 強度 q 均勻負載作用於點 B 及 C 之間。求在樞鈕之撓曲 B 及在支撐 A 之旋轉角 θ_A（註：樑具常數撓曲剛性 EI。）

【解析】

(一)內力分析



由AB自由體圖可求得 $F=\frac{2P}{3}$

(二)重疊原理

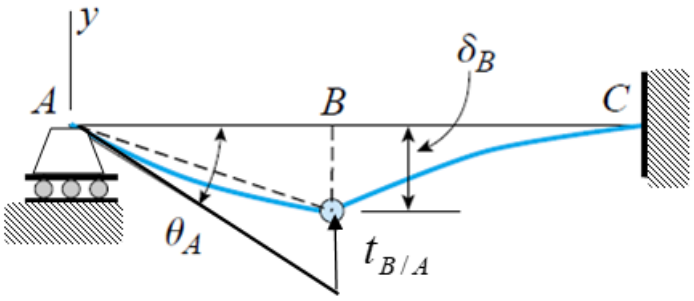
由BC自由體圖之重疊原理可得

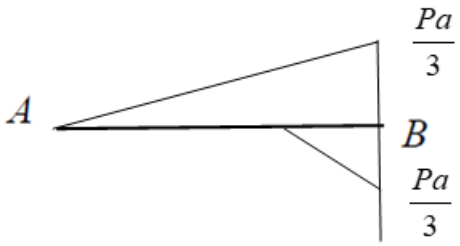
$$\delta_B = \frac{qb^4}{8EI} + \frac{Fb^3}{3EI}$$

$$\delta_B = \frac{qb^4}{8EI} + \frac{2Pb^3}{9EI}$$

(三) 由變形曲線圖

由力矩面積法第二定理





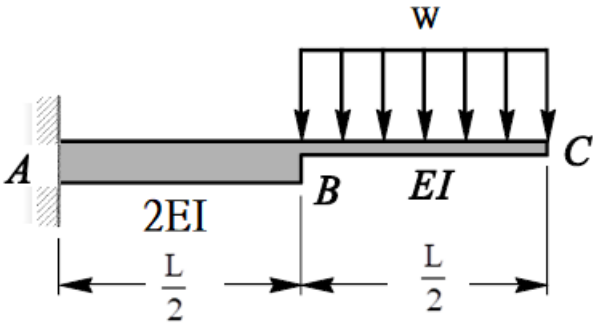
$$t_{B/A} = \frac{1}{EI}[\frac{1}{2} \times \frac{Pa}{3} \times a \times \frac{a}{3} + \frac{1}{2} \times \frac{Pa}{3} \times \frac{a}{3} \times \frac{a}{6}] = \frac{4Pa^3}{81EI}$$

$$a \times \theta_A = t_{B/A} + \delta_B$$

$$\theta_A = \frac{qb^4}{8aEI} + \frac{2Pb^3}{9aEI} + \frac{4Pa^2}{81EI}$$

範例 18

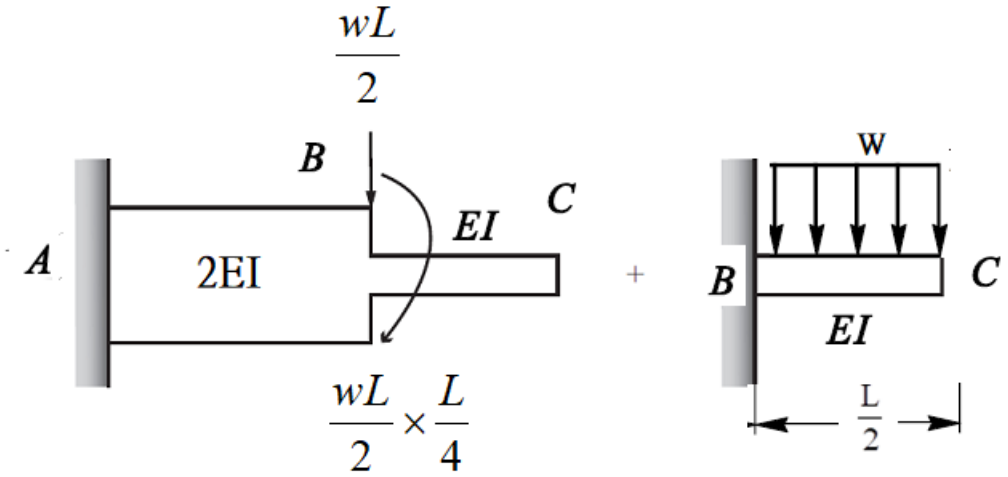
圖中懸臂梁 BC 間承受均佈負荷 q，試求 B、C 兩點之撓度。



【成大土木】

【解析

(一)由等效力系法內力分析



(二)由重疊原理

$$\delta_B = \frac{(\frac{wL}{2})\times(\frac{L}{2})^3}{3(2EI)} + \frac{(\frac{wL}{2}\times\frac{L}{4})\times(\frac{L}{2})^2}{2(2EI)} = \frac{7wL^4}{384EI}$$

$$\delta_C = \frac{(\frac{wL}{2})\times(\frac{L}{2})^3}{3(2EI)} + \frac{(\frac{wL}{2}\times\frac{L}{4})\times(\frac{L}{2})^2}{2(2EI)} + \theta_B\times\frac{L}{2} + \frac{(w)\times(\frac{L}{2})^4}{8(EI)}$$

其中

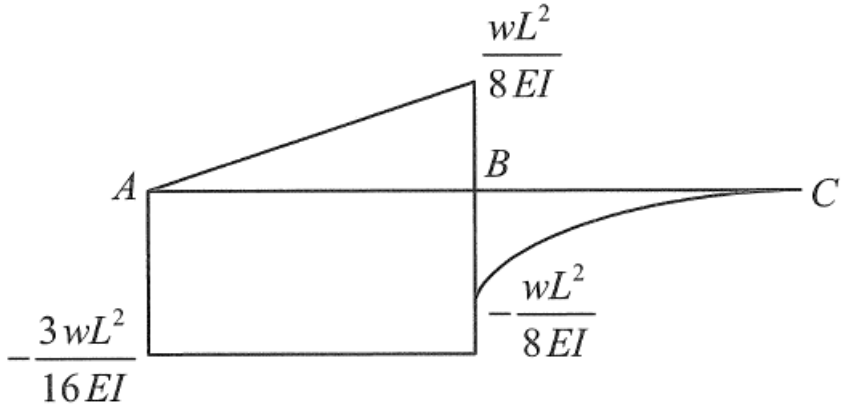
$$\theta_B = \frac{(\frac{wL}{2})\times(\frac{L}{2})^2}{2(2EI)} + \frac{(\frac{wL}{2}\times\frac{L}{4})\times(\frac{L}{2})}{(2EI)}$$ 帶入

$$\delta_C = \frac{11wL^4}{192EI}$$

【另解】：

(一)內力分析

$$R_A=\frac{wL}{2}(\uparrow)\,,\quad M_A=\frac{3wL^2}{8}(\circlearrowleft)$$



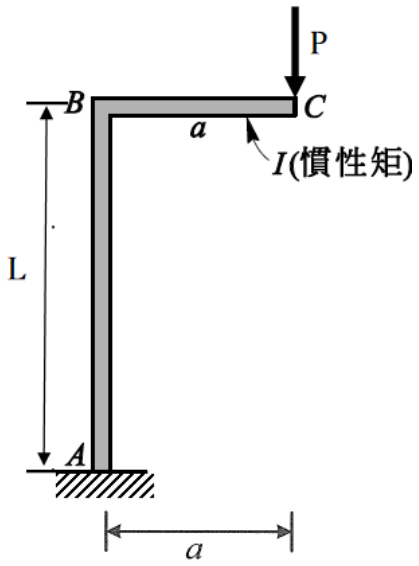
(二)由力矩面積法

$$\delta_B=(\frac{1}{2}\times\frac{wL^2}{8EI}\times\frac{L}{2})\times\frac{L}{6}+(-\frac{3wL^2}{16EI}\times\frac{L}{2})\times\frac{L}{4}=-\frac{7wL^4}{384EI}(\text{負表向下})$$

$$\delta_C=(\frac{wL^3}{32EI})\times\frac{2L}{3}+(-\frac{3wL^3}{32EI})\times\frac{3L}{4}+(-\frac{wL^3}{48EI})\times\frac{3L}{8}=-\frac{11wL^4}{192EI}$$

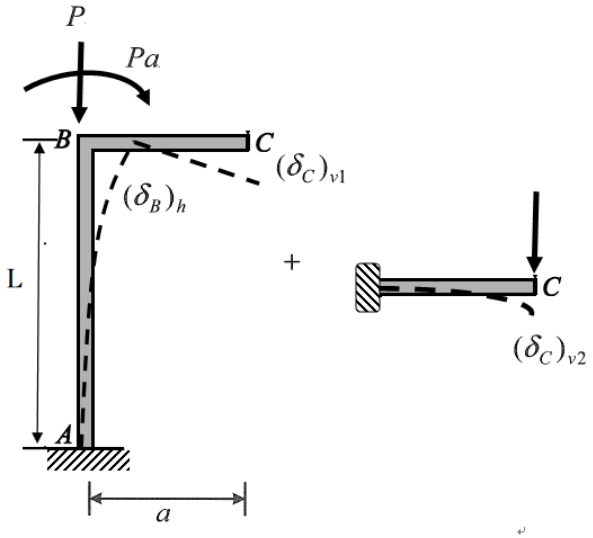
範例 19

圖中剛架在 BC 段承受集中負載 P，假設 EI 為定值，試求 C 端之水平及垂直撓度。設軸力所生之變形忽略不計。



【解析

(一) 由等效力系法內力分析



(二) 由重疊原理

$$(\delta_C)_h = (\delta_B)_h = \frac{PaL^2}{2EI}(\rightarrow) \quad ; \quad (\delta_C)_{v1} = \frac{PaL}{EI} \times a(\downarrow)$$

$$(\delta_C)_{v2} = \frac{Pa^3}{3EI}(\downarrow)$$

C 點之水平位移：

$$(\delta_C)_h = \frac{PaL^2}{2EI}(\rightarrow)$$

C 點之垂直位移：

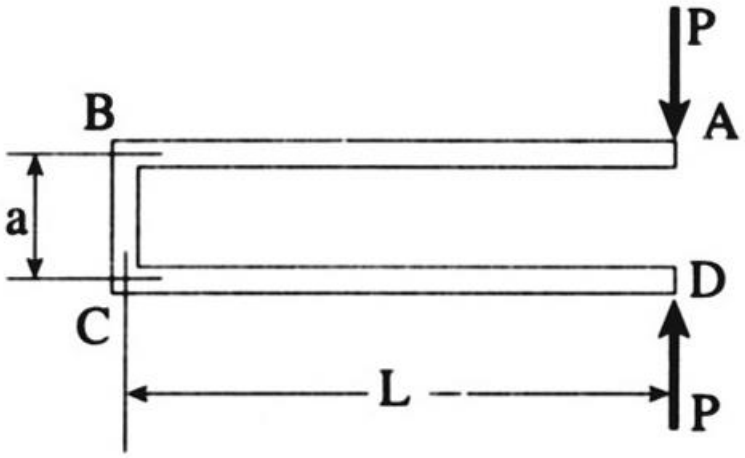
$$(\delta_C)_v = (\delta_C)_{v1} + (\delta_C)_{v2} = \frac{Pa^2}{3EI}(3L + a)(\downarrow)$$

範例 20

The frame ABCD shown in the figure is squeezed by two collinear forces P acting at points A and D. The flexural rigidity EI is constant throughout the frame. Disregard the effects of axial deformations and consider only the effects of bending due to the loads P.

(a)what is the angle of rotation at point B when the loads are applieds?

(b)what is the decrease d in the distance between points A and D when the loads P are applied?



【105 台大應力】

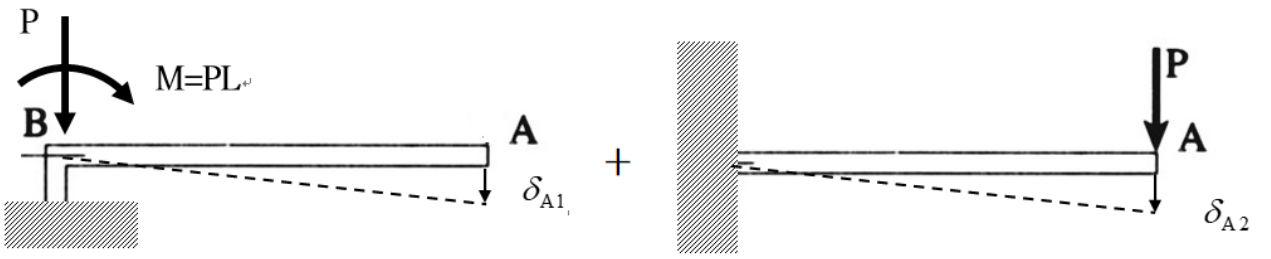
【中文翻譯】

示於圖之框架 *ABCD*，由兩作用於點 *A* 及 *D* 之共線的力 *P* 所擠壓。當負載 *P* 施加時，撓曲剛性 EI 個框架為常數。 註：忽略軸向變形的效應，只考慮由於負載 P 彎曲效應。試求(a)B 點的角度？(b)於點 A 及 D 的距離減少了多少？

【解析】

(一)桿件為對稱結構取上半部討論

由等效力法



$$\theta_B = \frac{ML}{EI} = \frac{PL \times (\frac{a}{2})}{EI} = \frac{PLa}{2EI}$$

$$\delta_{\mathrm{A}} = \delta_{A1} + \delta_{A2} = \theta_{\mathrm{B}} L + \frac{\mathrm{PL}^3}{3\mathrm{EI}} = \frac{PL^2}{6EI}(3a + 2L)$$

$$\delta = 2\delta_A = \frac{PL^2}{3EI}(3a + 2L)$$

12-5.靜不定樑

一、 懸臂樑之靜不定結構

(一) 樑靜不定度： n(靜不定度) $= b$(桿件數) $+ r$(反力數) $+ s$(剛接數) $- 2j$(節點)

(二) 解題方式：

1. 使用力矩面積法求解靜不定樑之未知贅力。
2. 先將各個分解力繪製分解之彎矩圖($\frac{M}{EI}$圖)，再利用力矩面積法以疊加的方式計算。
3. 採用每個已知之負載對應於彎矩圖內，利用位移諧和條件求解。

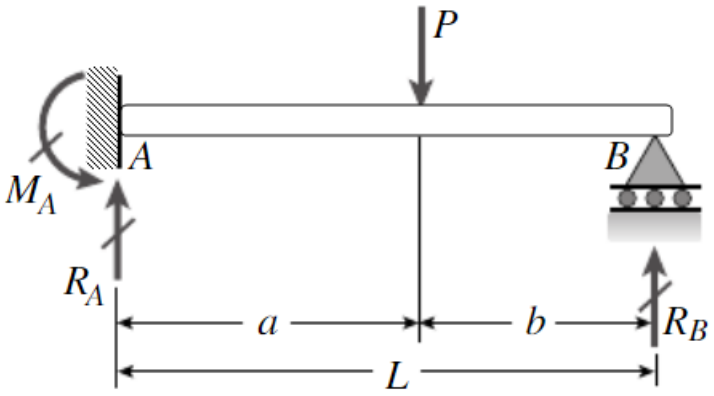
二、 連續樑

(一) 解題方式：

1. 取內支承處斷面彎矩為贅力，使用力矩面積法求解未知贅力。
2. 以疊加的方式繪製彎矩圖($\frac{M}{EI}$圖)。
3. 採用每個已知之負載對應於彎矩圖內，以內支承左右側旋轉角相同之位移諧和條件求解。

範 21

A propped cantilever beam AB of length L carries a concentrated load P acting at the position shown in the figure. Determine the reactions R_A, R_B, and M_A for this beam. Also, draw the shear-force and bending-moment diagrams, labeling all critical ordinates.



【105 台大工科】

【中文翻譯】

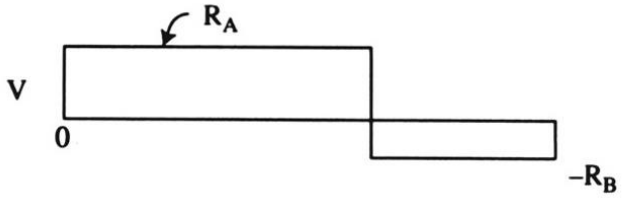
一支撐懸臂樑 *AB* 長 *L* 支持一集中負載 *P* 作用於如圖所示的位置。求此樑之反應力 R_A、R_B 及 M_A。此外，繪剪力及彎矩圖，標示所有關鍵的縱座標。

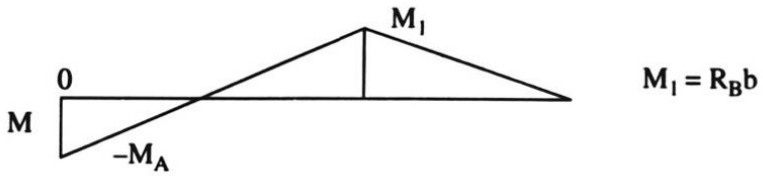
【解析】

(一)內力分析由靜平衡方程式

$R_A + R_B = P$

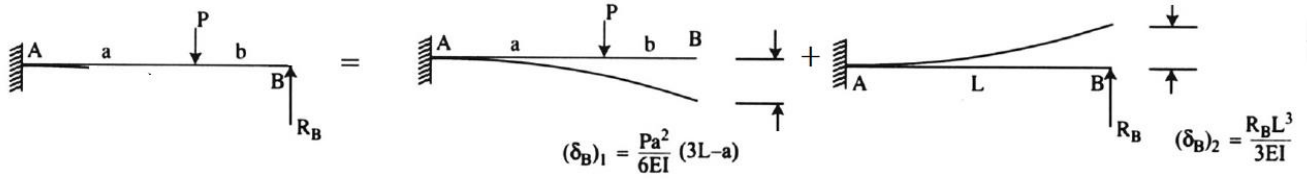
(二)剪力圖與彎矩圖





(三)變位一致性

假設 R_B 為贅力



由力矩面積法

$$(\delta_B)_1 = \frac{Pa^2}{6EI}(3L-a)$$

$$(\delta_B)_2 = \frac{R_BL^3}{3EI}$$

變位一致性

$$\delta = (\delta_B)_1 - (\delta_B)_2 = 0$$

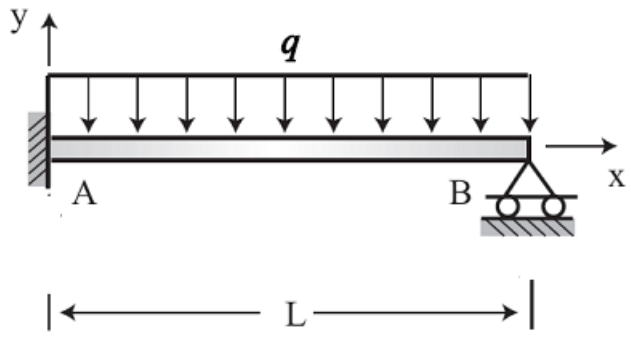
$$\delta_B = \frac{Pa^2}{6EI}(3L-a) - \frac{R_BL^3}{3EI} = 0$$

$$R_B = \frac{Pa^2}{2L^3}(3L-a)$$

$$R_A = \frac{Pb}{2L^3}(3L^2-b^2),\ M_A = \frac{Pab}{2L^2}(L-b)$$

範例 22

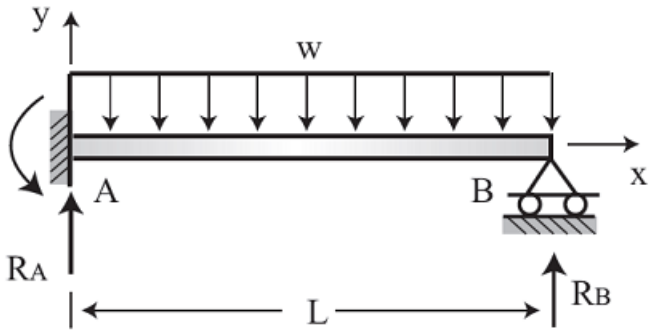
圖中長度為 L 之支撐懸臂梁 AB 承受均佈負荷 q，試求 (a)梁之支承反力，(b)撓度曲線方程式，(c)最大撓度及其位置，(d)剪力圖與彎矩圖。



【土木普考、四等】【中興機械】【105 鐵路高員】【成大機械】【101 、102 成大航太】

【解析】

內力分析



(a)設取 R_B 為贅力，由平衡方程式

$$\Sigma F_y = 0 \ ， \ R_A = qL - R_B \ .....①$$

$$\Sigma M_A = 0 \ ， \ M_A = \frac{qL^2}{2} - R_B L \ .....②$$

距 A 端為 x 處之彎矩方程式

$$M_x = R_A x - M_A - \frac{qx^2}{2} = qLx - R_B x - \frac{qL^2}{2} + R_B L - \frac{qx^2}{2}$$

$$EIy'' = qLx - R_B x - \frac{qL^2}{2} + R_B L - \frac{qx^2}{2}$$

$$EIy' = \frac{qL}{2}x^2 - \frac{R_B}{2}x^2 - \frac{qL^2}{2}x + R_B Lx - \frac{q}{6}x^3 + C_1 \ .....③$$

$$EIy = \frac{qL}{6}x^3 - \frac{R_B}{6}x^3 - \frac{qL^2}{4}x^2 + \frac{R_B L}{2}x^2 - \frac{q}{24}x^4 + C_1 x + C_2 \ .....④$$

$BC(1)：x=0$，$y'=0$，得 $C_1=0$

$BC(2)：x=0$，$y=0$，得 $C_2=0$

$BC(3)：x=L$，$y=0$，$\frac{qL}{6}(L^3)-\frac{R_B}{6}(L^3)-\frac{qL^2}{4}(L^2)+\frac{R_BL}{2}(L^2)-\frac{q}{24}(L^4)=0$

得 $R_B=\frac{3}{8}qL$ ，代入公式①及②得 $R_A=\frac{5}{8}qL$ ，$M_A=\frac{1}{8}qL^2$

(b)由公式④，$y=\frac{qL}{6EI}x^3-\frac{x^3}{6EI}\left(\frac{3qL}{8}\right)-\frac{qL^2}{4EI}x^2+\frac{Lx^2}{2EI}\left(\frac{3}{8}qL\right)-\frac{q}{24EI}x^4$

故 $y=-\frac{qx^2}{48EI}(3L^2-5Lx+2x^2)$

(c)最大撓度在 $y'=0$ 處，由公式③

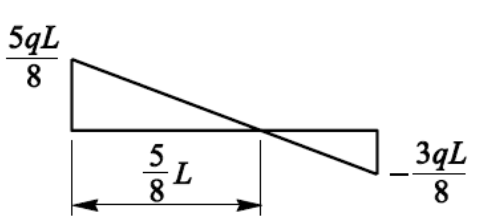
$\frac{qL}{2}x^2-\frac{x^2}{2}\left(\frac{3qL}{8}\right)-\frac{qL^2}{2}x+\left(\frac{3qL}{8}\right)Lx-\frac{q}{6}x^3=0$

$8x^2-15Lx+6L^2=0$，解得 $x=0.5785L$

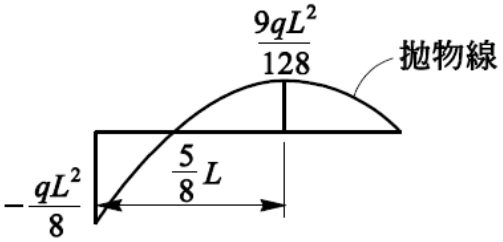
$y_{\max}=-\frac{q(0.5785L)^2}{48EI}[3L^2-5L(0.5785L)+2(0.5785L)^2]=-0.005416\frac{qL^4}{EI}$

或 $y_{\max}=0.005416\frac{qL^4}{EI}(\downarrow)$

(d)



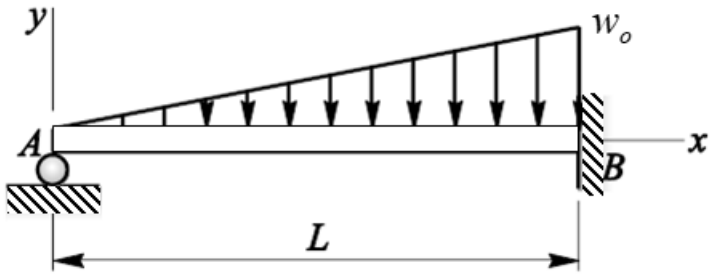
(剪力圖)



(彎矩圖)

範例 23

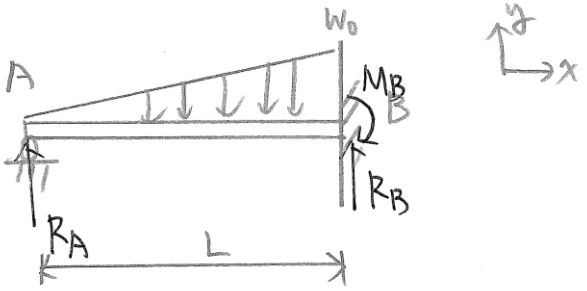
如圖所示，樑在 A 端為滾柱支持(roller support)，在 B 端為固定支持(fixed support)，受到線性分布負載（distributed loading），試求 A、B 端的反作用力（reactions）。假設軸向應力可忽略不計，且軸之撓曲剛度 EI 為常數。



【106 機械高考】【101 中興機械】【102 北科】

【解析】

一、內力分析

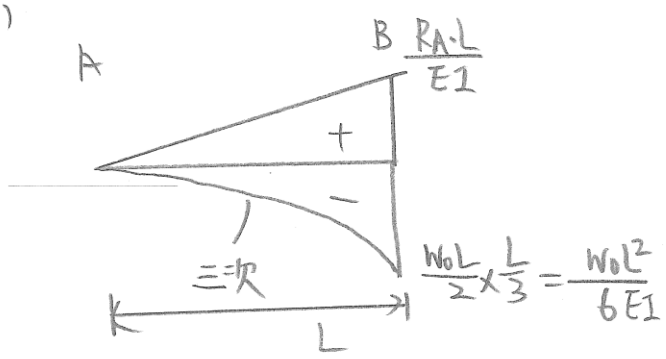


$+\uparrow \Sigma F_y=0 \Rightarrow R_A+R_B=\frac{1}{2}w_0L \quad -(1)$

$\curvearrowleft \Sigma M_A=0 \Rightarrow \frac{1}{2}w_0L\times\frac{2}{3}L-R_BL+M_B=0 \quad -(2)$

二. 變形量，由力矩面積法

(一)



$$t_{AB}=0=\frac{1}{EI}\left[-\frac{W_0L^2}{6}\times L\times\frac{1}{4}\times\frac{4}{5}L+R_A\cdot L\times L\times\frac{1}{2}\times\frac{2}{3}L\right]=0$$

$$\therefore R_A=\frac{1}{10}W_0L$$

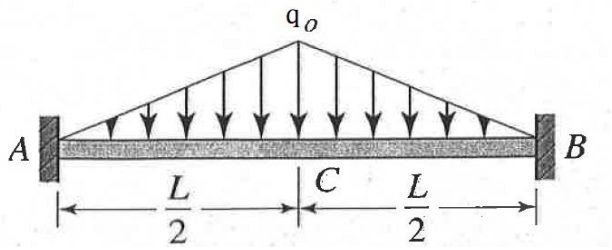
代入(1) 故 $R_B=\frac{2}{5}W_0L$

代入(2)

$$M_B=\frac{2}{5}W_0L^2-\frac{1}{3}W_0L^2=\frac{1}{15}W_0L^2$$

範例 24

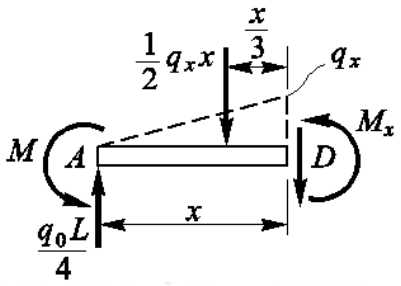
A fixed-ended beam AB (constant EI) is under a symmetric triangular load of maximum intensity w_o, as shown in Figure 3. Determine all reactions, the equation of elastic curve, and the maximum deflection. Use the second-order differential equation of the deflection. (15%)



【104 中興機械】

【解析】

(一)



(a) $R_A = R_B = \frac{q_0 L}{4}$ ， $M_A = M_B = M$ (左右對稱)

設取兩端之反力矩M為贅力

$$q_x = \frac{2x}{L} q_0$$

$$\Sigma M_D = 0$$

$$Mx = \frac{q_0 L}{4} x - \left(\frac{1}{2} q_x x\right)\left(\frac{x}{3}\right) - M = \frac{q_0 L}{4} x - \frac{q_0}{3L} x^3 - M$$

$$EIy'' = -\frac{q_0}{3L} x^3 + \frac{q_0 L}{4} x - M$$

$$EIy' = -\frac{q_0}{12L} x^4 + \frac{q_0 L}{8} x^2 - Mx + C_1$$

$$EIy = -\frac{q_0}{60L} x^5 + \frac{q_0 L}{24} x^3 - \frac{M}{2} x^2 + C_1 x + C_2$$

$BC(1)$：$x = 0$， $y' = 0$ ，得$C_1 = 0$

$BC(2)$：$x = 0$，$y = 0$，得$C_2 = 0$

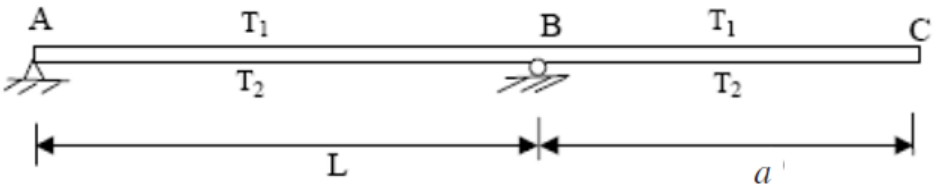
$BC(3)$：$x = \dfrac{L}{2}$，$y' = 0$，代入公式②

$$-\frac{q_0}{12L}\left(\frac{L}{2}\right)^4 + \frac{q_0 L}{8}\left(\frac{L}{2}\right)^2 - M\left(\frac{L}{2}\right) = 0 \quad , \quad M = \frac{5q_0 L^2}{96}$$

$$\delta_{\max} = y(\frac{L}{2}) = \frac{7q_0 L^4}{3840EI}$$

範例 25

An overhanging beam ABC of height h is heated to a temperature T1 on the top and T2 on the bottom (see figure). Determine the equation of the deflection curve of the beam, the angle of rotation C at end C, and the deflection C at end C.



【關務三等】

【中文翻譯】

如圖所示之矩形斷面樑，其樑頂面之溫度為 T_1，樑底部之溫度為 T_2 ，若樑斷面高為 h，α 為其熱膨脹係數，則該梁在此溫度變化下，C 點處之垂直撓度為何？(96 年機械關務三等)

【解析】

(一)撓曲線方程式

$$v'' = \frac{d^2v}{dx^2} = \frac{\alpha(T_2 - T_1)}{h}$$

$$v' = \frac{\alpha(T_2 - T_1)x}{h} + C_1$$

$$v = \frac{\alpha(T_2 - T_1)x^2}{2h} + C_1x + C_2$$

(二)帶入邊界條件

B.C. 1 $v(0) = 0 \quad \therefore C_2 = 0$

B.C. 2 $v(L) = 0 \quad \therefore C_1 = -\frac{\alpha(T_2 - T_1)L}{2h}$

$$v = \frac{\alpha(T_2 - T_1)}{2h}(x^2 - Lx)$$

$$v' = \frac{\alpha(T_2 - T_1)}{2h}(2x - L)$$

$$\theta_C = v'(L + a) = \frac{\alpha(T_2 - T_1)}{2h}(L + 2a)$$

$$\delta_C = v(L + a) = \frac{\alpha(T_2 - T_1)\,(L + a)\,(a)}{2h}$$

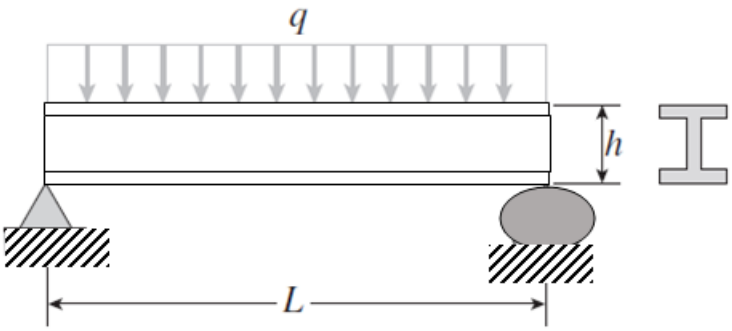
實戰演練

例 1

A uniformly loaded steel wide-flange beam with simple supports (see figure) has a downward deflection of 10 mm at the midpoint and angles of rotation equal to 0.01 radians at the ends. Calculate the height h of the beam if the maximum bending stress is 90 MPa and the modulus of elasticity is 200 GPa.

Hint:

$$\delta = \delta_{\max} = \frac{5qL^4}{384\,EI} \quad \theta = \theta_A = \frac{qL^3}{24\,EI}$$



【103 成大醫工】

【中文翻譯】

均勻負載簡支鋼樑 ，在中點具向下撓曲10mm，以及在端有0.01弧的旋轉角，如果最大彎曲應力為90 MPa，且彈性模數為200 GPa，求樑的高度 h 。

【解析】

$$\delta = \delta_{\max} = \frac{5qL^4}{384\,EI} \rightarrow q = \frac{384\,EI\delta}{5L^4}$$

$$\theta = \theta_A = \frac{qL^3}{24\,EI} \rightarrow q = \frac{24\,EI\theta}{L^3}$$

$$L = \frac{16\,\delta}{5\theta}$$

$$\sigma = \frac{Mc}{I} = \frac{Mh}{2I}$$

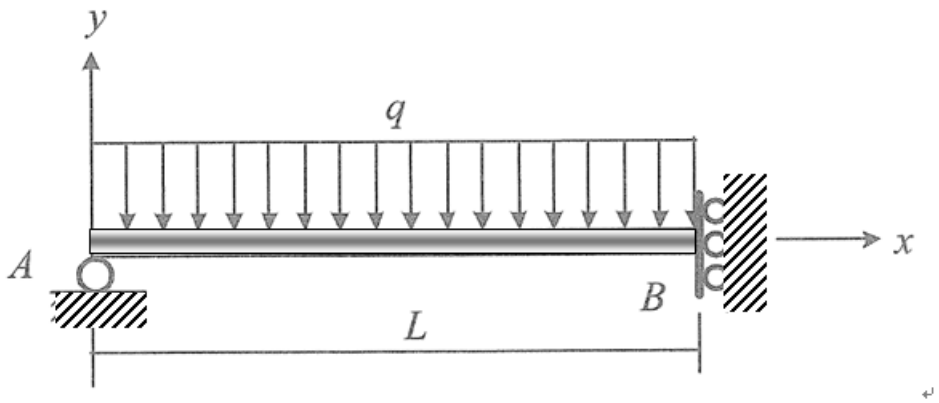
$$M = \frac{qL^2}{8} \qquad \therefore \sigma = \frac{qL^2h}{16I}$$

$$h = \frac{16I\sigma}{qL^2}$$

$$h = \frac{32(90\text{ MPa})(10\text{ mm})}{15(200\text{ GPa})(0.01\text{ rad})^2} = 96\text{ mm}$$

例 2

The beam shown in the figure has a roller support at A and a guided support at B. The guided support permits vertical movement but no rotation. Derive the equation of the deflection curve and determine the deflection δ_B at end B due to the uniform load of intensity q.

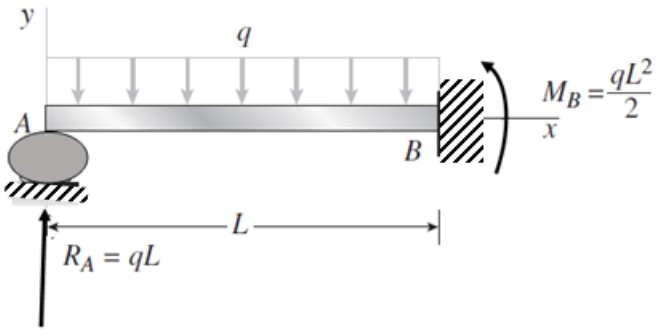


【101 清大動機】

【解析】

示於圖之樑在支撐 A 及 B 均具滾子。在 B 可以上下移動，但不能旋轉。導出撓曲曲線的，方程式及在端 B 由於強度 q 之均勻負載的撓曲 δ_B。

(一) 內力分析



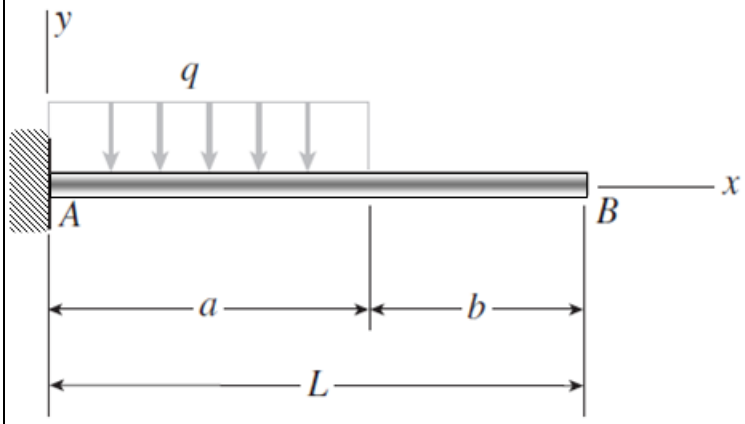
B.C. $v(0) = 0 \quad \therefore C_2 = 0$

$$v = -\frac{qx}{24EI}(8L^3 - 4Lx^2 + x^3)$$

$$\delta_B = -v(L) = \frac{5qL^4}{24EI}$$

例 3

Derive the equations of the deflection curve for a cantilever beam AB carrying a uniform load of intensity q over part of the span (see figure). Also, determine the deflection δ_B at the end of the beam.



【107 台大工科】

【中文翻譯】

求受有強度 q 作用於樑之一部份之懸臂樑 AB 之撓曲曲線之方程式（ 見圖 ）。此外，求在樑端之撓曲 δ_B 。

【解析】

$$EIv'' = M = -\frac{q}{2}(a-x)^2 = -\frac{q}{2}(a^2 - 2ax + x^2) \qquad (0 \le x \le a)$$

$$EIv' = -\frac{q}{2}\left(a^2x - ax^2 + \frac{x^3}{3}\right) + C_1 \quad (0 \le x \le a)$$

B.C. 1 $v'(0) = 0 \quad \therefore C_1 = 0$

$EIv'' = M = 0 \qquad (a \le x \le L)$

$EIv' = C_2 \qquad (a \le x \le L)$

B.C. 2 $(v')_{\text{Left}} = (v')_{\text{Right}}$ at $x = a$

$$\therefore C_2 = -\frac{qa^3}{6}$$

$$EIv = -\frac{q}{2}\left(\frac{a^2x^2}{2} - \frac{ax^3}{3} + \frac{x^4}{12}\right) + C_3$$

B.C. 3 $v(0) = 0 \quad \therefore C_3 = 0$

$$EIv = C_2 x + C_4 = -\frac{qa^3 x}{6} + C_4 \quad (\text{a} \le x \le L)$$

B.C. 4 $(v)_{\text{Left}} = (v)_{\text{Right}}$ at $x = a$

$$\therefore\ C_4 = \frac{qa^4}{24}$$

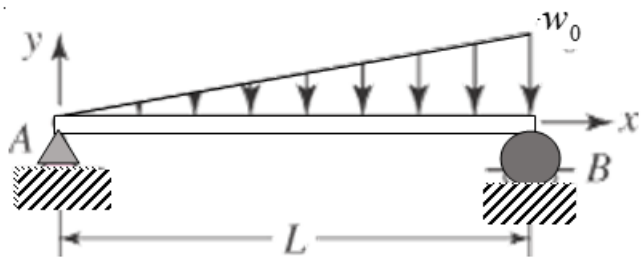
$$v = -\frac{qx^2}{24\,EI}(6a^2 - 4ax + x^2) \quad (0 \le x \le a)$$

$$v = -\frac{qa^3}{24\,EI}(4x - a)\ (\text{a} \le x \le L)$$

$$\delta_B = -v(L) = \frac{qa^3}{24\,EI}(4L - a)$$

例 4

一簡支樑 AB 承受一三角形分佈力，其最大密度為 w_o。(a)應用撓度之四階微分方程式求解樑之彈性曲線表示式。(b)試求最大撓度 v_{max}。



【清大動機】

【解析】

由撓度之四階微分方程式可得

$$EI\frac{d^4v}{dx^4}=EIv''''=-w_0\frac{x}{L},\qquad EIv'''=-\frac{w_0x^2}{2L}+c_1,\qquad EIv''=-\frac{w_0x^3}{6L}+c_1x+c_2$$

邊界條件：

$$v''(0)=0:\ c_2=0,\qquad v''(L)=0:\ c_1=\frac{w_0L}{6}$$

帶入上式，得

$$EIv'=-\frac{w_0x^4}{24L}+\frac{w_0Lx^2}{12}+c_3$$

$$EIv=-\frac{w_0x^5}{120L}+\frac{w_0Lx^3}{36}+c_3x+c_4$$

邊界條件：

$$v(0)=0:\quad c_4=0,\qquad v(L)=0:\quad c_3=-\frac{7w_0L^3}{360}$$

於是可得

$$v=-\frac{w_0x}{360EIL}(7L^4-10L^2x^2+3x^4)$$

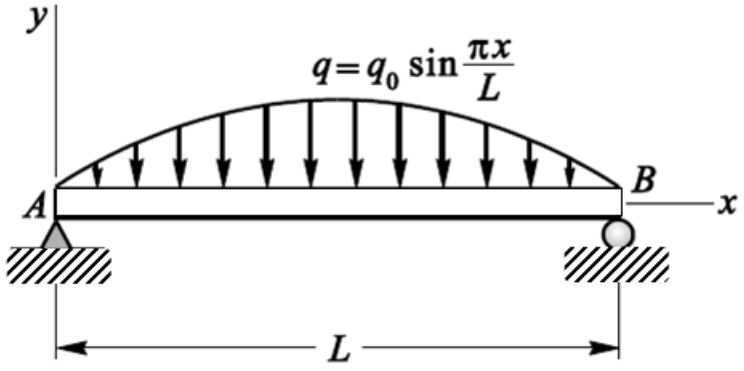
$$v'=-\frac{w_0}{360EIL}(7L^4-30L^2x_1^2+15x_1^4)=0$$

$$x_1=L\sqrt{1-\sqrt{\frac{8}{15}}}=0.5193L$$

$$v_{\max}=v(x_1)=0.006522\frac{w_0L^4}{EI}\downarrow$$

例 5

簡樑 AB 受有強度 $q = q_0 \sin \pi x / L$ 之分佈負載，於此 q_0 為負載之最大強度 （ 見圖 ）。導出撓曲曲線之方程式，再來求在樑 A 端之撓角及中點之撓曲 δ_{max} 。



【解析】

由撓度之四階微分方程式可得

$$EIy'''' = -q(x) = -q_0 \sin \frac{\pi x}{L}$$

$$EIy''' = V = q_0 \frac{L}{\pi} \cos \frac{\pi x}{L} + C_1$$

$$EIy'' = M = q_0 \frac{L^2}{\pi^2} \sin \frac{\pi x}{L} + C_1 x + C_2$$

邊界條件：

$$BC(1)：x = 0，M = 0，得\ C_2 = 0$$

$$BC(2)：x = L，M = 0，得\ C_1 = 0$$

帶入上式，得

$$EIy'' = q_0 \frac{L^2}{\pi^2} \sin \frac{\pi x}{L}$$

$$EIy' = -q_0 \frac{L^3}{\pi^3} \cos \frac{\pi x}{L} + C_3$$

$$EIy = -q_0 \frac{L^4}{\pi^4} \sin \frac{\pi x}{L} + C_3 x + C_4$$

邊界條件：

$$BC(3)：x = 0，\ y = 0\ ，得\ C_4 = 0$$

$$BC(4)：x = L，\ y = 0\ ，得\ C_3 = 0$$

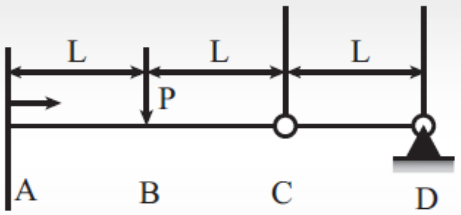
梁之撓度曲線：$y=-\frac{q_0L^4}{\pi^4EI}\sin\frac{\pi x}{L}$

A 端：$x=0$，$EIy_A'=-q_0\frac{L^3}{\pi^3}\cos(0)$，$\theta_A=y_A'=-\frac{q_0L^3}{\pi^3EI}$，或 $\theta_A=\frac{q_0L^3}{\pi^3EI}$ (cw)

$EIy_{\max}=-q_0\frac{L^4}{\pi^4}\sin\left(\frac{\pi}{2}\right)$，$y_{\max}=-\frac{q_0L^4}{\pi^4EI}$，或 $y_{\max}=\frac{q_0L^4}{\pi^4EI}(\downarrow)$

例 6

如圖所示，A 是固定支承，B 是 P 的施力點，C 是鉸接，D 是簡支承，AC 及 CD 具相同的慣性矩 I 及楊氏模數 E。試求：

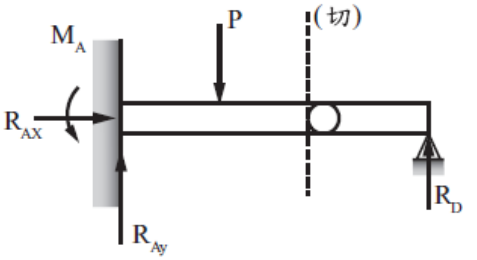


(1) AB 間的撓度（deflection）曲線。

(2) C 點的撓度及 D 點的轉角（rotation）。(103 年高考)

【解析】

解 (1) 求各支承反力

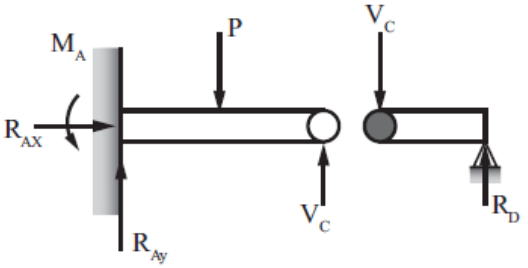


由右圖 $\Sigma F_y=0 \Rightarrow R_D=V_c=0$

由左圖 $\Sigma M_A=0(\overset{+}{\circlearrowleft}) \Rightarrow M_A=PL$

$\Sigma F_y=0(\uparrow +) \Rightarrow R_{Ay}=P$

$\Sigma F_x=0(\overset{+}{\rightarrow}) \Rightarrow R_{Ax}=0$

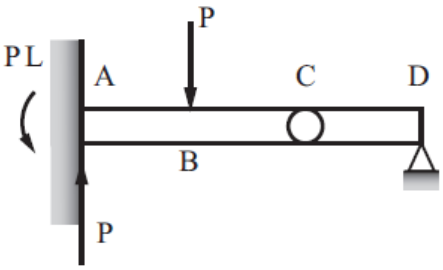


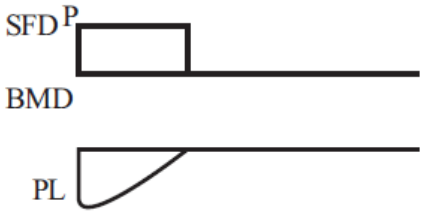
取整體 FBO

由 BMP 知

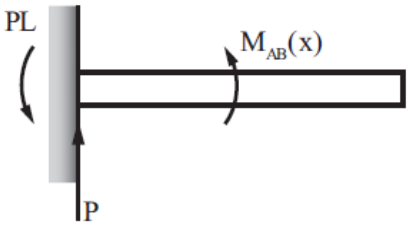
$\overline{BC}$ 和 CD 段內

彎矩為零 $\Rightarrow$ 變形曲線為直線





(2) 取 AB 段梁



$M_{AB}(x)=PX-PL \qquad 0 \le X \le L$

由 $EIy''(x)=M_{AB}(x)=PX-PL$

積分一次

$$EIy''(x)=\frac{PX^2}{2}-PLX+L_1$$

代入邊界條件 $y'(x=0)=0 \Rightarrow L_1=0$

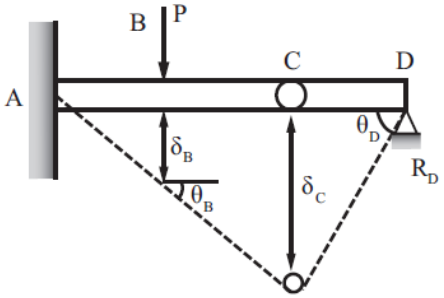
再積分一次

$$EIy(x)=\frac{PX^3}{6}-\frac{PL}{2}X^2+L_2$$

代入邊界條件 $y(x=0)=0 \Rightarrow L_2=0$

得 $$EIy(x)=\frac{PX^3}{6}-\frac{PLX^2}{2}$$

(3) 繪出 ABCD 梁變形曲線



(i) 求 δ_B

由 $$EIy(x)=\frac{PX^3}{6}-\frac{PLX^2}{2}$$

取 $X=L \Rightarrow \delta_B=y(x=L)=-\frac{PL^3}{3EI}(\downarrow)$

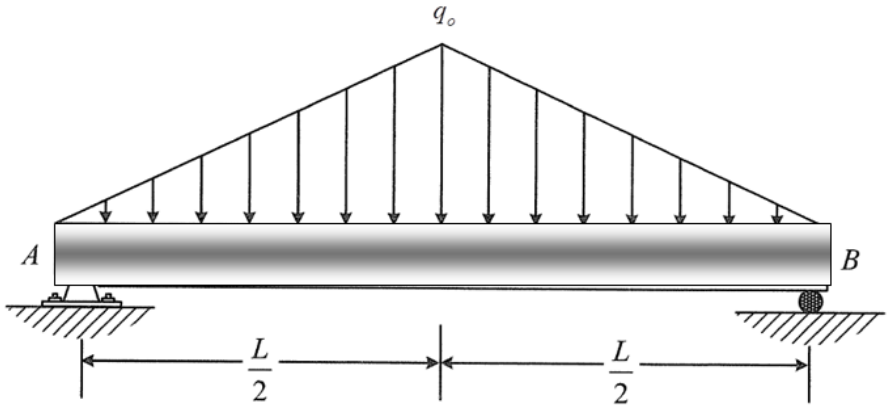
由 $EIy'(x)=\frac{PX^2}{2}-PLX \Rightarrow \theta_B=y'(X=L)=-\frac{PL^2}{2EI}$

(ii) $\delta_c=\delta_\theta+\theta_B L=\frac{5PL^3}{6EI}(\downarrow)$

(iii) $\theta_D=\frac{\delta_c}{L}=\frac{5PL^3}{6EI}(\circlearrowleft)$

例 7

The simply supported beam AB supports the triangular distributed loading as shown. Determine the maximum deflection δ_{max} for the beam. The beam has length L and constant rigidity EI



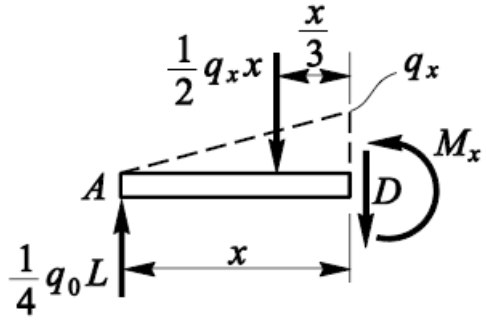
【成大土木】【雲林機械】

【中文翻譯】

圖中之簡支梁承受均變負荷，試求梁樑的最大撓度(材料常數為EI)

【解析】

(一)結構為對稱取半邊分析



$$q_x = \frac{2x}{L}q_0$$

$$\Sigma M_D = 0 \ , \ M_x = \frac{1}{4}q_0 Lx - \frac{1}{2}\left(\frac{2x}{L}q_0\right)x\left(\frac{x}{3}\right) = \frac{1}{4}q_0 Lx - \frac{q_0 x^3}{3L}$$

(二)積分法

$$EIy'' = \frac{q_0 L}{4}x - \frac{q_0}{3L}x^3$$

$$EIy' = \frac{q_0 L}{8}x^2 - \frac{q_0}{12L}x^4 + C_1$$

$$EIy = \frac{q_0 L}{24}x^3 - \frac{q_0}{60L}x^5 + C_1 x + C_2$$

(三)帶入邊界條件

$BC(1)：x=0$，$y=0$，得$C_2=0$

$BC(2)：x=\frac{L}{2}$，$y'=0$，$\frac{q_0 L}{8}\left(\frac{L}{2}\right)^2 - \frac{q_0}{12L}\left(\frac{L}{2}\right)^4 + C_1 = 0$，得$C_1 = -\frac{5q_0 L^3}{192}$

$$y = \frac{q_0 L}{24EI}x^3 - \frac{q_0}{60LEI}x^5 - \frac{5q_0 L^3}{192EI}x$$

(四)計算撓度

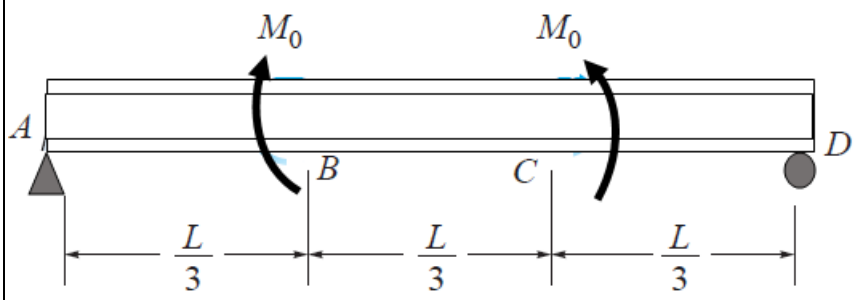
最大撓度在中點(C 點)：$x=\frac{L}{2}$

$$y_{\max} = \frac{q_0 L}{24EI}\left(\frac{L}{2}\right)^3 - \frac{q_0}{60LEI}\left(\frac{L}{2}\right)^5 - \frac{5q_0 L^3}{192EI}\left(\frac{L}{2}\right) = -\frac{q_0 L^4}{120EI}$$

$$y_{\max} = \frac{q_0 L^4}{120EI}(\downarrow)$$

例 8

Determine the eqution of the elastic curve, the slope at A, and the deflection at B of the simply supported beam. EI is constant.



【解析】

（三）由奇異函數法

$$M = -(-M_O)\left\langle x - \frac{L}{3}\right\rangle^0 - M_O\left\langle x - \frac{2}{3}L\right\rangle^0$$

$$= M_O\left\langle x - \frac{L}{3}\right\rangle^0 - M_O\left\langle x - \frac{2}{3}L\right\rangle^0$$

$$EI\frac{d^2v}{dx^2} = M$$

$$EI\frac{d^2v}{dx^2} = M_O\left\langle x - \frac{L}{3}\right\rangle^0 - M_O\left\langle x - \frac{2}{3}L\right\rangle^0$$

$$EI\frac{dv}{dx} = M_O\left\langle x - \frac{L}{3}\right\rangle - M_O\left\langle x - \frac{2}{3}L\right\rangle + C_1$$

$$EIv\frac{M_O}{2}\left\langle x - \frac{L}{3}\right\rangle^2 - \frac{M_O}{2}\left\langle x - \frac{2}{3}L\right\rangle^2 + C_1x + C_2$$

（四）由邊界條件

$$EI(0) = M_O\left(\frac{L}{2} - \frac{L}{3}\right) - 0 + C_1 \qquad C_1 = -\frac{M_OL}{6}$$

$$EI(0) = 0 - 0 + C_1(0) + C_2 \qquad C_2 = 0$$

$$\frac{dv}{dx} = \frac{M_O}{6EI}\left[6\left\langle x - \frac{L}{3}\right\rangle - 6\left\langle x - \frac{2}{3}L\right\rangle^2 - L\right]$$

（五）計算撓度

$$\theta_A = \left.\frac{dv}{dx}\right|_{x=0} = \frac{M_O}{6EI}\left[6(0) - 6(0) - L\right] = -\frac{M_O L}{6EI} = \frac{M_O L}{6EI}$$

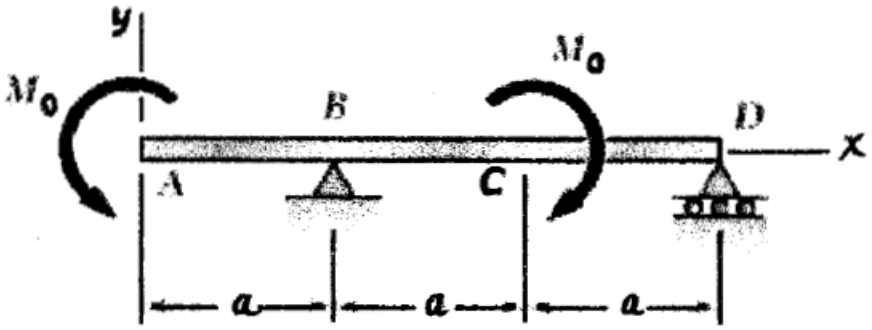
$$v = \frac{M_O}{6EI}\left[3\left(x - \frac{L}{3}\right)^2 - 3\left(x - \frac{2}{3}L\right)^2 - Lx\right]$$

$$v_B = v|_{x=\frac{L}{3}} = \frac{M_O}{6EI}\left[3(0) - 3(0) - L\left(\frac{L}{3}\right)\right]$$

$$= -\frac{M_O L^2}{18EI} = \frac{M_O L^2}{18EI} \quad \downarrow$$

例 9

(20pts) For the beam and loading shown, determine the deflection at end *A*, (b) the slope at end *D*.



【104 成大工科 】

【解析】

$R_B = 0 \qquad R_D = 0$

$0 < x < 2a \qquad M = -M_0$

$2a < x < 3a \qquad M = -M_0 + M_0 = 0$

$$EI\frac{d^2y}{dx^2} = M = -M_0 + M_0\langle x-2a\rangle^0$$

$$EI\frac{dy}{dx} = -M_0x + M_0\langle x-2a\rangle^1 + C_1$$

$$EIy = -\tfrac{1}{2}M_0x^2 + \tfrac{1}{2}M_0\langle x-2a\rangle^2 + C_1x + C_2$$

$$[x=3a,\ y=0] \qquad -\tfrac{1}{2}M_0(3a)^2 + \tfrac{1}{2}M_0a^2 + C_1(3a) + C_2 = 0$$

$$[x=a,\ y=0] \qquad -\tfrac{1}{2}M_0a^2 + 0 + C_1a + C_2 = 0$$

$$2aC_1 = \tfrac{7}{2}M_0a^2 \qquad C_1 = \tfrac{7}{4}M_0a$$

$$C_2 = \tfrac{1}{2}M_0a^2 - aC_1 = -\tfrac{5}{4}M_0a^2$$

$$y = \frac{M_0}{EI}\left\{-\tfrac{1}{2}x^2 + \tfrac{1}{2}\langle x-2a\rangle^2 + \tfrac{7}{4}ax - \tfrac{5}{4}a^2\right\}$$

$$\frac{dy}{dx} = \frac{M_0}{EI}\left\{-x + \langle x-a\rangle^1 + \tfrac{7}{4}a\right\}$$

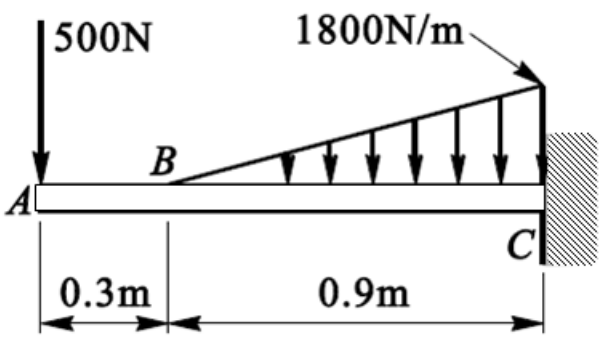
$$y_A = \frac{M_0 a^2}{EI}\left\{-0 + 0 + 0 - \frac{5}{4}\right\} = -\frac{5}{4}\frac{M_0 a^2}{EI}, \quad y_A = \frac{5}{4}\frac{M_0 a^2}{EI}\downarrow$$

$$y_C = \frac{M_0 a^2}{EI}\left\{-\frac{1}{2}(2)^2 + 0 + \frac{7}{4}(2) - \frac{5}{4}\right\} = \frac{1}{4}\frac{M_0 a^2}{EI}\uparrow$$

$$\theta_D = \frac{M_0 a}{EI}\left\{-3 + 1 + \frac{7}{4}\right\} = -\frac{1}{4}\frac{M_0 a}{EI}, \quad \theta_D = \frac{1}{4}\frac{M_0 a}{EI}\circlearrowleft$$

例 10

如圖所示一懸臂樑支持一集中負載 500N 及三角形分佈負載最大強度為 1800(N/m)，試求圖中懸臂梁在 B 點之斜角與 A 端之撓度。E = 200GPa， $I = 3.204 \times 10^{-6} m^4$



【106 中央土木】

【解析】

（一）由奇異函數法

$$EIy'' = -500x - \frac{2000}{6} < x - 0.3 >^3$$

$$EIy' = -250x^2 - \frac{250}{3} < x - 0.3 >^4 + C_1 \quad \cdots\cdots ①$$

$$EIy = -\frac{250}{3}x^3 - \frac{50}{3} < x - 0.3 >^5 + C_1 x + C_2 \quad \cdots\cdots ②$$

（二）由邊界條件

$BC(1)：x = 1.2\ \text{m}$，$y' = 0$，代入公式①，$-250(1.2)^2 - \frac{250}{3}(0.9)^4 + C_1 = 0$，$C_1 = 414.675$

$BC(2)：x = 1.2\ \text{m}$，$y = 0$，代入公式②，$-\frac{250}{3}(1.2)^3 - \frac{50}{3}(0.9)^5 + (414.675)(1.2) + C_2 = 0$

$$C_2 = -343.77$$

（三）計算撓角及撓度

$$EI = (200 \times 10^9\ \text{N/m}^2)\ (3.204 \times 10^{-6}\ \text{m}^4) = 640.8 \times 10^3\ \text{N-m}^2$$

B 點$(x = 0.3\ \text{m})$斜角，由公式①

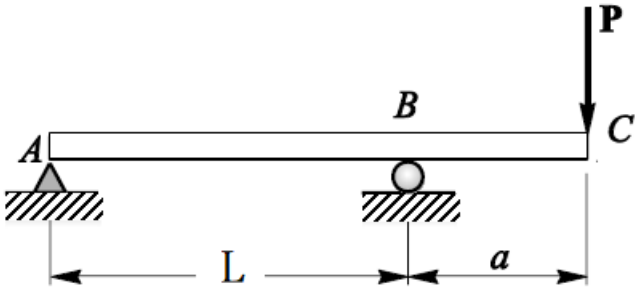
$$(640.8 \times 10^3)\theta_B = -250(0.3)^2 + 414.675 \quad ， \quad \theta_B = 0.612 \times 10^{-3}\ \text{rad (ccw)}$$

A 端$(x = 0)$撓度，由公式②

$$(640.8 \times 10^3)y_A = -343.77 \quad ， \quad y_A = -0.536 \times 10^{-3}\ \text{m}，y_A = 0.536\ \text{mm}\ (\downarrow)$$

例 11

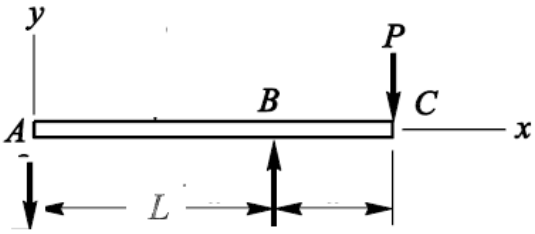
如圖所示一外伸樑支持一集中負載，試求圖中外伸樑最大撓度及發生位置。



【機械技師】【清大動機】

【解析】

(一) 內力分析



$$R_A=\frac{a}{L}P(\downarrow) \qquad R_B=(\frac{L+a}{L})P(\uparrow)$$

(二) 由奇異函數法

$$EIy''=-\frac{a}{L}P<x>^1+(\frac{L+a}{L})P<x-L>^1$$

$$EIy'=-\frac{a}{2L}P<x>^2+(\frac{L+a}{2L})P<x-L>^2+C_1$$

$$EIy=-\frac{a}{6L}P<x>^3+(\frac{L+a}{6L})P<x-L>^3+C_1x+C_2$$

(三) 由邊界條件

$$x=0\ ,\ y=0 \qquad \therefore C_2=0$$

$$x=L\ ,\ y=0\ :\ -\frac{a}{6L}PL^3+C_1L=0 \qquad \therefore C_1=\frac{a}{6}PL$$

(四) 計算撓角及撓度

梁中 AB 段$(0<x<L)$內最大撓度發生於轉角爲零之處

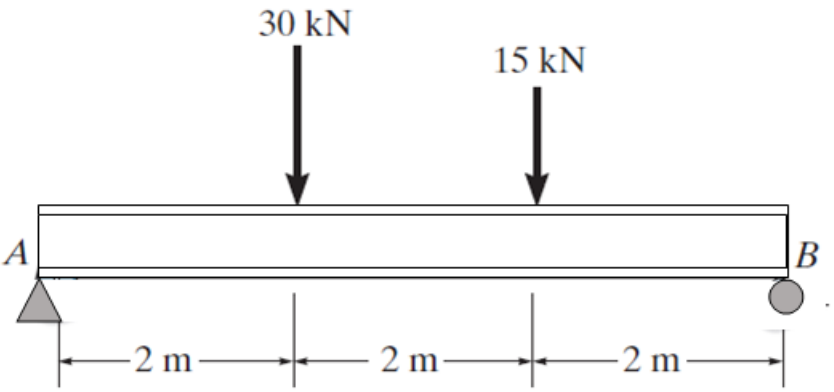
$$-\frac{a}{2L}Px^2+\frac{a}{6}PL=0 \qquad \therefore x=\frac{L}{\sqrt{3}}$$

$$\delta_{\max} = y\left(\frac{L}{\sqrt{3}}\right) = \frac{1}{EI}\left[-\frac{Pa}{6L}\left(\frac{L}{\sqrt{3}}\right)^3 + \frac{a}{6}PL\left(\frac{L}{\sqrt{3}}\right)\right]$$

$$= \frac{\sqrt{3}PaL^2}{27EI}(\uparrow)$$

例 12

Determine the maximum deflection of the simply supported beam. E = 200 GPa and



【中文翻譯】

試求出簡支樑之最大位移

【解析】

(一)由奇異函數法

$$M = -(-25)(x - 0) - 30(x - 2) - 15(x - 4)$$

$$= 25x - 30(x - 2) - 15(x - 4)$$

$$EI\frac{d^2v}{dx^2} = M$$

$$EI\frac{d^2v}{dx^2} = 25x - 30(x - 2) - 15(x - 4)$$

$$EI\frac{dv}{dx} = 12.5x^2 - 15(x - 2)^2 - 7.5(x - 4)^2 + C_1$$

$$EIv = 4.1667x^3 - 5(x - 2)^3 - 2.5(x - 4)^3 + C_1x + C_2$$

(二)帶入邊界條件

$$x = 0, v = 0.$$

$$0 = 0 - 0 - 0 + C_1(0) + C_2$$

$$x = 6\text{ m}, v = 0.$$

$$0 = 4.1667\left(6^3\right) - 5(6 - 2)^3 - 2.5(6 - 4)^3 + C_1(6) + C_2$$

$$C_1 = -93.333\text{ kN}\cdot\text{m}^3$$

$$v = \frac{1}{EI}\left[4.1667x^3 - 5(x-2)^3 - 2.5(x-4)^3 - 93.333x\right]$$

(三)計算撓角與撓度

撓角=0 時有最大撓度

$$\frac{dv}{dx} = \frac{1}{EI}\left[12.5x^2 - 15(x-2)^2 - 7.5(x-4)^2 - 93.333\right]$$

$$\frac{dv}{dx} = 0 = \frac{1}{EI}\left[12.5x^2 - 15(x-2)^2 - 93.333\right]$$

$$12.5x^2 - 15(x-2)^2 - 93.333 = 0$$

$$2.5x^2 - 60x + 153.333 = 0$$

$$x = 2.9079$$

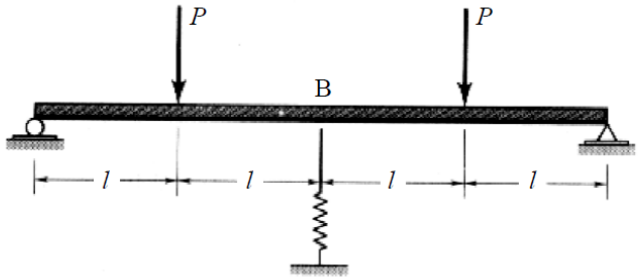
$$v_{\max} = \frac{1}{EI}\left[4.1667\left(2.9079^3\right) - 5(2.9079-2)^3 - 0 - 93.333(2.9079)\right]$$

$$= -\frac{172.69\text{kN}\cdot\text{m}^3}{EI} = -\frac{172.69\left(10^3\right)}{200\left(10^9\right)\left[65.0\left(10^{-6}\right)\right]}$$

$$= -0.01328\text{ m} = 13.3\text{ mm}$$

例 13

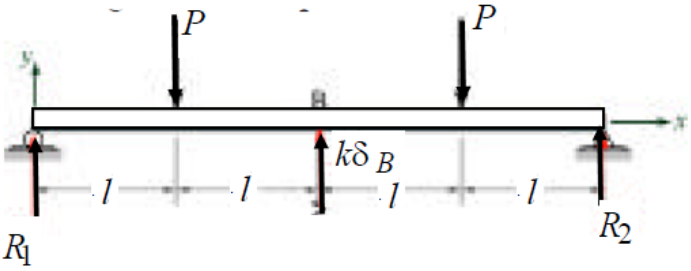
有一樑之負荷狀態如下圖所示，求彈簧之彈性常數 k，以使 B 點之彎矩為 0，假設樑的慣性矩（I）與彈性模數（E）的乘積 EI 為常數。（25 分）



【106 地特三等】

【解析】

（一）內力分析



$$R_1 = R_2 = P - \frac{1}{2}k\delta_B$$

（二）由奇異函數法

$$-M(x) = R_1\langle x\rangle^1 - P\langle x-l\rangle^1 + k\delta_B\langle x-2l\rangle^1 - P\langle x-3l\rangle^1$$

$M(x=2l)=0$

$$-M(x=2l) = R_1(2l) - Pl = 0;\quad R_1 = \frac{P}{2} = P - \frac{1}{2}k\delta_B;\quad k = \frac{P}{\delta_B}$$

$$-M(x) = \frac{P}{2}\langle x\rangle^1 - P\langle x-l\rangle^1 + P\langle x-2l\rangle^1 - P\langle x-3l\rangle^1$$

$$= P\left[\frac{1}{2}\langle x\rangle^1 - \langle x-l\rangle^1 + \langle x-2l\rangle^1 - \langle x-3l\rangle^1\right]$$

$$EI\frac{d^2y}{dx^2} = -M(x) = P\left[\frac{1}{2}\langle x\rangle^1 - \langle x-l\rangle^1 + \langle x-2l\rangle^1 - \langle x-3l\rangle^1\right]$$

$$EI\frac{dy}{dx} = \frac{P}{2}\left[\frac{1}{2}\langle x\rangle^2 - \langle x-l\rangle^2 + \langle x-2l\rangle^2 - \langle x-3l\rangle^2\right] + C_1$$

$$EIy = \frac{P}{6}\left[\frac{1}{2}\langle x\rangle^3 - \langle x-l\rangle^3 + \langle x-2l\rangle^3 - \langle x-3l\rangle^3 + \right] + C_1x + C_2$$

（三）由邊界條件

y=0 at x=0 and y=0 at x=4l

$$0 = \frac{P}{6}\left[\frac{1}{2}(4l)^3 - (3l)^3 + (2l)^3 - l^3\right] + C_1(4l); \quad C_1 = -\frac{Pl^2}{2}$$

C_2=0

（四）計算撓角及撓度

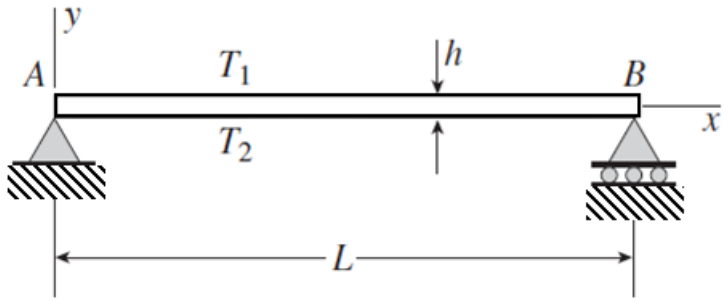
$$EIy = \frac{P}{6}\left[\frac{1}{2}\langle x\rangle^3 - \langle x-l\rangle^3 + \langle x-2l\rangle^3 - \langle x-3l\rangle^3 + \frac{1}{2}\langle x-4l\rangle^3 - 3l^2x\right]$$

$$EIy = EI\delta_B = \frac{P}{6}\left[\frac{1}{2}(2l)^3 - l^3 - 6l^3\right]; \quad \delta_B = -\frac{Pl^3}{2EI}$$

$$k = \frac{P}{\left(\frac{Pl^3}{2EI}\right)} = \frac{2EI}{l^3}$$

例 14

A simple beam AB of length L and height h undergoes a temperature change such that the bottom of the beam is at temperature T_2 and the top of the beam is at temperature T_1 (see figure). Determine the equation of the deflection curve of the beam, the angle of rotation θ_A at the left-hand support, and the deflection δ_{max} at the midpoint.



【清大動機】

【中文翻譯】

簡樑 AB 長 L 高 h，受溫度變化，使得樑底為 T_2，而樑頂為 T_1（ 見圖 ）。求樑的撓曲曲線方程式，在左支撐之旋轉角 θ_A，及中間點之撓曲 δ_{max}。

【解析】

（一）積分分析

$$v'' = \frac{d^2v}{dx^2} = \frac{\alpha(T_2 - T_1)}{h}$$

$$v' = \frac{dv}{dx} = \frac{\alpha(T_2 - T_1)x}{h} + C_1$$

（二）由邊界條件

$$v'\left(\frac{L}{2}\right) = 0$$

$$C_1 = -\frac{\alpha L(T_2 - T_1)}{2h}$$

$$v = \frac{\alpha(T_2 - T_1)x^2}{2h} - \frac{\alpha L(T_2 - T_1)x}{2h} + C_2$$

$$v(0) = 0 \qquad \therefore C_2 = 0$$

$$v = -\frac{\alpha(T_2 - T_1)(x)(L - x)}{2h}$$

$$v' = -\frac{\alpha(T_2 - T_1)(L - 2x)}{2h}$$

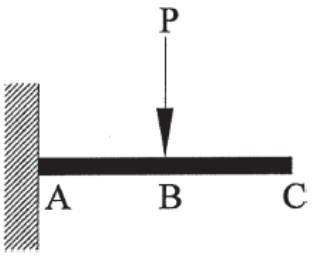
(三) 計算撓角及撓度

$$\theta_A = -v'(0) = \frac{\alpha L(T_2 - T_1)}{2h}$$

$$\delta_{\max} = -v\left(\frac{L}{2}\right) = \frac{\alpha L^2(T_2 - T_1)}{8h}$$

例 15

5. 當一力 P 作用於右圖懸臂梁中點 B 時，請求出 B 點之垂直位移為何？假設樑長為 L，其勁度為 EI：



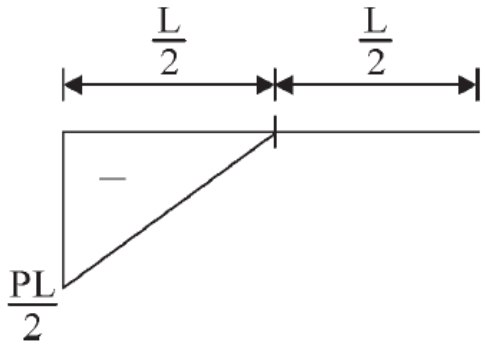
(A) $\frac{PL^3}{24EI}$ (B) $\frac{PL^3}{4EI}$ (C) $\frac{PL^3}{6EI}$ (D) $\frac{PL^3}{12EI}$。

(96 年經濟部)

6. 承上題，C 點之垂直位移為何？ (A) $\frac{PL^3}{3EI}$ (B) $\frac{PL^3}{16EI}$ (C) $\frac{5PL^3}{24EI}$ (D) $\frac{5PL^3}{48EI}$。(96 年經濟部)

【解析】

5. (A)。取彎矩圖



利用力矩面積法第二定理 $\Delta_{BA} = \delta_B = \frac{1}{EI}[\frac{PL}{2} \times \frac{L}{2} \times \frac{1}{2} \times \frac{L}{2} \times \frac{2}{3}]$

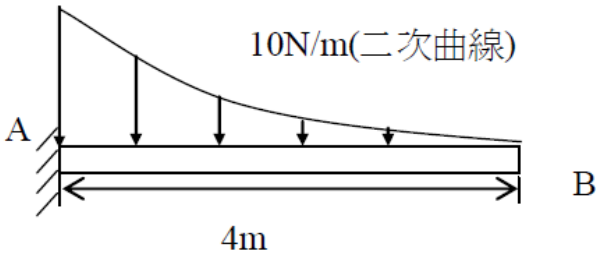
$= \frac{PL^3}{24EI}$

6. (D)。承上題彎矩圖

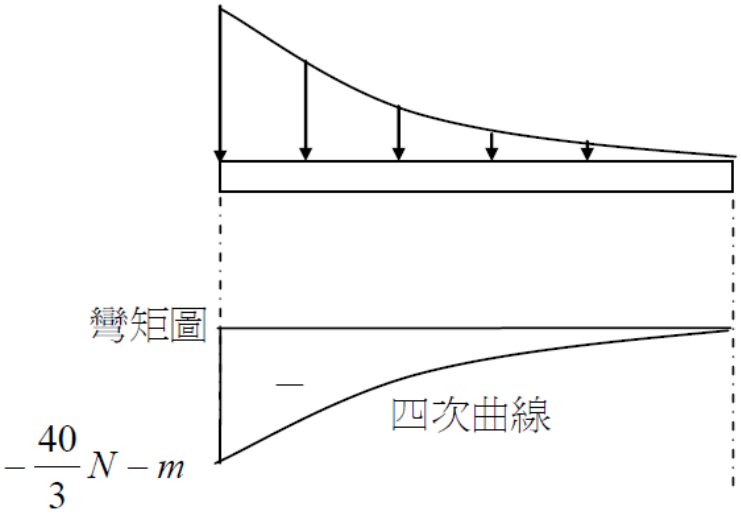
$\Delta_{CA} = \delta_C = \frac{1}{EI}[\frac{PL}{2} \times \frac{L}{2} \times \frac{1}{2} \times (\frac{L}{2} \times \frac{2}{3} + \frac{L}{2})] = \frac{5PL^3}{48EI}$

例 16

懸臂梁(材料特性：EI)受如下圖之分佈荷重(二次曲線)後，B 點之垂直變位與旋轉角為何？



【解析】



利用第一面積力矩法

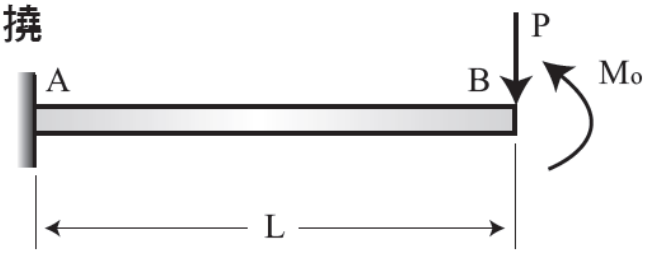
旋轉角 $\theta_B=(\frac{40}{3EI}\times 4)\times\frac{1}{5}=\frac{32}{3EI}$ (順時針)

利用第二面積力矩法

垂直變位 $v_B=(\frac{40}{3EI}\times 4)\times\frac{1}{5}\times(4\times\frac{5}{6})=\frac{32}{3EI}$ (↓)

範例 17

如圖所示，試求在B點之角度與撓度。(EI為樑的撓度剛性)。



【解析】

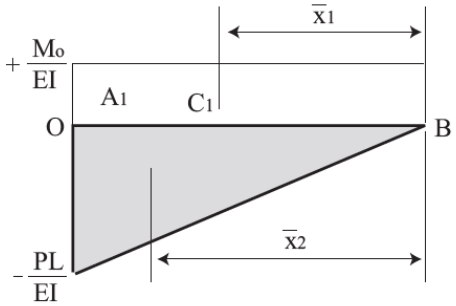
答：(1) 力矩面積法第一定理

$$A_1=\frac{M_0L}{EI}\quad \bar{x}_1=\frac{L}{2}\quad A_2=-\frac{PL^2}{2EI}\quad \bar{x}_2=\frac{2L}{3}$$

$$A_0=A_1+A_2=\frac{M_0L}{EI}-\frac{PL^2}{2EI}$$

$$\theta_{B/A}=\theta_B-\theta_A=A_0\quad \theta_A=0$$

$$\theta_B=A_0=\frac{M_0L}{EI}-\frac{PL^2}{2EI}$$



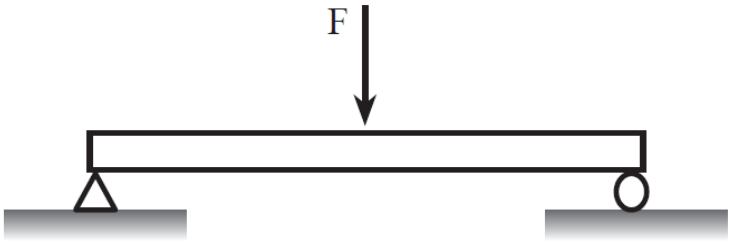
(2) 力矩面積法第二定理

$$Q=A_1\bar{x}_1+A_2\bar{x}_2=\frac{M_0L^2}{2EI}-\frac{PL^3}{3EI}$$

$$t_{B/A}=Q=\delta_B\quad \delta_B=\frac{M_0L^2}{2EI}-\frac{PL^3}{3EI}\quad \theta_B=\frac{PL^2}{2EI}-\frac{M_0L}{EI}\quad \delta_B=\frac{PL^3}{3EI}-\frac{M_0L^2}{2EI}$$

例 18

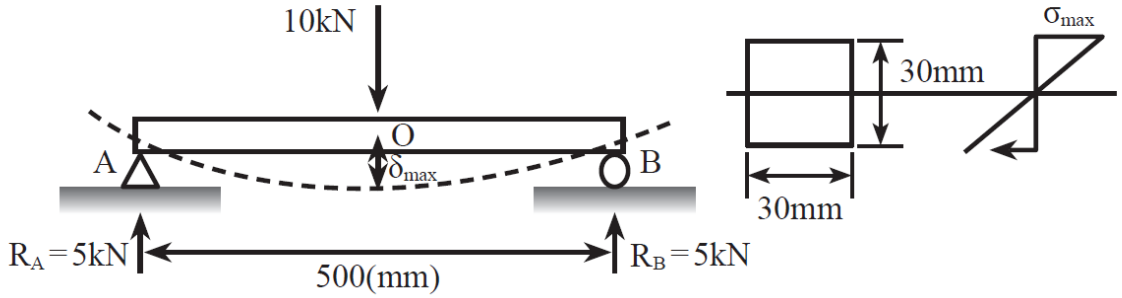
請計算下圖長 500 mm 簡支臂樑中央受力 F = 10 kN 樑之最大變形及最大應力（不考慮應力集中現象）。樑之截面為一長 30 mm 寬 30 mm 之正方形。楊氏模數為 200 GPa。

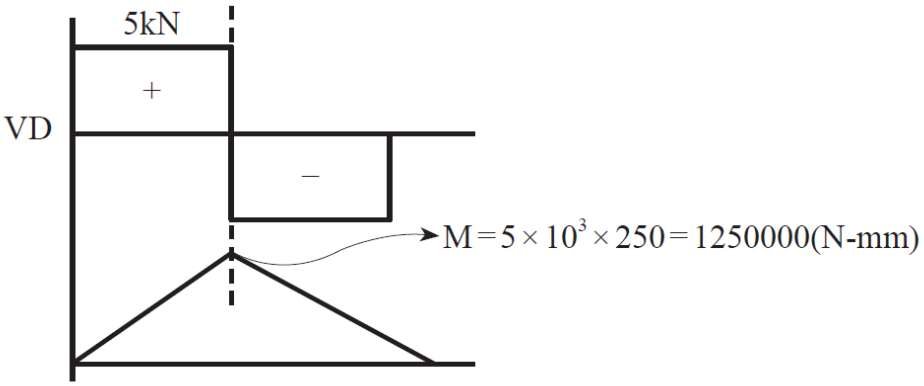


【104 普考】

【解析】

解





$$\sigma_{max}=\frac{My}{I}=\frac{1250000\times 15}{\frac{1}{12}\times(30)^4}=277.78(MPa)$$

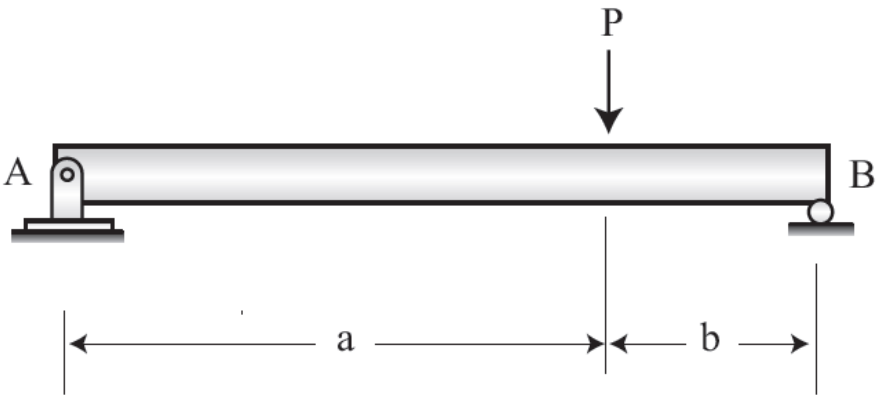
$$\delta_{max}=\frac{PL^3}{48EI}=\frac{10\times 10^3\times 500^3}{48\times 200\times 10^3\times\frac{1}{12}\times(30)^4}=1.929(mm)$$

例 19

As shown in **Fig. 3**, a simply supported beam is loaded by a noncentral load *P*. Suppose that the flexural rigidity of the beam *EI* has been known already.

(a) Construct the bending moment diagram for the beam. **(10%)**

(b) **Use the moment-area method** (sometimes called area-moment method) to determine the end slopes, the displacement at the load and the maximum displacement. **(15%)**



【104 中央機械】

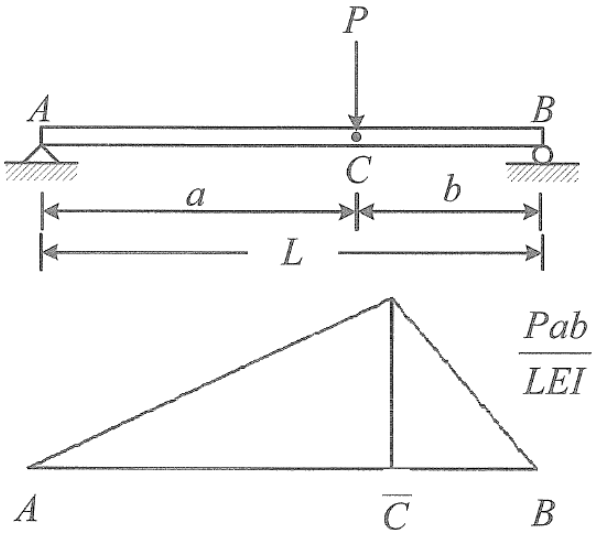
【解析】

(一)

由靜力平衡可得支承反力爲：

$$R_A = \frac{Pb}{L}(\uparrow);\quad R_B = \frac{Pa}{L}(\uparrow)$$

選取 *C* 爲參考點，繪製組合的 *M/EI* 圖



(二)

$$\triangle_{B/A}=\frac{1}{2}(\frac{Pab}{LEI})(a)\times(\frac{a}{3}+b)+\frac{1}{2}(\frac{Pab}{LEI})(b)\times(\frac{2b}{3})$$

$$=\frac{Pab}{6EI}(L+b)(\uparrow)$$

$$\Rightarrow\theta_A=\frac{\triangle_{B/A}}{L}=\frac{Pab}{6LEI}(L+b)\,(\circlearrowright)$$

同理：$\theta_B=\frac{Pab}{6LEI}(L+a)\,(\circlearrowleft)$

$$\delta_C=a\times\theta_A-\triangle_{C/A}$$

$$=\frac{Pa^2b(L+b)}{6LEI}-(\frac{Pa^2b}{2LEI})(\frac{a}{3})$$

$$=\frac{Pa^2b}{6LEI}(L+b-a)=\frac{Pa^2b^2}{3LEI}(\downarrow)$$

設最大撓度發生於 D 點（距 A 點 x 處），則 $\theta_D=0$

由第一矩面理論

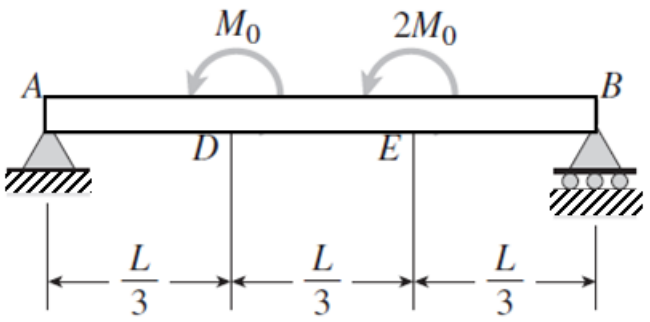
$$\theta_D=\theta_A+\frac{1}{2}(\frac{Pbx}{LEI})x=0$$

$$\Rightarrow x=\sqrt{\frac{a}{3}(L+b)}$$

$$\Rightarrow\delta_{\max}=(\frac{Pbx^2}{2LEI})(\frac{2x}{3})=\frac{Pb}{3LEI}(\sqrt{\frac{a}{3}(L+b)})^3(\downarrow)$$

例 20

A simple beam AB is subjected to couples M_0 and $2M_0$ acting as shown in the figure. Determine the angles of rotation θ_A and θ_B at the ends of the beam and the deflection δ at point D where the load M_0 is applied.



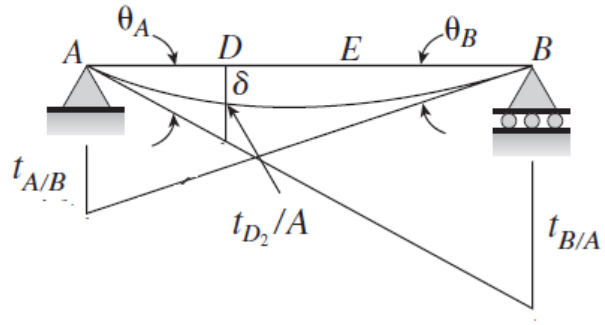
【103 台大土木】

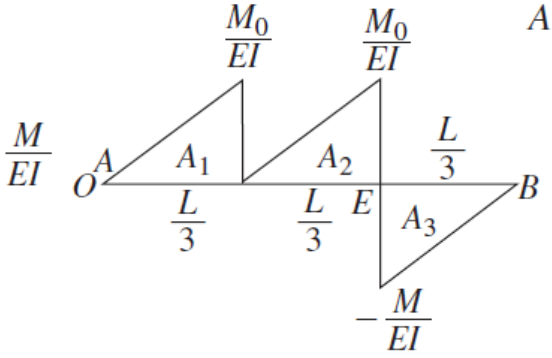
【中文翻譯】

簡樑 AB 受一力偶 M_0 及一力偶 $2M_0$ 作用在圖示的地方。求在樑端之旋轉角 θ_A 及 θ_B，以及在負載 M_0 所施加的點之撓曲 δ。

【解析】

(一)由力矩面積法





$$A_1 = A_2 = \frac{1}{2}\left(\frac{M_0}{EI}\right)\left(\frac{L}{3}\right) = \frac{M_0 L}{6EI} \qquad A_3 = -\frac{M_0 L}{6EI}$$

$$t_{B/A} = A_1\left(\frac{2L}{3} + \frac{L}{9}\right) + A_2\left(\frac{L}{3} + \frac{L}{9}\right) + A_3\left(\frac{2L}{9}\right) = \frac{M_0 L^2}{6EI}$$

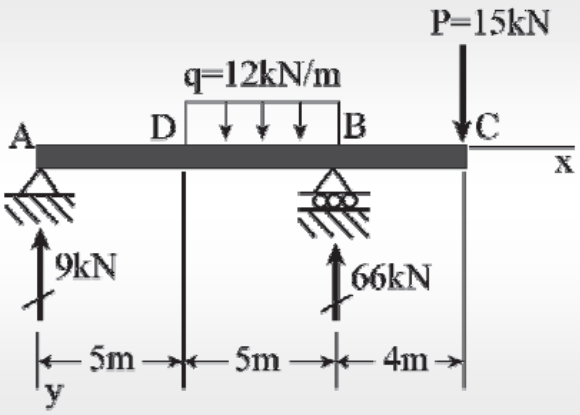
$$\theta_A = \frac{BB_1}{L} = \frac{M_0 L}{6EI}$$

$$t_{D_2}/A = A_1\left(\frac{L}{9}\right) = \frac{M_0 L^2}{54EI}$$

$$\delta = \frac{M_0 L^2}{18\,EI} - \frac{M_0 L^2}{54EI} = \frac{M_0 L^2}{27\,EI}$$

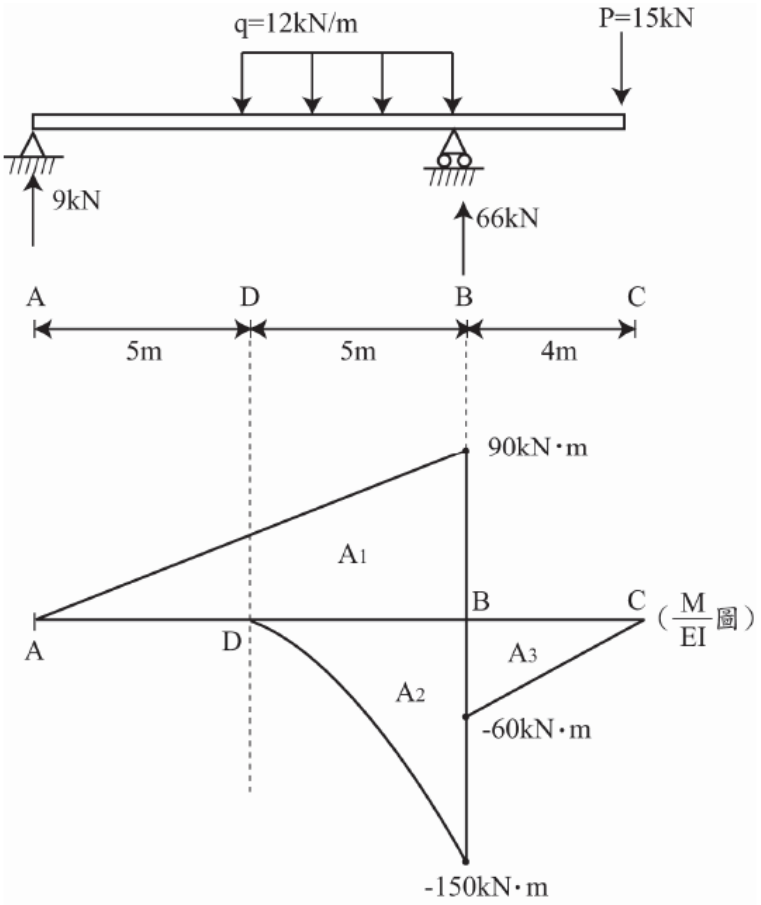
例 21

如圖示的梁 ABC 包含跨距 AB 的簡支部分及外伸的 BC 部分，求在 C 及 D 上的垂直撓度 δ_C 及 δ_D 。彈性模數 $E = 200\,GPa$ ，慣性矩 $I = 118 \times 10^6\ mm^4$ 。(98 年專利師)



【解析】

解 (1)



$$A_1=\frac{1}{EI}[90\times10\times\frac{1}{2}\times10^3]=\frac{450000}{200\times10^9\times118\times10^{-6}}=0.0191$$

$$A_2=\frac{-1}{EI}[150\times5\times\frac{1}{3}\times10^3]=\frac{250000}{23600000}=-0.0106$$

$$A_3=\frac{-1}{EI}[60\times4\times\frac{1}{2}\times10^3]=\frac{120000}{23600000}=-0.0051$$

$$\triangle_{CB}=A_3\times\frac{2}{3}\times4=-0.0136$$

$$\triangle_{AB}=A_1\times\frac{2}{3}\times10+A_2\times(5+\frac{3}{4}\times5)=0.0346$$

$$\theta_B=\frac{\triangle_{AB}}{10}=0.00346$$

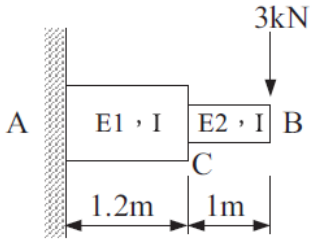
$$\delta_C=-(4\times\theta_B+\triangle_{CB})=0.152(m)$$

(2) $$\triangle_{DB}=A_2\times\frac{3}{4}\times5+\frac{45\times10^3\times5\times\frac{5}{2}+\frac{1}{2}\times45\times10^3\times5\times\frac{2}{3}\times5}{23600000}=0$$

故 $\delta_D=5\times\theta_B=0.0173(m)$

例 22

如右圖所示之複合材料懸臂樑，已知彈性模數 E_1 為 200GPa、E_2 為 100GPa，慣性矩 I 為 $1152cm^4$，今樑端 B 處受到 3KN 之集中荷重作用，則 B 點之垂直變位為何？

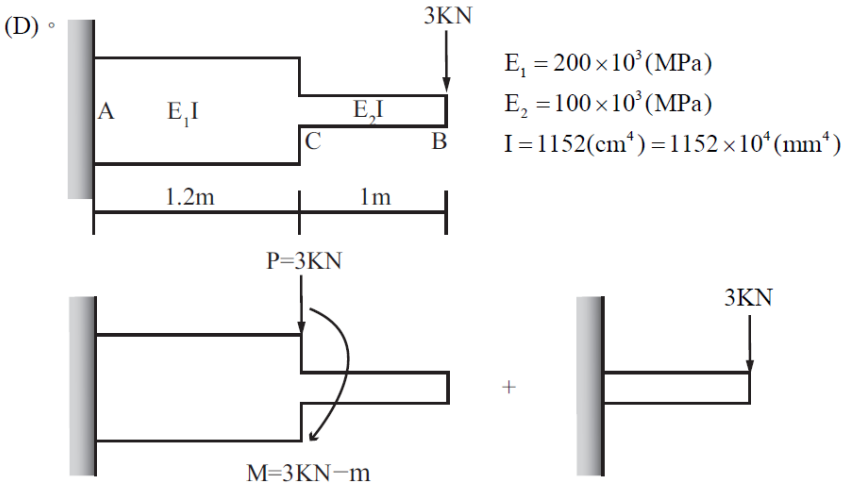


(A)2.56mm　(B)3.16mm　(C)4.42mm　(D)5.06mm。

【103 台電】

【解析】

(D)。



$E_1 = 200\times10^3(MPa)$

$E_2 = 100\times10^3(MPa)$

$I = 1152(cm^4) = 1152\times10^4(mm^4)$

$$\delta_C = \frac{PL^3}{3E_1I} + \frac{ML^2}{2E_1I} = \frac{30000\times1200^3}{3\times200\times10^3\times1152\times10^4} + \frac{3000\times1000\times1200^2}{2\times200\times10^3\times1152\times10^4}$$

$$= \frac{5184}{6912} + \frac{4320}{4608} = 0.75 + 0.9375 = 1.6875$$

$$\theta_C = \frac{PL2}{2E_1I} + \frac{ML}{E_1I}$$

$$= \frac{1}{200\times10^3\times1152\times10^4}\times\left(\frac{30000\times1200^2}{2} + 3000\times1000\times1200\right)$$

$$= \frac{432}{460800} + \frac{36}{23040} = 0.009375 + 1.5625\times10^{-3} = 2.5\times10^{-3}$$

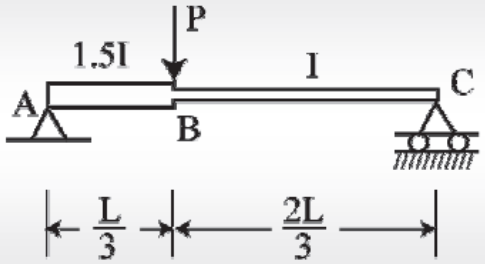
$$\delta_C^* = Q_C\times1000 = 2.5$$

$$\delta^* = \frac{PL^3}{3E_2I} = \frac{3000\times1000^3}{3\times100\times10^3\times1152\times10^4} = 0.868$$

$$\delta_B = \delta_C + \delta_C^* + \delta^* = 1.6875 + 2.5 + 0.868 = 5.05(mm)$$

例 23

簡樑 ABC，從 A 至 B 段具慣性矩 1.5I，而從 B 至 C 段具慣性矩 I (見圖)。集中負載 P 作用在點 B。求在支撐處之旋轉角 θ_A 及 θ_C，以及在點 B 之撓曲 δ_B。



【解析】

解 $E(3I/2)v'' = M = 2Px/3,\ (0 \le x \le L/3)$

(1) $EIv'' = M = PL/3 - Px/3,\ (L/3 \le x \le L)$

(2) Slopes $EIv'' = 4Px^2/ + C_1,\ (0 \le x \le L/3)$

(3) $EIv'' = \dfrac{PLx}{3} - \dfrac{Px^2}{6} + C_2,\ \left(\dfrac{L}{3} \le x \le L\right)$

當 $x = \dfrac{L}{3}$ (B.C.1)：$\dfrac{4P}{18}\left(\dfrac{L}{3}\right)^2 + C_1 = \dfrac{PL}{3}\left(\dfrac{L}{3}\right) - \dfrac{P}{6}\left(\dfrac{L}{3}\right)^2 + C_2 \Rightarrow C_2 = C_1 - \dfrac{11PL^2}{162}$

(4) 位移

$$EIv = \frac{4Px^3}{54} + C_1X + C_3,\ \left(0 \le x \le \frac{L}{3}\right)$$

$$EIv = \frac{PLx^2}{6} - \frac{Px^3}{18} + C_2X + C_4,\ \left(\frac{L}{3} \le x \le L\right)$$

B.C.2: $v(0) = 0 \quad \therefore C_3 = 0$

(5) B.C.3: $v(L) = 0 \quad \therefore C_4 = -\dfrac{PL^3}{9} - C_2L$

B.C.4: $x = \frac{L}{3}$

$$\frac{4P}{54}\left(\frac{L}{3}\right)^3 + C_1\left(\frac{L}{3}\right) = \frac{PL}{6}\left(\frac{L}{3}\right)^2 - \frac{P}{18}\left(\frac{L}{3}\right)^3 + C_2\left(\frac{L}{3}\right) + C_4$$

$$C_1 L = \frac{10PL^3}{243} + C_2 L + 3C_4$$

$$C_1 L = -\frac{38PL^2}{729}, C_2 = -\frac{175PL^2}{1458}, C_3 = 0, C_4 = \frac{13PL^3}{1458}$$

$$\nu' = -\frac{2PL}{729EI}\left(19L^2 - 81X^2\right), \left(0 \le x \le \frac{L}{3}\right)$$

$$\nu' = -\frac{P}{1458EI}\left(175L^2 - 486Lx + 243x^2\right), \quad \left(\frac{L}{3} \le x \le L\right)$$

$$\theta_A = -\nu'(0) = \frac{38PL^2}{729EI} \Leftarrow \quad \theta_C = \nu'(L) = \frac{34PL^2}{729EI}$$

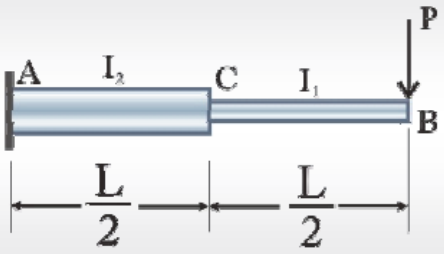
位移 $\nu = -\frac{2Px}{729EI}\left(19L^2 - 27x^2\right), \quad \left(0 \le x \le \frac{L}{3}\right)$

$$\nu = \frac{P}{1458EI}\left(13L^3 - 175x^2 + 243Lx^2 - 81x^3\right), \quad \left(\frac{L}{3} \le x \le L\right)$$

$$\delta_B = -\nu\left(\frac{L}{3}\right) = \frac{32PL^3}{2187EI}$$

例 24

示於圖之懸臂樑 ACB，AC 及 CB 段之慣性矩分別為 I_2 及 I_1，求由於負載 P 在自由端之撓曲 δ_B 。



【108 台大應力】【108 台大土木】

【解析】

解 (1) BC 段

$$\delta_1 = \frac{P(L/2)^3}{3EI_1} = \frac{PL^3}{24EI_1}$$

(2) AC 段

$$\delta_c = \frac{P(L/2)^3}{3EI_2} = \frac{(PL/2)(L/2)^2}{EI_2} = \frac{52PL^2}{48EI_2}$$

(3) B 變位

$$\delta_B = \delta_1 + \delta_c + \theta_c\left(\frac{L}{2}\right) = \frac{PL^3}{24EI_1}\left(1 + \frac{7I_1}{I_2}\right)$$

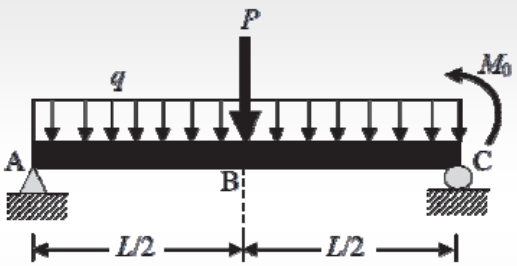
例 25

如圖三所示，均勻簡支梁 ABC，梁的撓曲剛度為 EI。其中 L=2m，q=12kNm，P=36kN，M0=48kN·m。

(1) 試求 A 及 C 端之反作用力。

(2) 試繪製梁 ABC 之剪力與彎矩分布圖，並標示最大剪力及最大彎矩，以及剪力及彎矩不連續處之值為何。

(3) 若 EI=135kN·m^2，試求 C 端處之轉角。

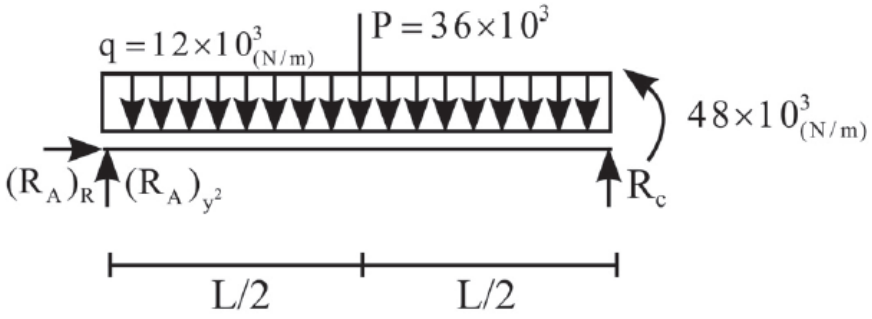


圖三

【104 機械高考】

【解析】

解 (1)

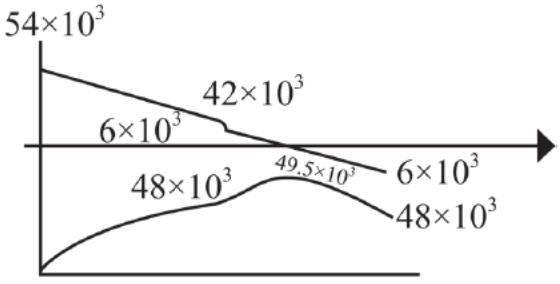


$\circlearrowright$ 之 $M_A=0$

$2R_C=(-48+36+12\times2)\times10^3\rightarrow R_C=6\times10^3$

$\uparrow F_y=0$

$(R_A)_y=(-6+36+12\times2)\times10^3\rightarrow(R_A)_y=54\times10^3$



$\theta_C=(\theta_C)_1+(\theta_C)_2+(\theta_C)_3$

其中

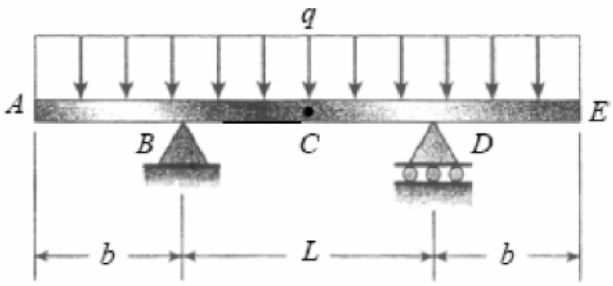
$$(\theta_C)_1 = \frac{PL^2}{EI16} \mid P = 36\times10^3 = \frac{9}{EI}$$

$$(\theta_C)_2 = \frac{M_oL}{3EI} \mid M_o = 48\times10^3 = \frac{32}{EI} \text{ 故 } \theta_C = \frac{35}{145} = \frac{1}{3}$$

$$(\theta_C)_3 = \frac{qL^3}{24EI} \mid q = 12\times10^3 = \frac{4}{EI} \text{ 。}$$

例 26

樑 ABCDE 由 B、D 兩點簡支，中間段長度為 L，兩邊外伸長度為 b，樑上承載均佈載重其強度為 q。(一)若樑中心點之撓度（deflection）δ_C 與兩端點之撓度 δ_A 與 δ_E 相等時，試求 b/L 之比值。(二)此時 δ_C 之值為多少？（20 分）(97 年專利師)



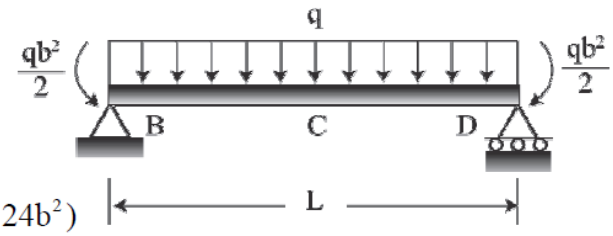
【台大應力】

【解析】

解 $\theta_B = \dfrac{qL^3}{24EI} - \dfrac{qL^2}{2}\left(\dfrac{L}{2EI}\right)$

$= \dfrac{qL}{24EI}(L^2 - 6b^2)$ (順時針為正)

$\delta_C = \dfrac{5qL^4}{384EI} - \dfrac{qL^2}{2}\left(\dfrac{L^2}{8EI}\right) = \dfrac{qL^2}{384EI}(5L^2 - 24b^2)$



B_{EAM} AB：

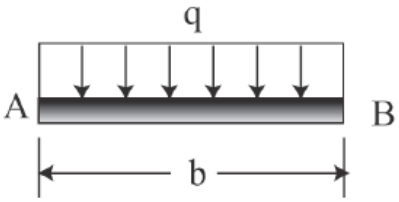


Table G-1, Case 1：

$$\delta_A = \frac{qb^4}{8EI} - \theta_B b = \frac{qb^4}{8EI} - \frac{qL}{24EI}(L^2 - 6b^2)b = \frac{qb}{24EI}(3b^3 + 6b^2L - L^3)$$ (向下為正)

$$\frac{qL^2}{384EI}(5L^2 - 24b^2) = \frac{qb}{24EI}(3b^3 + 6b^2L - L^3)$$

$$\Rightarrow 48b^4 + 96b^3L + 24b^2L^2 - 16bL^3 - 5L^4 = 0$$

$$\text{or} \Rightarrow 48\left(\frac{b}{L}\right)^4 + 96\left(\frac{b}{L}\right)^3 + 24\left(\frac{b}{L}\right)^2 - 16\left(\frac{b}{L}\right) - 5 = 0$$

(a) $R_{ATIO}\ \dfrac{b}{L}$

$$\frac{b}{L} = 0.40301$$

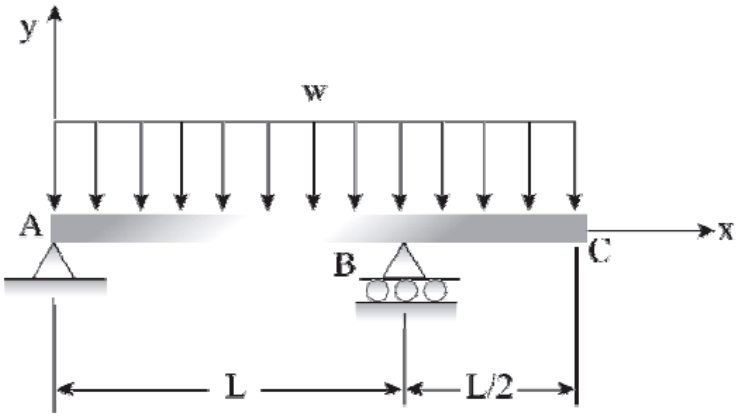
(b) $$\delta_C = \frac{qL^2}{384EI}(5L^2 - 24b^2)$$

$$= \frac{qL^2}{384EI}[5L^2 - 24(0.40301\,L)^2]$$

$$= 0.002870\frac{qL^4}{EI}$$

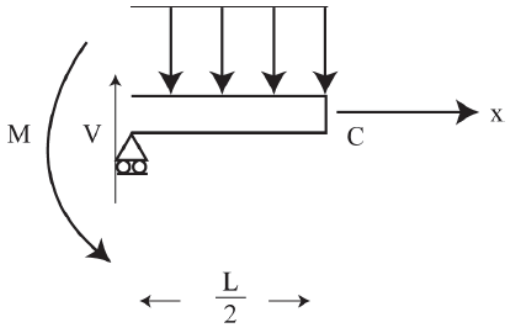
例 27

右圖簡支梁原長 L，但另超出原有跨距外懸 L/2。已知該梁斷面慣性矩為 I、材料的楊氏係數為 E。假設梁承受圖中所示之均佈荷重 w，在線彈性範圍的假設下，請問 A、B 兩支撐點的變位角（slope）各為何？（請列出算式，並以前述符號表示答案）。（99 關三）



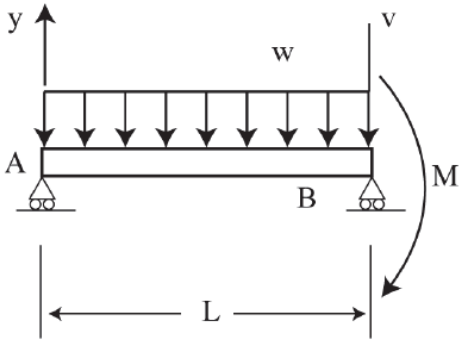
【解析】

解 (1) 取 BC 自由體圖



$$v=\frac{wL}{2}\ 、\ \sum M_B=0\Rightarrow M=\frac{wL^2}{8}$$

(2) 取 AB 自由體圖



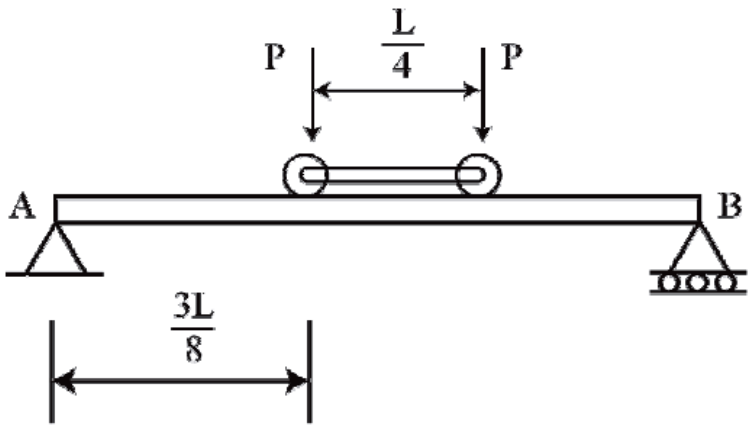
利用基本撓度公式線性疊加

$$\theta_A = \frac{wL^3}{24EI} - \frac{ML}{6EI} = \frac{wL^3}{24EI} - \frac{\left(\frac{wL^2}{8}\right)L}{6EI} = \frac{wL^3}{48EI}$$

$$\theta_B = \frac{wL^3}{24EI} - \frac{ML}{6EI} = \frac{wL^3}{24EI} - \frac{\left(\frac{wL^2}{8}\right)L}{3EI} = 0$$

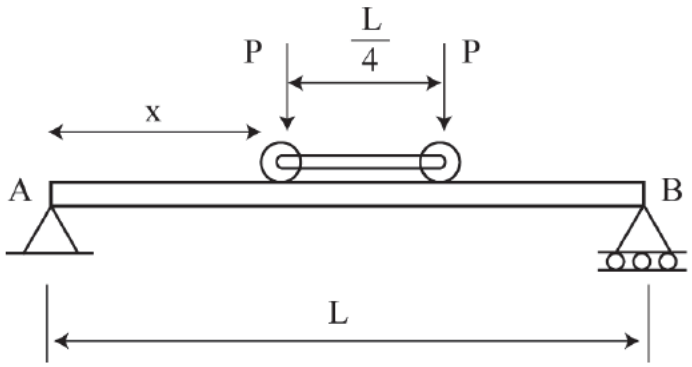
例 28

兩相等的相距 L/4 的兩輪，各荷載負載 P，慢慢地沿樑 AB 的跨距 L 移動（見圖）。求在樑的中點之最大撓曲的值 δ_c。(97 年土木地特三等)



【解析】

解



(1) 當 $x=\frac{L}{4}\text{or}\frac{L}{2}$

$$\delta_c=\frac{1}{48EI}\left(\frac{PL}{4}\right)\left[3L^2-4\left(\frac{L}{4}\right)^2\right]+\frac{PL^3}{48EI}=\frac{9PL^3}{256EI}$$

(2) 當 $\frac{L}{4} < x < \frac{L}{2}$

$$\delta_c = \frac{Px\left(3L^2 - 4x^2\right)}{48EI} + \frac{P\left(3L/4 - x\right)\left[3L^2 - 4\left(3L/4 - x\right)^2\right]}{48EI}$$

$$= \frac{3PL}{256EI}\left(L^2 + 12Lx - 16x^2\right)$$

(3) 最大位移 $d\delta_c/dx = 0$ or $12L - 32x = 0$

$$x = \frac{3L}{8}$$

$$\delta_c = \frac{39PL^3}{1024EI}$$

$$\left(\delta_c\right)_{max} = \frac{39PL^3}{1024EI}$$

例 29

某左端固定之懸臂梁，長度為 L、撓性剛度為 EI。其上承受往下均布負載 w 作用，且其右端同時承受一順時針彎矩 $M_o = wL^2/2$ 作用。試求

(1)梁最大之旋轉角（撓角），（以 w、L、EI 表之）。

(2)梁最大之撓度，（以 w、L、EI 表之）。(98 年地方特考四等)

【解析】

解 $M_0 = \dfrac{WL^2}{2}$

(1) 利用基本撓度公式

均布負載 W 作用之撓角 $\theta_1 = \dfrac{WL^3}{6EI}$ (↻)

彎矩 M_0 作用之撓角 $\theta_2 = \dfrac{M_0 L}{EI} = \dfrac{WL^3}{2EI}$ (↻)

$$\theta = \theta_1 + \theta_2 = \frac{2WL^3}{3EI} \ (\circlearrowright)$$

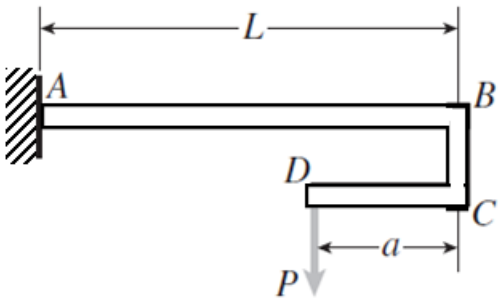
(2) 均布負載 W 作用之撓度 $\delta_1 = \dfrac{WL^4}{8EI}$ (↓)

彎矩 M_0 作用之撓度 $\delta_2 = \dfrac{M_0 L^2}{2EI} = \dfrac{WL^4}{4EI}$

$$\delta = \delta_1 + \delta_2 = \frac{3WL^4}{8EI}$$

例 30

The cantilever beam AB shown in the figure has an extension BCD attached to its free end. A force P acts at the end of the extension. (a) Find the ratio a/L so that the vertical deflection of point B will be zero. (b) Find the ratio a/L so that the angle of rotation at point B will be zero.



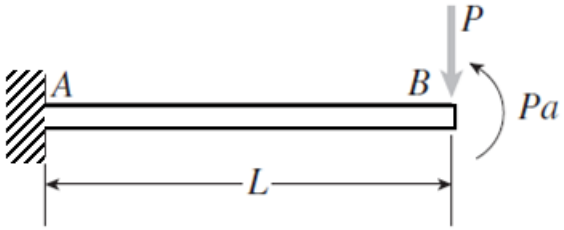
【104 台大應力】

【中文翻譯】

示於圖之懸臂樑 *AB* 有一外伸 *BCD* 依附於自由端。力 *P* 作用於外伸之自由端。(a) 求 a/L 之比率，使點 *B* 的垂直撓曲為零。(b) 求比率 a/L ，使點 *B* 的旋轉角為零。

【解析】

(一)由等效力系法



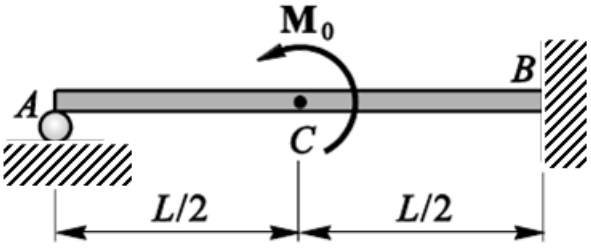
(二)由重疊原理

$$\text{(a)}\ \delta_B = \frac{PL^3}{3EI} - \frac{PaL^2}{2EI} = 0 \quad \frac{a}{L} = \frac{2}{3}$$

$$\text{(b)}\ \theta_B = \frac{PL^2}{2EI} - \frac{PaL}{EI} = 0 \quad \frac{a}{L} = \frac{1}{2}$$

例 31

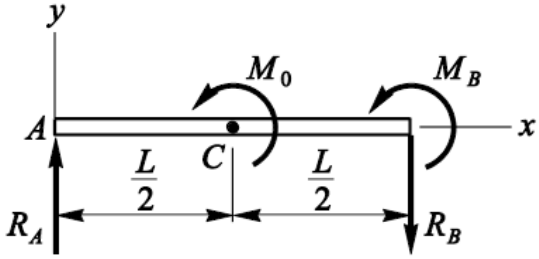
圖中支撐懸臂梁 AB 在中點 C 承受一力偶矩 M_0，試求梁在支承 A 之反力與中點 C 之撓度。



【101 成大工科】

【解析】

(一)內力分析



(一)由奇異函數法

設取 R_A 為贅力

$$EIy'' = R_A x - M_0\left\langle x-\frac{L}{2}\right\rangle^0$$

$$EIy' = \frac{R_A}{2}x^2 - M_0\left\langle x-\frac{L}{2}\right\rangle^1 + C_1 \quad .....①$$

$$EIy = \frac{R_A}{6}x^3 - \frac{M_0}{2}\left\langle x-\frac{L}{2}\right\rangle^2 + C_1 x + C_2 \quad .....②$$

(二)邊界條件

$BC(1)：x=0，\ y=0，\ 得\ C_2=0$

$$BC(2)：x=L，\ y'=0，\ \frac{R_A}{2}(L^2) - M_0\left(\frac{L}{2}\right) + C_1 = 0 \quad .....③$$

$$BC(3)：x=L，\ y=0，\ \frac{R_A}{6}(L^3) - \frac{M_0}{2}\left(\frac{L}{2}\right)^2 + C_1(L) = 0 \quad .....④$$

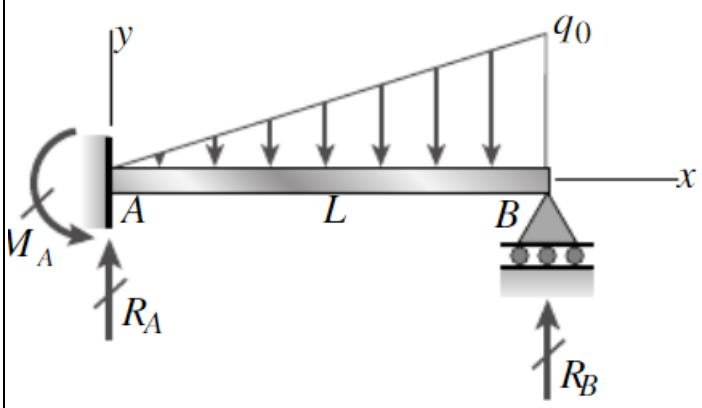
由③④兩式解得：$R_A = \dfrac{9M_0}{8L}$、$C_1 = -\dfrac{M_0 L}{16}$

$$EIy_C = \frac{1}{6}\left(\frac{9M_0}{8L}\right)\left(\frac{L}{2}\right)^3 - \frac{M_0 L}{16}\left(\frac{L}{2}\right) = -\frac{M_0 L^2}{128}$$

$$y_C = \frac{M_0 L^2}{128EI}(\downarrow)$$

例 32

A cantilever beam of length L and loaded by a triangularly distributed load of maximum intensity q_0 at B.DDetermine the reactions forces at A and B and the angle of the rotation θ_B



【106 台大土木】

【中文翻譯】

一支撐懸臂樑 *AB* 長 *L* 支持一三角形分佈負載，最大強度為 q_0 （ 見圖 ）。，求樑的反應力及撓曲曲線的方程式。

【解析】

(一)由四重積分法

$$q = q_0 \frac{x}{L}$$

$$EIv'''' = -q = -q_0 \frac{x}{L}$$

$$EIv''' = -q_0 \frac{x^2}{2L} + C_1$$

$$EIv'' = M = -q_0 \frac{x^3}{6L} + C_1 x + C_2$$

$$EIv' = -q_0 \frac{x^4}{24L} + C_1 \frac{x^2}{2} + C_2 x + C_3$$

$$EIv = -q_0 \frac{x^5}{120L} + C_1 \frac{x^3}{6} + C_2 \frac{x^2}{2} + C_3 x + C_4$$

(二)帶入邊界條件求出撓曲線方程式

B.C. 1 $v''(L) = 0 \qquad \therefore\ C_1 L + C_2 = q_0 \frac{L^2}{6}$

B.C. 2 $v'(0) = 0 \qquad \therefore\ C_3 = 0$

B.C. 3 $v(0) = 0 \qquad \therefore\ C_4 = 0$

B.C. 4 $v(L) = 0 \qquad \therefore\ C_1 \frac{L}{3} + C_2 = q_0 \frac{L^2}{60}$

$$C_1 = \frac{9}{40} q_0 L$$

$$C_2 = -\frac{7}{120} q_0 L^2$$

$$V = -q_0 \frac{x^2}{2L} + \frac{9}{40} q_0 L$$

$$R_A = V(0) = \frac{9}{40} q_0 L$$

$$R_B = -V(L) = \frac{11}{40} q_0 L$$

$$M_A = \frac{7}{120} q_0 L^2$$

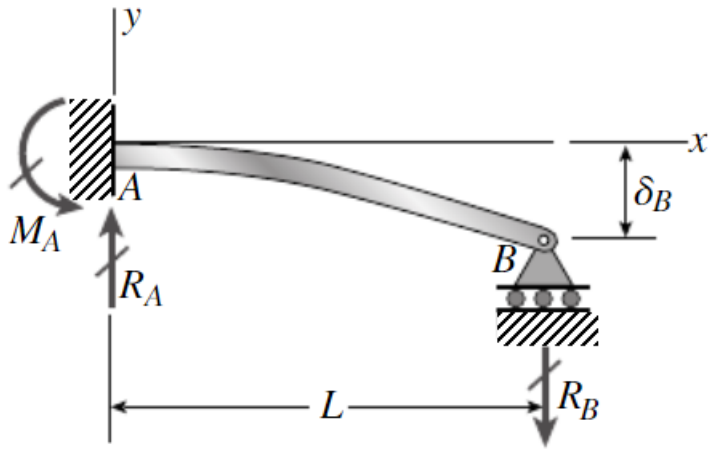
$$EIv = -q_0 \frac{x^5}{120L} + \frac{9}{40} q_0 L \frac{x^3}{6} - \frac{7}{120} q_0 L^2 \frac{x^2}{2}$$

$$v = \frac{1}{240LEI}(-2q_0 x^5 + 9q_0 L x^3 - 7q_0 L^2 x^2)$$

$$\theta_B = v'(L) = \left(\frac{1}{EI}\right)\left[\left(-q_0 \frac{x^4}{24L}\right) + \left(q_0 L \frac{9x^2}{80}\right) + \left(-q_0 L^2 \frac{7x}{120}\right)\right] = \frac{q_0 L^3}{80EI}$$

例 33

A cantilever beam AB of length L has a fixed support at A and a roller support at B (see figure). The support at B is moved downward through adistance δ_B Determine the reactions of the beam and the equation of the deflectioncurve. (Note: Express all results in terms of the imposed displacement .



【100 清大動機】

【中文翻譯】

一懸臂樑 *AB* 長 *L* 具固定支撐於 *A*，及一滾子支撐於 *B* （ 見圖 ）。支撐 *B* 向下移了一距離 δ_B，求樑之反應力及撓曲曲線的方程式。(註：將所有的結果以位移 δ_B 為項表示)。

【解析】

(一)由四重積分法

$$R_A = R_B \qquad (1) \qquad\qquad M_A = R_B L$$

$$EIv'''' = -q = 0$$

$$EIv''' = V = C_1$$

$$EIv'' = M = C_1 x + C_2$$

$$EIv' = C_1 x^2/2 + C_2 x + C_3$$

$$EIv = C_1 x^3/6 + C_2 x^2/2 + C_3 x + C_4$$

(二)帶入邊界條件求出撓曲線方程式

$$1\ v(0) = 0 \qquad \therefore C_4 = 0$$

$$2\ v'(0) = 0 \qquad \therefore C_3 = 0$$

-

$$3\ v''(L) = 0 \qquad \therefore\ C_1L + C_2 = 0$$

$$4\ v(L) = -\delta_B$$

$$C_1L + 3C_2 = -6EI\delta_B/L^2$$

$$C_1 = \frac{3EI\delta_B}{L^3} \qquad C_2 = -\frac{3EI\delta_B}{L^2}$$

$$V = \frac{3EI\delta_B}{L^3} \qquad R_A = V(0) = \frac{3EI\delta_B}{L^3}$$

$$R_A = R_B = \frac{3EI\delta_B}{L^3}$$

$$M_A = R_BL = \frac{3EI\delta_B}{L^2}$$

$$v = -\frac{\delta_B x^2}{2L^3}(3L - x)$$

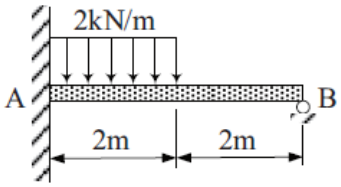
$$v' = -\frac{3\delta_B x}{2L^3}(2L - x)$$

例 34

39. 如右圖所示，樑之固接端 A 處之彎矩大小為何？

(A)1.95KN-m　(B)2.25KN-m

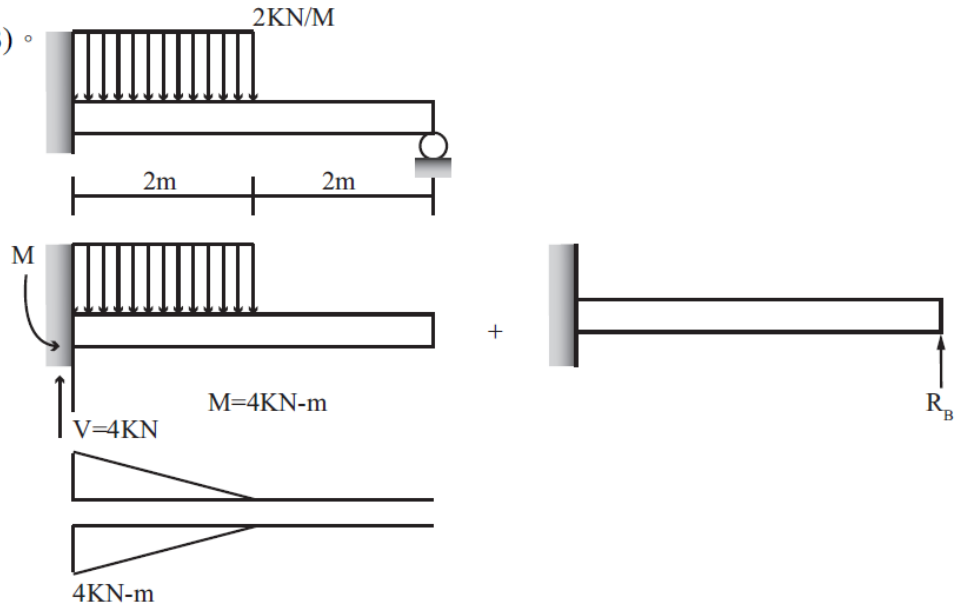
(C)3.25KN-m　(D)3.85KN-m。



【103 台電】

【解析】

39. (B)。



$\delta_B=(\delta_B)_1+(\delta_B)_2=0$ ，其中，

$(\delta_B)_1=\frac{1}{EI}\cdot\left(4\times\frac{1}{3}\times2\times\left(\frac{3\times2}{4}+2\right)\right)=\frac{9.33}{EI}$ ，$(\delta_B)_2=\frac{-PL^3}{3EI}\Big|_{P=R_B}^{L=4m}=\frac{-R_B4^3}{3EI}$

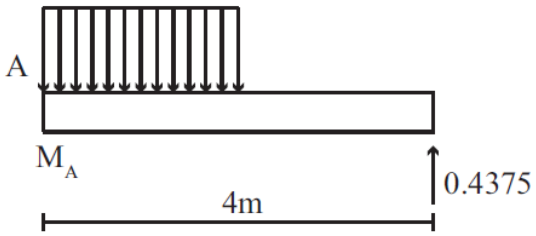
代入可得

$\delta_B=\frac{9.33}{EI}-\frac{R_B4^3}{3EI}=0\Rightarrow R_B=0.4375$

取桿 FBD

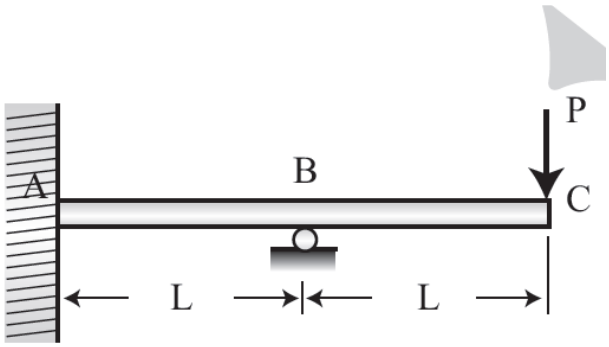
$\sum M_A=0$

$M_A=4-0.4375\times4=2.25$ (KN-m)



例 35

有一懸臂樑如圖所示，在中點有一支點B，在自由端C受到一向下的力P，試問：(1)此樑在此負荷與拘束狀態之下稱為什麼狀態？(2)求各支點之反作用力與力矩。(3)繪出剪力圖與彎曲力矩圖。(95年機械技師)



【台大機械】

【解析】

答：(1) 靜不定樑結構。

(2) 假設B為贅力x，釋放結構體：

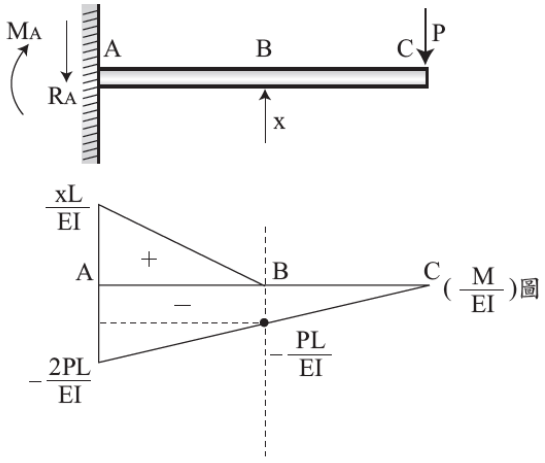
利用力矩面積法第二定理

$$\Delta_{BA} = \delta_B = 0$$

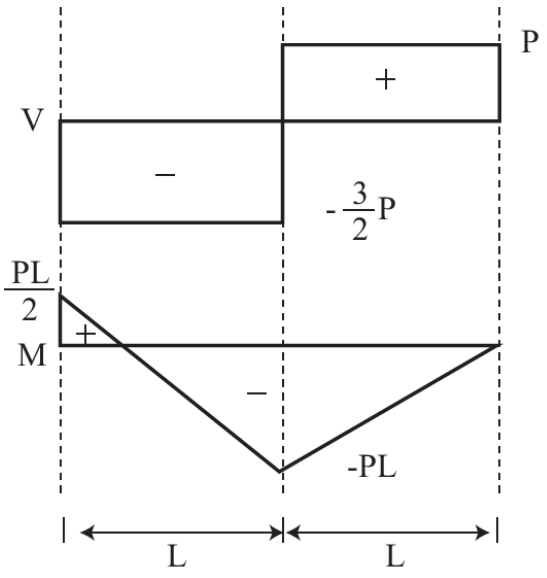
$$\Rightarrow \frac{1}{EI}\left[-PL\times L\times\frac{L}{2}-PL\times L\times\frac{1}{2}\times\frac{2}{3}L+xL\times L\times\frac{1}{2}\times\frac{2}{3}L\right]=0$$

$$\Rightarrow \frac{5PL^3}{6}=\frac{xL^3}{3}\Rightarrow R_B=x=\frac{5}{2}P(\uparrow)\quad \Sigma F_y=0\quad R_A=\frac{3P}{2}(\downarrow)$$

$$\Sigma M_A=0\Rightarrow M_A=\frac{PL}{2}(\circlearrowleft)$$

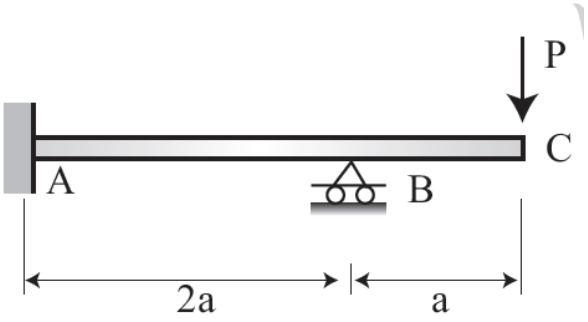


(3) 畫剪力圖與彎矩圖：



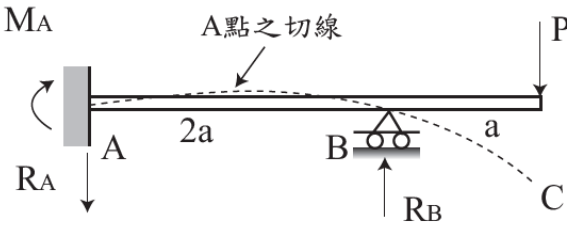
例 36

由一端懸空與另一段為連續樑組合而成之簡支樑，其支撐與負載情形如圖所示。利用力矩面積法求解支撐之反作用力。



【解析】

答：另 R_B 為贅力，則



$\theta_A = 0$

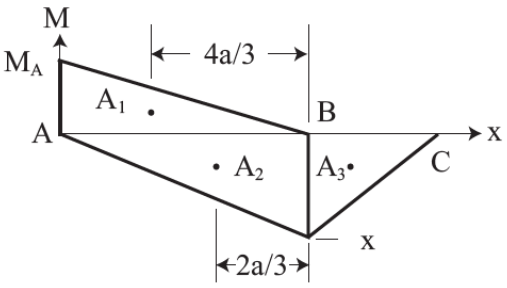
$$A_1 = \frac{1}{2}M_A(2a) = M_A a$$

$$A_2 = -\frac{1}{2}Pa(2a) = -Pa^2$$

$$A_3 = -\frac{1}{2}Pa^2$$

$$v_B = t_{BA} = 0 = \frac{1}{EI}[A_1(\frac{4a}{3}) + A_2(\frac{4a}{3})]$$

或 $M_A = \frac{1}{2}Pa$

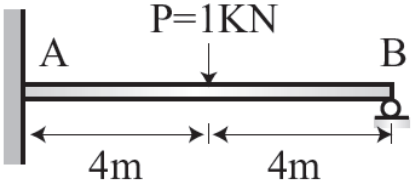


靜力平衡：$R_B = \frac{7}{4}P \uparrow$　　$R_A = \frac{3}{4}P \downarrow$

例 37

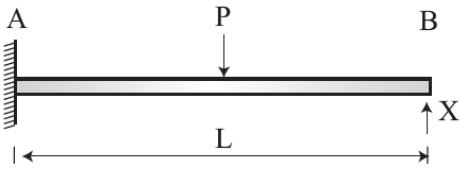
一根橫樑受力情形如圖所示。求B處的反作用力為何？已知 $E=200\ GPa$ 及 $I=100\ cm^4$ 。

註：懸臂樑端點受P力時，端點的撓度及斜率分別為 $y=\frac{PL^3}{3EI}$ 及 $\theta=\frac{PL^2}{2EI}$ (97年關務三等)



【解析】

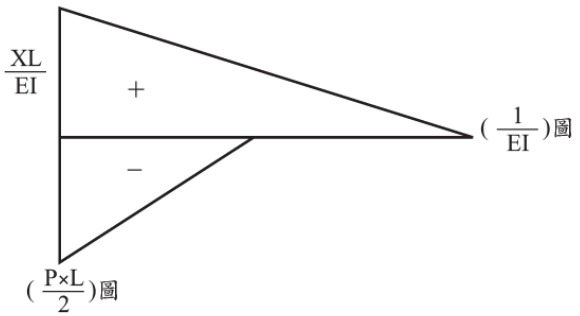
答：假設B點受力為x，靜定釋放



$$\Delta_{BA}=0=\frac{1}{EI}\{[-\frac{P\times L}{2}\times\frac{L}{2}\times\frac{1}{2}\times(\frac{2}{3}\times\frac{L}{2}+\frac{L}{2})]+[xL\times\frac{1}{2}\times L\times\frac{2}{3}L]\}$$

$$\Rightarrow\frac{xL^3}{3EI}=\frac{5PL^3}{48EI}$$

$$\Rightarrow x=\frac{5}{16}P=\frac{5}{16}\times 1\times 10^3=312.5(N)$$

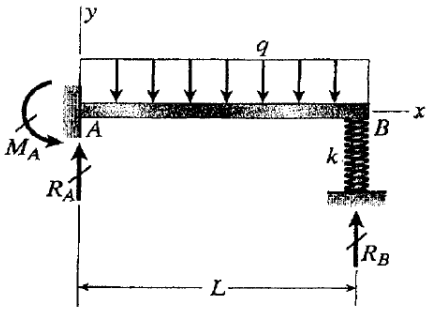


例 38

(25%) A uniform cantilever beam AB of length L and flexure rigidity EI has a fixed support at A and a linear-spring support at B (as shown). If a uniform load of intensity q acts on the beam,

(a) what is the displacement δ_B of end B of the beam?

(b) Find all the support reactions when the spring constant $k \to \infty$.



【成大航太】【清大動機】【土木高考】

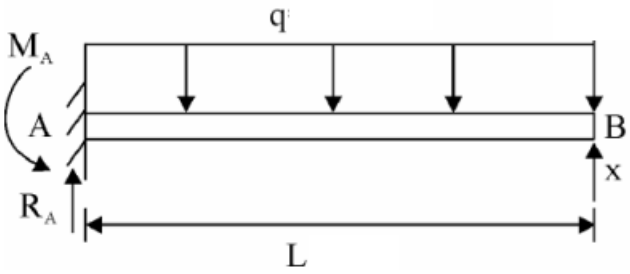
【中文翻譯】

一懸臂樑 AB 長 L 具一固定支撐於 A 及一彈簧支撐於 B （ 見圖 ）。彈簧係線性彈性的，具勁度 K 。如果一強度 q 的均勻負載作用在樑上，樑之端 B 的向下位移為何？

【解析】

(一)重疊原理

1. **靜不定結構假設 B 為贅力 x：**



利用基本撓度公式

$$\delta_B = \frac{qL^4}{8EI} - \frac{xL^3}{3EI}$$

又彈簧 $x = k\delta_B$

$$x = R_B = \frac{3qL^4}{8(KL^3 + 3EI)}$$

5.若 $K \to \infty$

$$0 = \frac{qL^4}{8EI} - \frac{xL^3}{3EI} \Rightarrow x = R_B = \frac{3}{8}qL$$

$$R_A = \frac{5}{8}qL \text{ 、 } M_A = \frac{1}{8}qL^2$$

(二)另解:積分法

$$R_A = qL - R_B$$

$$M_A = \frac{qL^2}{2} - R_B L$$

$$M = R_A x - M_A - \frac{qx^2}{2}$$

$$EI\nu'' = M = R_A x - M_A - \frac{qx^2}{2}$$

$$EI\nu' = R_A \frac{x^2}{2} - M_A x - \frac{qx^3}{6} + C_1$$

$$EI\nu = R_A \frac{x^3}{6} - M_A \frac{x^2}{2} - \frac{qx^4}{24} + C_1 x + C_2$$

B.C.1: $\nu'(0) = 0, \ \therefore \ C_1 = 0$

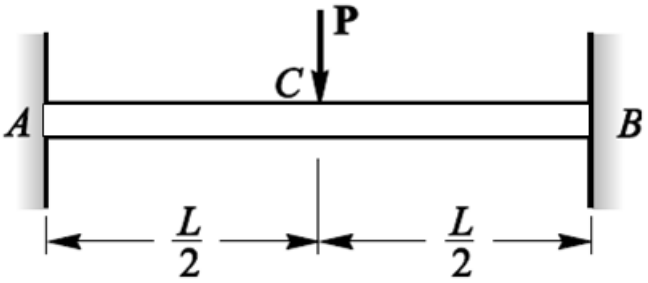
B.C.2: $\nu(0) = 0, \ \therefore \ C_2 = 0$

B.C.3: $\nu(L) = -\delta_B, \ \therefore \ -EI\delta_B = \frac{R_A L^3}{6} - \frac{M_A L^2}{2} - \frac{qL^4}{24}$

$$\delta_B = \frac{3qL^4}{24EI + 8kL^3}$$

例 39

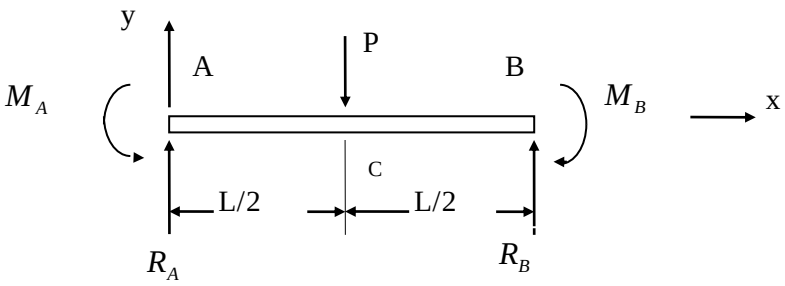
一兩端固定之樑於跨距中點承受一集中負載 P(如圖所示)。試求所有反作用力與彈性曲線之方程式。



【107 成大航太】

【解析】

(一)內力分析



由對稱性可得：

$$M_A = -M_B$$

$$R_A = R_B = \frac{P}{2}$$

<u>AC 段</u>

$$EIv'''' = 0, \qquad EIv''' = c_1, \qquad EIv'' = c_1 x + c_2$$

$$EIv' = \tfrac{1}{2} c_1 x^2 + c_2 x + c_3, \quad EIv = \tfrac{1}{6} c_1 x^3 + \tfrac{1}{2} c_2 x^2 + c_3 x + c_4$$

(二)邊界條件：

$$EIv'''(0) = c_1 = V = -\tfrac{P}{2}, \quad EIv''(0) = M_A, \quad c_2 = M_A$$

$v'(0) = 0: \quad c_3 = 0, \qquad v'(L/2) = 0: \quad M_A = -M_B = \dfrac{\mathrm{PL}}{8}$

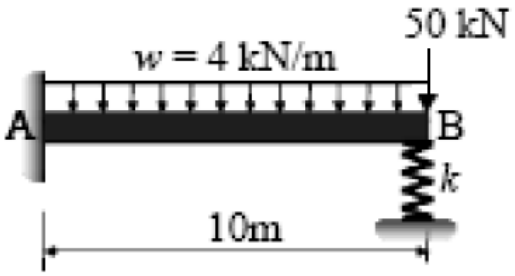
$v(0) = 0: \quad c_4 = 0.$

帶入方程式

$$v = -\frac{Px^2}{48EI}(3L - 4x)$$

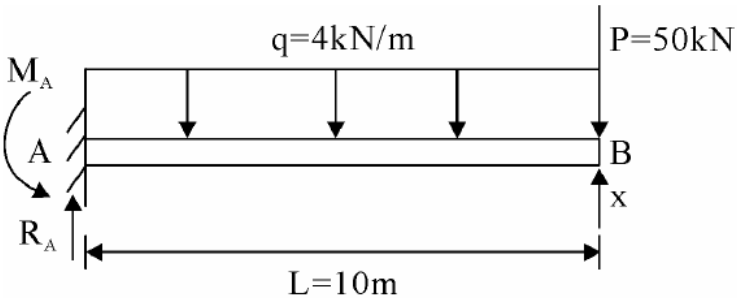
例 40

樑 AB 其左端固定於 A 點，B 點為一彈簧支撐，如圖所示。在承受圖示之荷重時，試求(一)A、B 兩支撐之反作用力(二)樑之最大剪力及最大彎矩。假設樑之彎曲剛度(flexural rigidity)EI=10MN-m^2，彈簧的彈簧常數為 2kN/m。(97 年機械鐵路特考三等)



【解析】

1. **靜不定結構假設 B 為贅力 x：**



利用基本撓度公式

$$\delta_B = \frac{qL^4}{8EI} - \frac{xL^3}{3EI} + \frac{PL^3}{3EI}(\downarrow)$$

$$= \frac{4\times10^4}{8\times10\times10^3} - \frac{x\times10^3}{3\times10\times10^3} + \frac{50\times10^3}{3\times10\times10^3} = 2.167 - \frac{x}{30}$$

又彈簧 $x = k\delta_B = 2\times(2.167 - \frac{x}{30}) \Rightarrow x = 4.063(kN)$

$R_A = 50 + 4\times10 - 4.063 = 85.94(kN)$

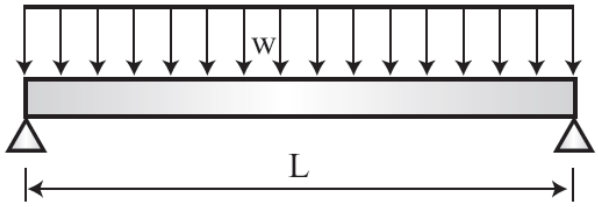
$\sum M_A = 0 \Rightarrow M_A = 659.37kN\cdot m(\curvearrowleft)$

2. **最大剪力 $V_{max} = R_A = 85.94(kN)$：**

最大彎矩 $M_{max} = M_A = 659.37kN\cdot m$

例 41

工程人員為了怕建築用的石柱變髒而將石柱放在兩塊石頭間（如下所示，石柱長L；兩石頭支點之距離亦為L；石柱單位長度之重量為W），但卻發現石柱常因本身之重量而從中斷裂。工程人員為了避免此事發生，在兩支點正中間（L/2處）加另一個支撐的石頭以增加支撐；請列出石柱最大的彎矩（bending moment）

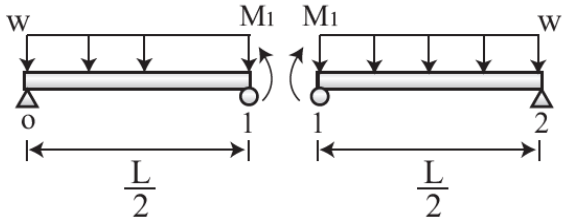


【解析】

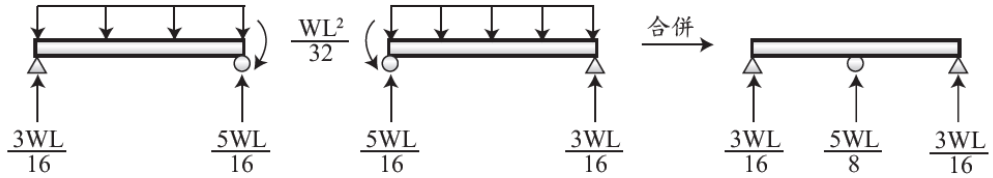
(1) 若中間有支點，假設m_1為贅力，利用三力矩公式

$$\overline{01}\text{段}\ \theta_1=\frac{-W(\frac{L}{2})^3}{24EI}-\frac{M_1(\frac{L}{2})}{2EI}\ (\curvearrowright)$$

$$\overline{12}\text{段}\ \theta_2=\frac{W(\frac{L}{2})^3}{24EI}+\frac{M_1(\frac{L}{2})}{3EI}\ (\curvearrowright)$$



$$\theta_1=\theta_2\ \Rightarrow\frac{-W(\frac{L}{2})^3}{24EI}-\frac{M_1(\frac{L}{2})}{3EI}=\frac{W(\frac{L}{2})^3}{24EI}+\frac{M_1(\frac{L}{2})}{3EI}\Rightarrow M_1=-\frac{WL^2}{32}$$

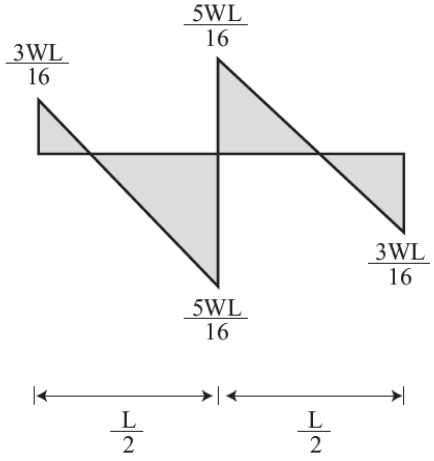


(2) 故VD圖為

$$M^+=\frac{3WL}{16}\times\frac{3}{8}\times\frac{L}{2}=\frac{9WL^2}{256}$$

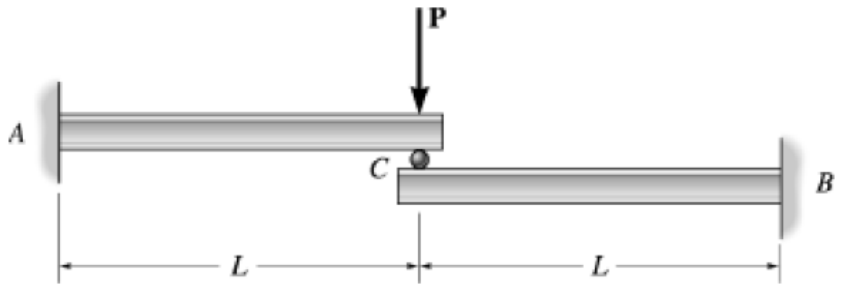
$$M^-=\frac{5WL}{16}\times\frac{5}{8}\times\frac{L}{2}-\frac{9WL^2}{256}=\frac{1WL^2}{16}$$

故三塊石頭最大彎矩$=\frac{1WL^2}{16}$



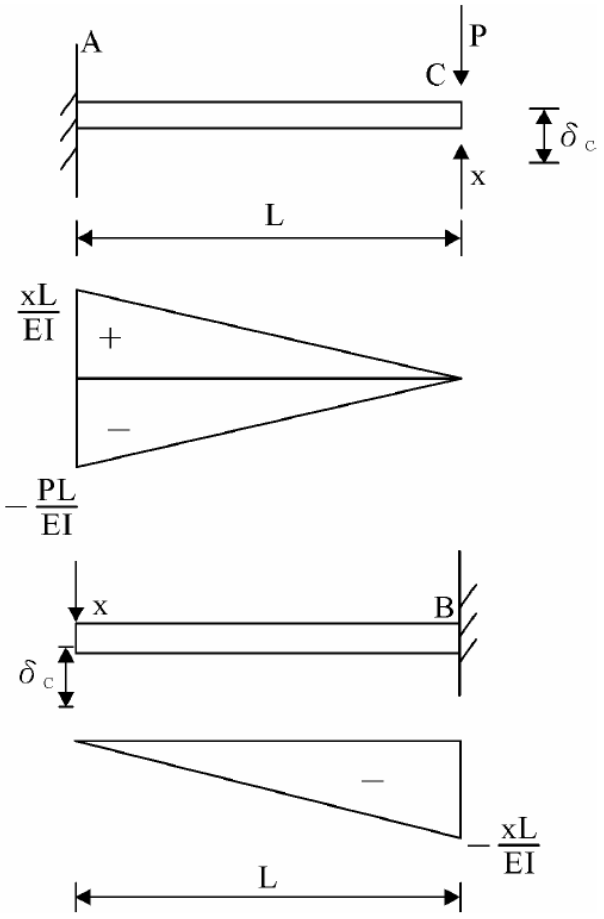
例 42

當負載 P 施加時，求固定支撐之 C 之反力(假設 EI 為常數)



【解析】

1. 假設 C 處之支承受力為 x，則取 AC 及 BC 之自由體圖且畫出彎矩之分解圖：



2. (1) 如圖 BC 圖，根據力矩面積法第二定理：

$$\delta_C = \frac{1}{EI}\left[-xL \times L \times \frac{1}{2} \times \frac{2}{3}L\right] = \frac{-xL^3}{3EI}$$

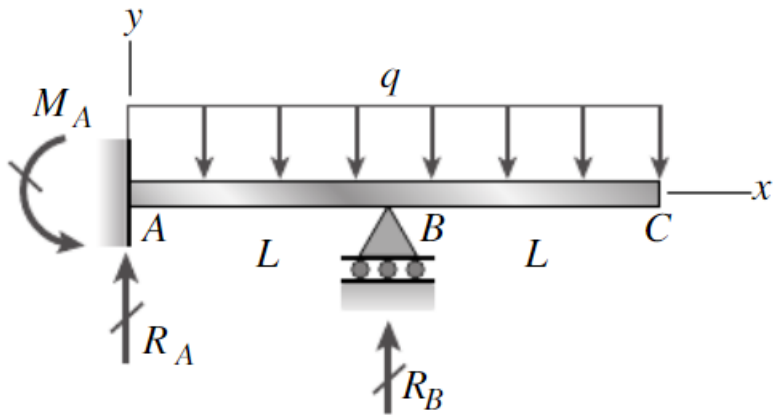
(2) 如圖 AC 圖，根據力矩面積法第二定理

$$\delta_C = \frac{1}{EI}\left[x \times L \times \frac{1}{2} \times \frac{3}{2}L + (-PL) \times L \times \frac{1}{2} \times \frac{2}{3}L\right] = \frac{xL^3}{3EI} - \frac{PL^3}{3EI}$$

$$\Rightarrow \frac{-xL^3}{3EI} = \frac{xL^3}{3EI} - \frac{PL^3}{3EI} \quad \Rightarrow x = \frac{P}{2}$$

例 43

A propped cantilever beam of length 2L with support at B is loaded by a uniformly distributed load with intensity q. Use the method of superposition to solve for all reactions. Also draw shear-force and bending-moment diagrams, labeling all critical ordinates



【105 台大土木】【106 台大工科】

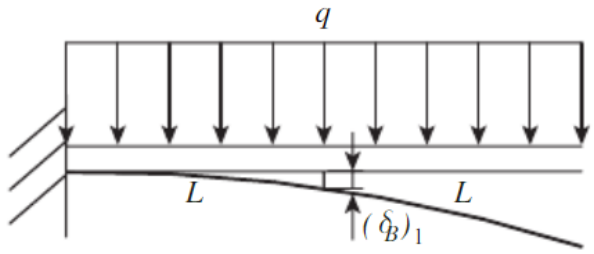
【解析】

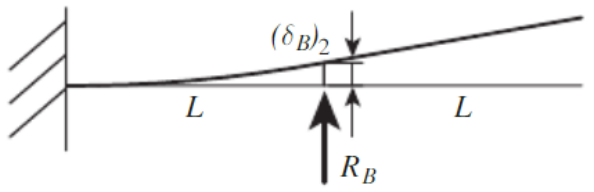
(一)內力分析由靜平衡方程式

$$R_A + R_B = qL$$

(二)變位一致性

假設 R_B 為贅力





由力矩面積法

$$(\delta_B)_1 = \frac{17qL^4}{24EI} \qquad (\delta_B)_2 = \frac{R_B L^3}{3EI}$$

變位一致性

$$(\delta_B)_1 = (\delta_B)_2 \qquad R_B = \frac{17}{8}qL$$

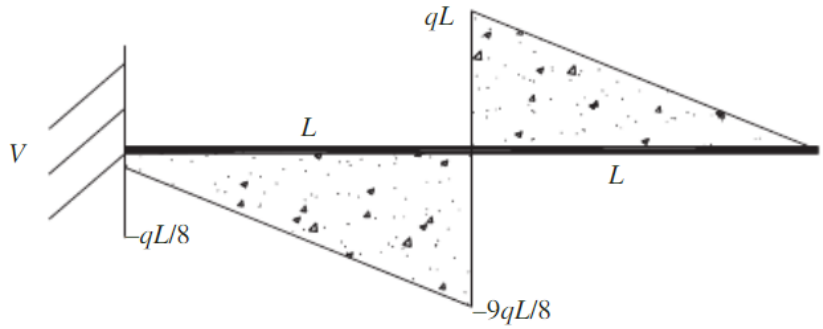
$$R_A = 2qL - R_B = -\frac{1}{8}qL$$

$$M_A = 2qL^2 - R_B L = -\frac{1}{8}qL^2$$

(三)剪力圖與彎矩圖

$$V = R_A - qx = -\frac{qL}{8} - qx \qquad (0 \le x \le L)$$

$$V = 2qL - qx \qquad (L \le x \le 2L)$$

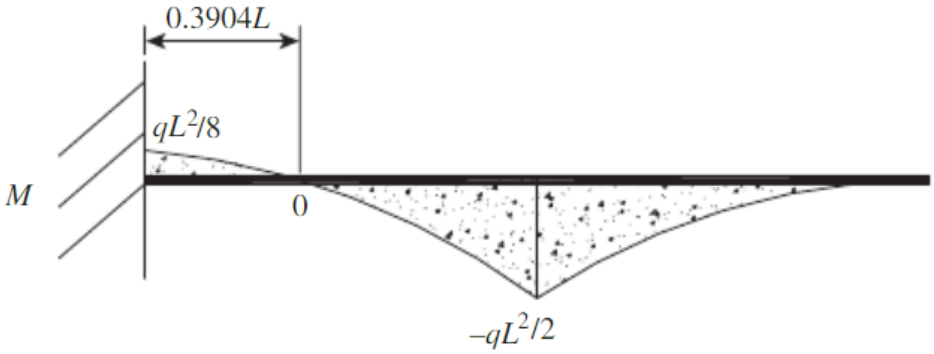


$$M = R_A x - M_A - q\frac{x^2}{2} = -\frac{qx^2}{2} - \frac{qLx}{8} + \frac{qL^2}{8}$$

$$(0 \le x \le L)$$

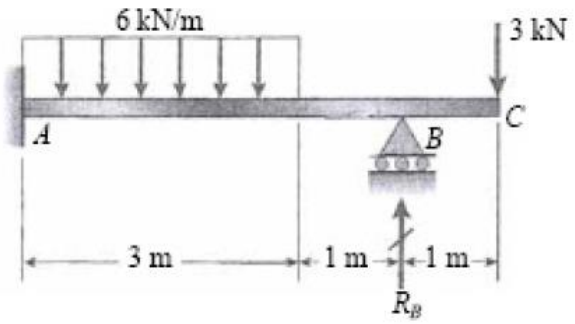
$$M = R_A x - M_A - q\frac{x^2}{2} + R_B x = \frac{-qx^2}{2} + 2qLx - 2qL^2$$

$$(L \le x \le 2L)$$



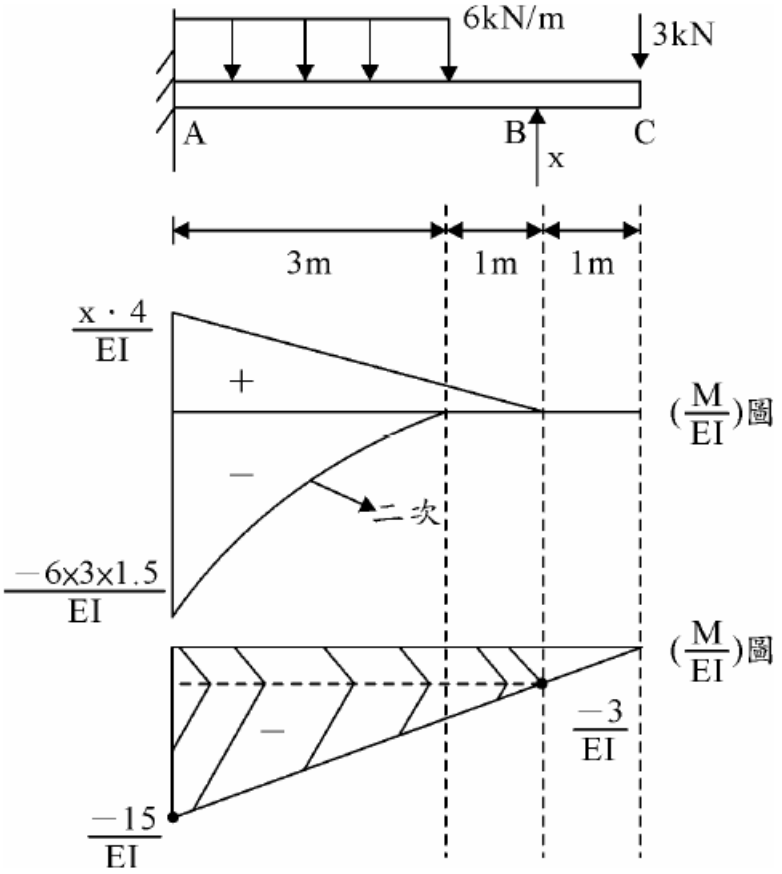
例 44

樑 ABC 其左端固定於 A 點，B 點為一簡支點，如圖所示。在承受圖示之荷重時，試求 A、B 兩支撐之反作用力。假設樑之彎曲剛度(flexural rigidity)為 EI = 4 MN-m^2。(96 年機械專利特考)



【解析】

1. 靜不定結構假設 B 處贅力 x，畫出各自分解力之$(\frac{M}{EI})$圖：



$$\delta_B = 0 = \Delta_{BA} = \frac{1}{EI}[x \cdot 4 \times 4 \times \frac{1}{2} \times \frac{8}{3} - 6 \times 3 \times 1.5 \times 3 \times \frac{1}{3} \times (3 \times \frac{3}{4} + 1)$$

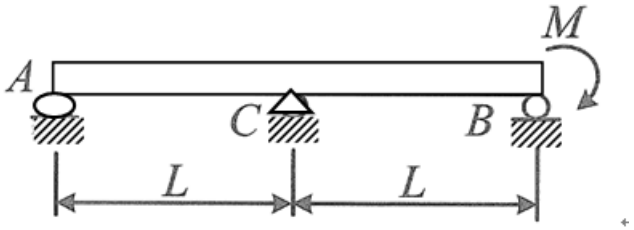
$$\underbrace{-3 \times 4 \times 4 \times \frac{1}{2}}_{\text{斜線長方形}} \underbrace{-4 \times 12 \times \frac{1}{2} \times 4 \times \frac{2}{3}}_{\text{斜線三角形}}]$$

$$\Rightarrow R_B = x = 8.24(kN)$$

$$\sum F_y = 0 \Rightarrow R_A = 12.76(kN)$$

例 45

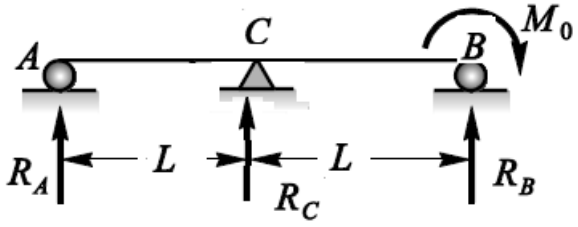
試求各支點之支承反力



【清大動機】【中正機械】

【解析】

（四）內力分析



$$\sum M_B = 0 \quad -2R_A L - R_C L - M = 0$$

$$\therefore R_C = -2R_A - \frac{M}{L}$$

$$\sum F_y = 0 \quad R_A + R_C + R_B = 0$$

$$R_B = R_A + \frac{M}{L}$$

（五）由奇異函數法

$$R_A < x >^1 + R_C < x - L >^1$$

$$EIy' = \frac{R_A}{2} < x >^2 + \frac{R_C}{2} < x - L >^2 + C_1$$

$$EIy = \frac{R_A}{6} < x >^3 + \frac{R_C}{6} < x - L >^3 + C_1 x + C_2$$

（六）帶入邊界條件

$x=0 \quad y=0 \quad C_2=0$

$x=L \quad y=0 \quad \frac{R_A L^3}{6}+C_1 L=0 \quad C_1=-\frac{R_A}{6}L^2$

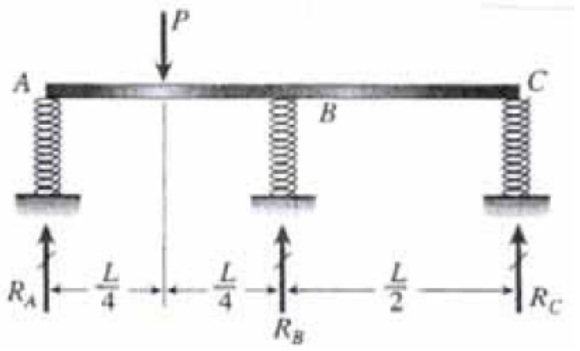
$x=2L \quad y=0 \quad \frac{4R_A L^3}{3}+\frac{R_C L^3}{6}+2C_1 L=0$

(七) 求反力

$R_A=\frac{M}{4L} \quad R_C=\frac{-3M}{2L}$

例 46

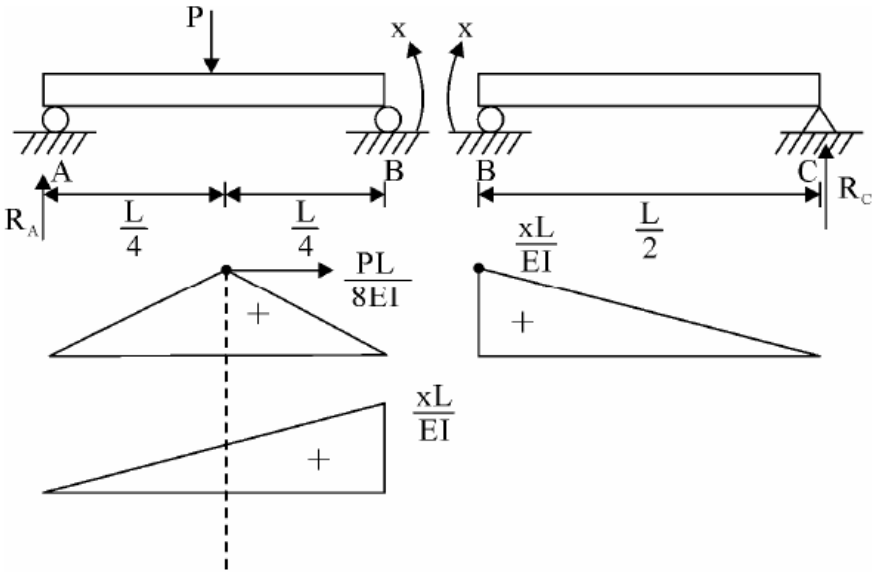
樑 ABC 放置於三根彈簧常數相同之彈簧上，如圖所示。若樑之長度 L＝4 m，其彎曲剛度 (flexural rigidity) 為 EI $=10\times10^6$ N-m^2，彈簧常數(spring constan t) 為 k $=100\times10^6$ N /m，若荷重 P＝40 kN，試求此三彈簧之反作用力 R_A、R_B、R_C。(95 年機械地方特考三等)



【清大動機所】

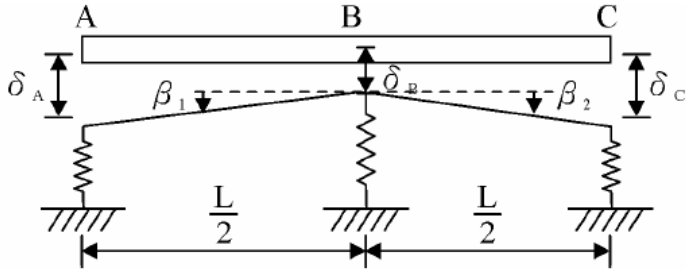
【解析】

1. **先不考慮彈簧效應，假設 A、B、C 支承均為固定支承，假設 B 受贅力彎矩 x，取 AB 段及 BC 段自由體圖：**



由 AB 段得 $\theta_1 = \dfrac{-P(\frac{L}{2})^2}{16EI} - \dfrac{x(\frac{L}{2})}{3EI}(\curvearrowright)$　由 BC 段得 $\theta_2 = \dfrac{x(\frac{L}{2})}{3EI}$

2. **考慮彈簧效應：**



則 $\theta_B^{左} = \theta_1 - \beta_1(\curvearrowright)$，$\theta_B^{右} = \theta_1 + \beta_2(\curvearrowright)$

$$\Rightarrow \theta_B^{左} = \frac{-P(\frac{L}{2})^2}{16EI} - \frac{x(\frac{L}{2})}{3EI} - \beta_1 = \theta_B^{右} = \frac{x(\frac{L}{2})}{3EI} + \beta_2 \text{——(1)}$$

且 $\beta_1 = \dfrac{\frac{\delta_A - \delta_B}{L}}{2}$，$\beta_2 = \dfrac{\frac{\delta_C - \delta_B}{L}}{2}$ ——(2)

3. **由 AB 段** $\Rightarrow R_A = k \cdot \delta_A = (x + \dfrac{PL}{4})\dfrac{2}{L}$ ——(3)

由 BC 段 $\Rightarrow R_C = k \cdot \delta_C = \dfrac{2x}{L}$ ——(4)

$R_B = k \cdot \delta_B = \dfrac{1}{L}(\dfrac{PL}{2} - 4x)$——(5)

由(1)(2)(3)(4)(5)得

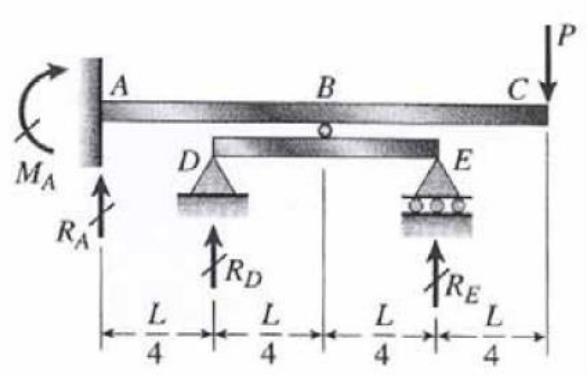
$$x = \frac{(192EI - 3kL^3) \times PL}{32(144EI + 2kL^3)} = \frac{(192 \times 10 \times 10^6 - 3 \times 100 \times 10^6 \times 4^3) \times 40 \times 4}{32 \times (144 \times 10 \times 10^6 + 2 \times 100 \times 10^6 \times 4^3)}$$

$= -6.0674 kN \cdot m$

$\Rightarrow R_A = 16.967kN(\uparrow)$，$R_B = 26.067kN(\uparrow)$，$R_C = -3.0337kN(\downarrow)$

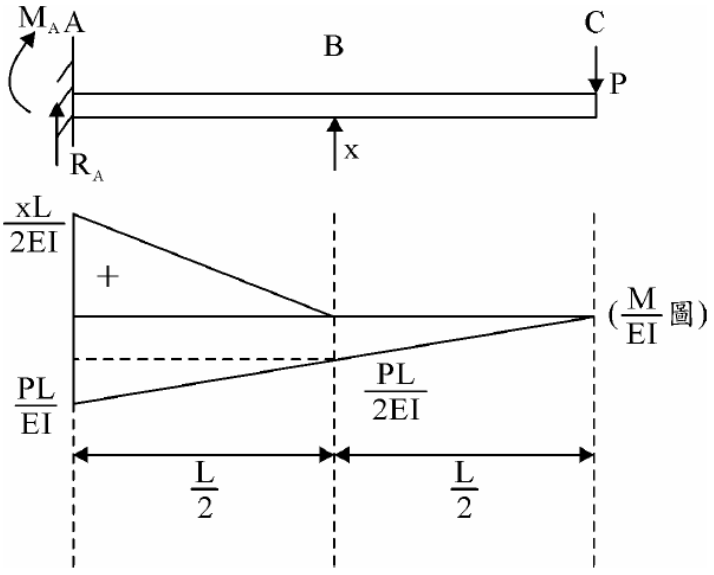
例 47

如下圖所示，懸臂樑 ABC 固定於 A 端，同時支撐於樑 DBE 之 B 點上，在 C 點處承受一垂直負載 P，如圖所示。二樑之彎曲剛度(flexural rigidity)均為 EI。試求：(1)各支撐點之反作用力(2)二樑之最大彎矩各為多少。(3)B 點之撓度(deflection)δB 之大小。(95 年機械關務三等)



【解析】

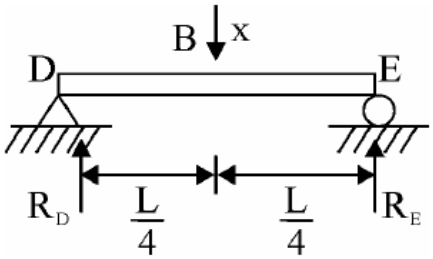
(1) 假設 B 點贅力為 x，取 ABC 自由體圖：



由力矩面積法第二定理

$$\delta_B = \Delta_{BA} = \frac{1}{EI}\left[\frac{xL}{2}\times\frac{L}{2}\times\frac{1}{2}\times\frac{L}{2}\times\frac{2}{3} - \frac{PL}{2}\times\frac{L}{2}\times\frac{L}{2}\times\frac{1}{2} - \frac{1}{2}\times\frac{PL}{2}\times\frac{L}{2}\times\frac{L}{2}\times\frac{2}{3}\right] = \frac{xL^3}{24EI} - \frac{5PL^3}{48EI}(\uparrow) \text{——①}$$

(2) 取 DE 自由體圖，由基本撓度公式：



$$\delta_B = \frac{x(\frac{L}{2})^3}{48EI} = \frac{x(L)^3}{384EI} \text{——②}$$

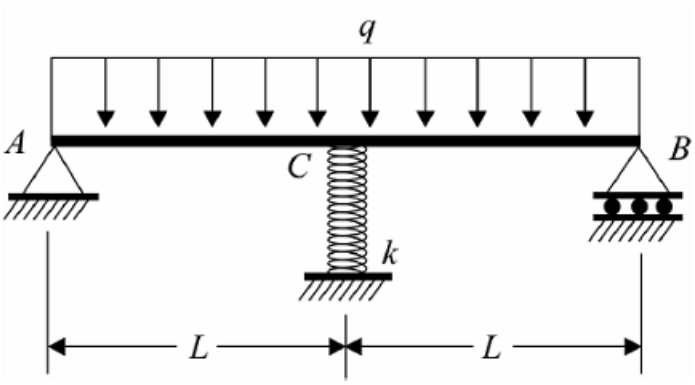
由①②可知 $\frac{xL^3}{384EI} = \frac{-xL^3}{24(EI)} + \frac{5PL^3}{48EI}(\downarrow) \Rightarrow x = \frac{40}{17}P$

(3) $R_D = R_E = \frac{20}{17}P$，$R_A = -\frac{23}{17}P(\downarrow)$，$M_A = \frac{3PL}{17}$：

$$\delta_B = \frac{x(L)^3}{384EI} = \frac{40PL^3}{6528EI}(\downarrow)$$

例 48

The beam AB shown in the figure is simply supported at A and B and supported on a spring of stiffness k at its midpoint C. The beam has flexural rigidity EI and length 2L.What should be the stiffness k of the spring in order that the maximum bending moment in the beam (due to the uniform load) will have the smallest possible value?



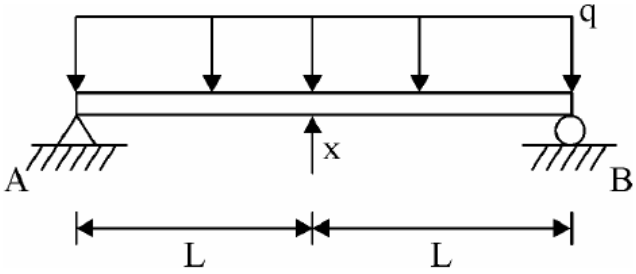
【北科】【108 台大生機】

【中文翻譯】

樑 AB 其長度為 2L，彎曲剛度(flexural rigidity)為 EI，承受一均勻分佈負荷(uniform distributed loading) q，樑在 A、B 處為簡支撐(simply supports)，在 C 處為一彈簧支撐(彈簧係數為 k)。若使產生在樑中之最大彎矩減至最低，試求 k 之值為多少？並求此時 C 點之撓度(deflection)。(以 EI、q、L 表示)。(93 年機械關務三等)

【解析】

(1) **假設 C 點贅力 x：可由基本撓度變形得知：**



A. 由 q 造成 C 點的位移：$\delta_{C_1} = \frac{5q(2L)^4}{384EI} = \frac{5qL^4}{24EI}(\downarrow)$

B. 由贅力造成 C 點的位移：$\delta_{C_2} = \frac{x(2L)^3}{48EI} = \frac{xL^3}{6EI}(\uparrow)$

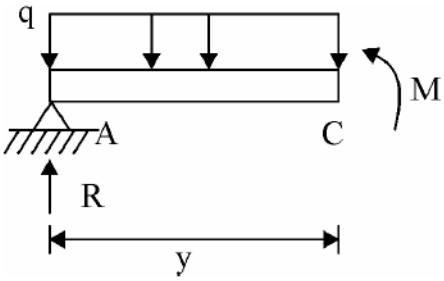
(2) **C 點位移** $\delta_C = \delta_{C_1} + \delta_{C_2} = \frac{5qL^4}{24EI} = \frac{xL^3}{6EI}$ ——①；

又 $x = K\delta_C$ ——②

由①②可得 $\delta_C = \frac{5qL^4}{4(kL^3 + 6EI)}$ $\Rightarrow x = \frac{5kqL^4}{4(kL^3 + 6EI)}$

(3) **求支承反力：**

$R_A = R_B = \frac{1}{2}[q \cdot 2L - x] = qL - \frac{x}{2}$ 令 $R = qL - \frac{x}{2}$ 取 V = 0 之自由體圖



$\Sigma F_y = 0 \Rightarrow y = \frac{R}{q}$　$\Sigma M_C = 0 \Rightarrow R \cdot y - qy \times \frac{y}{2} = M$　$\Rightarrow M = \frac{R^2}{2q}$ ——①

又因為樑內的彎矩為均佈負載所造成　$M = qL \times \frac{L}{2} - RL$——②

由①②

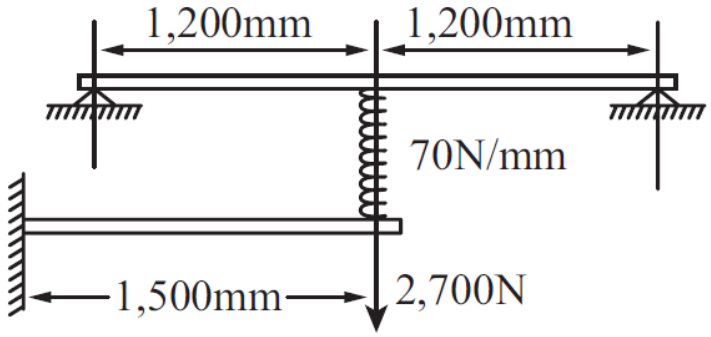
$\frac{R^2}{2q} = \frac{qL^2}{2} - RL \Rightarrow R^2 - q^2L^2 + 2qRL = 0 \Rightarrow R = (\sqrt{2} - 1)qL$

將 $R = qL - \frac{x}{2}$，$x = \frac{5kqL^4}{4(kL^3 + 6EI)}$ 代入上式

$k = 89.63\frac{EI}{L^3}$　$\delta_C = \frac{x}{k} = \frac{5qL^4}{4(kL^3 + 6EI)} = \frac{5qL^4}{382.52EI}$

例 49

如圖所示各鋼樑寬 30 mm 及厚 45 mm，試求各樑之最大撓曲，已知各樑的彈性模數 E=206900 MPa

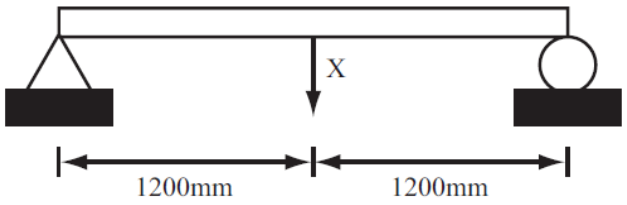


【專利特考】

【解析】

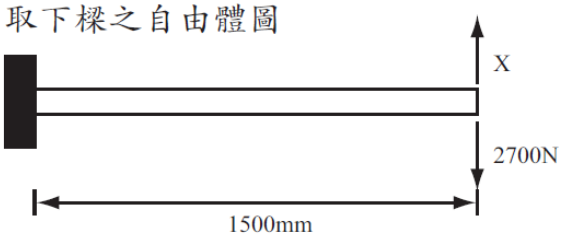
假設 y_1 為上樑之撓曲，y_2 為下樑之撓曲

取上樑之自由體圖



$$y_1=\frac{xL_1^3}{48EI}\ (\downarrow)$$

取下樑之自由體圖



$$y_2=\frac{(2700-x)\,L_2^3}{3EI}$$

由變位諧和條件 $y_2=y_1+\frac{x}{k}$

$$\Rightarrow \frac{(2700-x)\,L_2^3}{3EI}=\frac{xL_1^3}{48EI}+\frac{x}{70}\cdots\cdots(1)$$

已知 E=206900Mpa

$L_1 = 1200mm + 1200mm = 2400mm$

$L_2 = 1500mm$

$$I = \frac{x}{12} \times 30 \times 45^3 = 227812.5mm^4$$

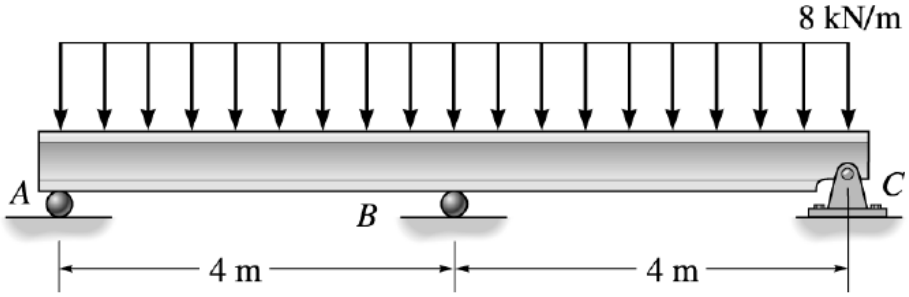
代入(1)可得 x＝1455.89N

$$y_1 = \frac{1455.89 \times (2400)^3}{48 \times 206900 \times 227812.5} = 8.896mm$$

$$y_2 = \frac{(2700 - 1455.89) \times (1500)^3}{3 \times 206900 \times 227812.5} = 26.69mm$$

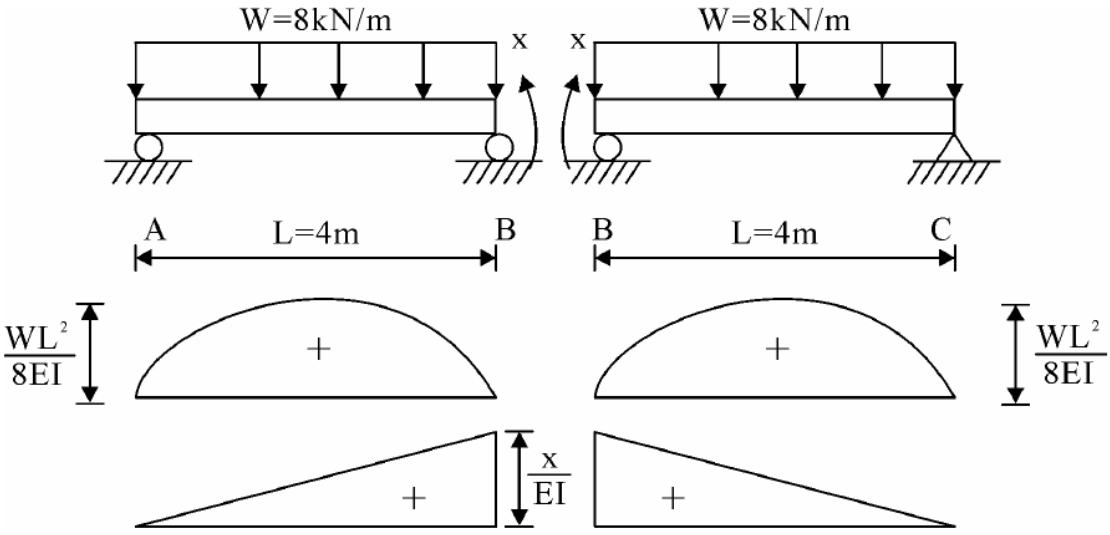
例 50

樑 ABC 其右端固定於 C 點，A、B 點為一簡支點，如圖所示。在承受圖示之荷重時，試求 A、B、C 兩支撐之反作用力。假設樑之彎曲剛度(flexural rigidity)為 EI。



【解析】

解析 1. 假設 B 處所受力矩為 x，取 AB 及 BC 自由體圖：



由 AB 段得　$\theta_B^{左} = \dfrac{-WL^3}{24EI} - \dfrac{xL}{3EI}(\curvearrowright)$——(1)

由 BC 段得　$\theta_B^{右} = \dfrac{WL^3}{24EI} + \dfrac{xL}{3EI}(\curvearrowright)$——(2)

由(1)(2)　$\theta_B^{左} = \theta_B^{右} \Rightarrow x = -\dfrac{1}{8}WL^2$

2. **求支承反力：**

由 AB 段得

$\sum M_A = 0 \Rightarrow R_B^{左} = \dfrac{5}{8}WL = \dfrac{5}{8} \times 8 \times 4 = 20kN(\uparrow)$

$\sum F_y = 0 \Rightarrow R_A = \dfrac{3}{8}WL = \dfrac{3}{8} \times 8 \times 4 = 12kN(\uparrow)$

由 BC 段得　$\sum M_B = 0 \Rightarrow R_B^{右} = \dfrac{5}{8}WL(\uparrow) = \dfrac{5}{8} \times 8 \times 4 = 20kN(\uparrow)$

$\sum F_y = 0 \Rightarrow R_C = \dfrac{3}{8}WL(\uparrow) = \dfrac{3}{8} \times 8 \times 4 = 12kN(\uparrow)$

$R_B = R_B^{右} + R_B^{左} = 20 + 20 = 40kN(\uparrow)$